The Fabius Function and Rvachev's Up-Function

Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics

Human-readable synthesis of the formal development

Vladimir Reshetnikov's `ProveIt` repository

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Abstract

This article is the human-readable exposition of the formally verified Fabius-function development in the `ProveIt` repository. It keeps three related functions separate: the bounded distribution function F, Rvachev's even compactly supported bump up, and the signed global extension $\mathcal{F}$. Their construction and uniqueness, differential and refinement laws, strict shape properties, optimal Lipschitz constants, derivative bounds, effective flatness, and precisely formalized analytic and non-elementarity results are presented together with direct pointers to their Lean proofs.

The probabilistic part identifies F and up with exact probability expressions for an infinite weighted sum of independent uniform coordinates. Finite atomic convolution laws, weak convergence, half-weighted step approximants, and centered uniform splines give proved finite models and quantitative approximation results. The analytic part develops the formal sinc and cosine products, Fourier inversion and Poisson summation, exact moment recurrences, and the Fourier–Legendre expansion. The arithmetic part gives reciprocal-power values, Thue–Morse formulas for arbitrary dyadic arguments, executable binary reduction, two-adic information, and the exact half-base q-binomial identities present in Lean.

The endpoint chapters report only the negative-Laplace decompositions, periodic correction, lower-Lambert formulas, saddle estimates, and correction terms that have named proved counterparts in the formal source. Plausible extensions, numerical observations, generic q-analogues, and other arguments not yet represented by an exact Lean theorem are kept in the companion notebooks under `docs/non-formalized-research-frontiers/`, not in this primary exposition.

Contents

1 Historical note and scope

The probability distribution now attached to the name of Fabius is older than Fabius's two-page note. It occurs in the 1935 work of Jessen and Wintner on infinite convolutions of independent random variables [1], was rediscovered several times, and entered a different mathematical culture through the compactly supported "finite function" of V. A. Rvachev. Fabius used it in 1966 as a particularly economical probabilistic construction of a nowhere analytic C^∞ function [2]. Independently, and for entirely different reasons, the Rvachevs introduced the same object in 1971 as the canonical compactly supported solution of a functional–differential equation [3], and V. A. Rvachev later developed it into the theory of *atomic functions* used in approximation theory and numerical analysis [4, 5, 6]. Rvachev's normalization emphasizes the even compactly supported bump, whereas the Fabius normalization emphasizes its cumulative distribution function. Arias de Reyna's 1982 paper gave a systematic treatment of the compactly supported function, including its Fourier product, probability interpretation, partitions of unity, derivatives, and dyadic values; an English translation appeared in 2017 [7]. His later arithmetic paper organized the exact rational sequences and denominator questions [8]. Haugland's note [9] gives a practical algorithm for evaluating F at dyadic rationals, closely related to the bit-recursion of Section 8.

The present article is not merely a paraphrase of those papers. It follows the proof dependency graph of the Lean development in the `ProveIt` repository [21]. That development first constructs the bounded distribution function, then derives the Rvachev and signed-global forms, and finally builds the arithmetic and asymptotic layers on top. This order has two advantages. It prevents identities valid for the signed extension from being silently applied to the bounded distribution, and it makes the exact dyadic algorithms independent of any numerical approximation.

The formalization also settles several questions that are only implicit, conjectural, or incorrectly normalized in the source material. Most importantly, the sharp endpoint asymptotic contains a nonconstant periodic function of an exact lower-Lambert phase. The periodic term is not a cosmetic improvement: every nonzero Fourier mode is explicit and nonzero, and an elementary formula without it fails at the error scale usually claimed. The development further proves a full inverse-Lambert expansion, establishes an exact bit-recursive evaluator for every dyadic rational, and supplies an absolutely and uniformly convergent Fourier–Legendre expansion of up.

Arias de Reyna's arithmetic paper [8] enumerates the independent occurrences of this function in chronological order, and two of them deserve mention beside Jessen–Wintner, Fabius, and the Rvachevs. In 1981 G. Kh. Kirov and G. A. Totkov arrived at it in a study of the distribution of the zeros of the derivatives of the function they wrote $\lambda(x)$ (*Godishnik Vissh. Uchebn. Zaved. Prilozhna Mat.* **17** (1981), no. 4, 157–165, in Bulgarian with English and Russian summaries); related dyadic derivative phenomena are represented in the formal Taylor and exact-evaluation APIs used below. In 1985 R. Schnabl gave a further independent construction, "Über eine C^∞-Funktion," in *Zahlentheoretische Analysis*, Lecture Notes in Mathematics 1114, Springer, Berlin (1985), 134–142. Neither paper is used in the proofs below, which is why their data are given here rather than in the bibliography; they are recorded because a reader tracing the function through the literature will meet both names.

Our convention is fixed throughout:

- F is the bounded cumulative distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported bump on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$.

All logarithms are natural logarithms. We write $L = \log 2$ in the asymptotic sections, $\mathrm{wt}_2(n)$ for the sum of the binary digits of n, and $\operatorname{sinc} z = \sin z/z$ with the removable value 1 at the origin.

1.1 A guide to the main results

The exposition follows the dependency order of the formal development. The opening chapters separate the bounded function F, the bump up, and the signed extension $\mathcal{F}$; construct the canonical function; and record the exact uniqueness and shape theorems. The probability chapter then gives the weighted-uniform representation and its three finite approximations: atomic convolution measures, half-weighted step functions, and centered uniform splines.

The Fourier and global-extension chapters collect the proved product, inversion, Poisson, derivative, bound, and flatness APIs. They lead to the exact analytic locus of F, the proved nonanalyticity regions for up and $\mathcal{F}$, the totalized inverse on $[0, 1]$, and the formal non-elementarity results. The moment and arithmetic chapters then derive the rational recurrences, reciprocal-power identities, path and composition formulas, parity and two-adic statements, exact dyadic evaluation, binary reduction, and the specialized half-base q-binomial formulas.

The later analytic chapters summarize the formal endpoint machinery: the exact negative-Laplace product and logarithmic decomposition, the periodic Fourier correction, the lower-Lambert coordinate, the proved saddle asymptotics and correction coefficients, and the rigorous obstructions to deleting the periodic term. The Fourier–Legendre, summation, and computability chapters finish the proved story. The final module guide names the Lean declarations supporting each retained theorem.

A statement appears here only when its complete mathematical content has an exact proved Lean counterpart. Material removed from earlier drafts because that test is not yet met is preserved in the research-frontier notebooks. Thus the division between this document and those notebooks is a proof-status boundary, not a judgment about mathematical plausibility or importance.

One formal object, three global functions. The bounded F is the cumulative distribution function used for probabilities and dyadic values. The compactly supported up is obtained by folding F, and its Fourier transform has the proved sinc and cosine product descriptions. The signed $\mathcal{F}$ is the global Thue–Morse continuation and satisfies the pure dilation equation on all of $\mathbb{R}$. Domains and normalizations are stated explicitly whenever a theorem passes from one presentation to another.

2 Three functions that must not be conflated

The literature often uses the same letter F for three related, but globally different, functions. That economy of notation is harmless in a local calculation and dangerous in a long proof. We therefore begin by separating them.

2.1 The bounded Fabius distribution function

Definition 2.1 (Bounded Fabius function)**.** A function $F : \mathbb{R} \to [0, 1]$ is called a *bounded Fabius function* if

$$\begin{aligned} F(x) &= 0 && (x \le 0), && (1)\\ F(x) &= 1 && (x \ge 1), && (2)\\ F(1-x) &= 1 - F(x) && (0 \le x \le 1), && (3)\\ F'(x) &= 2F(2x) && (0 \le x \le \tfrac{1}{2}), && (4) \end{aligned}$$

and F is infinitely differentiable on $\mathbb{R}$.

The derivative equation on the right half follows from symmetry:

$$F'(x) = 2F(2-2x), \qquad \tfrac{1}{2} \le x \le 1. \tag{5}$$

In particular $F'(x) > 0$ for $0 < x < 1$, once positivity in the interior is known. Thus F is a genuine cumulative distribution function, not merely a smooth interpolation between 0 and 1.

Because F is constant on both exterior half-lines and is C^∞, every derivative of positive order vanishes at both endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0 \qquad (m \ge 1). \tag{6}$$

This flatness is the source of both its extraordinary endpoint decay and its failure to be analytic there.

2.2 Rvachev's compactly supported bump

Definition 2.2 (Rvachev up-function)**.** Given F, define

$$\mathrm{up}(x) = \begin{cases} F(x+1), & x \le 0, \\ F(1-x), & x > 0. \end{cases} \tag{7}$$

Equivalently,

$$\mathrm{up}(x) = F(1-|x|). \tag{8}$$

Here the right side is interpreted using the globally bounded function F.

It follows immediately that up is even, nonnegative, supported on $[-1,1]$, and $\mathrm{up}(0) = 1$. On the unit interval the two functions are simply translates:

$$F(x) = \mathrm{up}(x-1), \qquad 0 \le x \le 1. \tag{9}$$

The graph of F is an infinitely flat sigmoid; the graph of up is its symmetric fold.

The primary exposition omits sampled plots: their decimal coordinates are not themselves Lean declarations. Exact point evaluation is developed below through rational dyadic formulas and executable evaluators.

2.3 The signed global extension

Let $\mathrm{wt}_2(n)$ denote the sum of the binary digits of n, and put

$$\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}. \tag{10}$$

The binary recurrences are

$$\mathrm{wt}_2(2n) = \mathrm{wt}_2(n), \qquad \mathrm{wt}_2(2n+1) = \mathrm{wt}_2(n) + 1, \tag{11}$$

$$\varepsilon_{2n} = \varepsilon_n, \qquad \varepsilon_{2n+1} = -\varepsilon_n. \tag{12}$$

The sequence (ε_n) is the $\{\pm 1\}$ form of the Thue–Morse sequence [14].

Definition 2.3 (Signed extension)**.** Define

$$\mathcal{F}(x) = \sum_{n \ge 0} \varepsilon_n \, \mathrm{up}(x - 2n - 1). \tag{13}$$

At a fixed x only finitely many summands can be nonzero, so this is a locally finite sum. On every block $[2b, 2b+2]$ only the b-th translate can be nonzero (and at the endpoints it too vanishes):

$$\mathcal{F}(x) = \varepsilon_b \operatorname{up}(x - 2b - 1), \qquad 2b \le x \le 2b + 2. \tag{14}$$

It vanishes for $x \le 0$, and it agrees with the bounded F for every $x \le 1$ (both functions vanish on $(-\infty, 0]$). It does *not* remain in $[0, 1]$ to the right of that interval. This is the global object that satisfies the cleanest differential equation:

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \qquad (x \in \mathbb{R}). \tag{15}$$

Arias de Reyna's arithmetic paper uses this extension. Statements about its values outside $[0, 1]$ must therefore not be transferred blindly to the bounded cumulative distribution.

Notation used in this article. F always means the bounded distribution function; up always means the compactly supported even bump; and $\mathcal{F}$ always means the signed global extension. We have $F = \mathcal{F}$ throughout $(-\infty, 1]$ (hence on $[0, 1]$), but globally they are different functions.

2.4 Exact shape: positivity, bijectivity, and unimodality

The three defining equations (1)–(4) already determine the qualitative picture, but the statements that are actually used later are the sharp ones: not $F \ge 0$ but $F(x) > 0$ exactly for $x > 0$; not $\operatorname{up} \le 1$ but $\operatorname{up}(x) = 1$ exactly at the origin. We record them here, in the form of equivalences, because the strict versions are what make the convexity and Lipschitz theorems below sharp.

Everything rests on a single formula that merges (4) and (5) into one identity valid on the whole line.

Lemma 2.4 (Unified derivative formula)**.** *For every real x,*

$$F'(x) = 2 \operatorname{up}(2x - 1). \tag{16}$$

Proof. By (8), $\operatorname{up}(2x-1) = F(1 - |2x-1|)$. For $0 \le x \le \frac{1}{2}$ one has $|2x-1| = 1 - 2x$, so $\operatorname{up}(2x-1) = F(2x)$ and (16) is (4). For $\frac{1}{2} \le x \le 1$ one has $|2x-1| = 2x - 1$, so $\operatorname{up}(2x-1) = F(2-2x)$ and (16) is (5). For $x < 0$ or $x > 1$ one has $|2x-1| > 1$, hence $1 - |2x-1| < 0$ and $\operatorname{up}(2x-1) = 0$ by (1), while $F'(x) = 0$ because F is constant on each exterior half-line by (1) and (2). □

Since $\operatorname{up} \ge 0$, formula (16) shows at once that F is nondecreasing on all of $\mathbb{R}$. It is also convenient to note that the reflection symmetry (3), stated on the unit interval, in fact holds globally:

$$F(1 - x) = 1 - F(x) \qquad (x \in \mathbb{R}). \tag{17}$$

Indeed, for $x < 0$ both sides equal 1 by (1) and (2), and for $x > 1$ both sides equal 0.

Theorem 2.5 (Positivity, strict monotonicity, bijectivity)**.** *The bounded Fabius function satisfies the following.*

1. *$F(x) > 0$ if and only if $x > 0$.*
2. *$F(x) < 1$ if and only if $x < 1$.*
3. *$F'(x) > 0$ for $0 < x < 1$, and F is strictly increasing on $[0, 1]$.*
4. *F maps $[0, 1]$ bijectively onto $[0, 1]$.*

Proof. The heart of the matter is the following doubling step: if $0 \le x \le \frac{1}{2}$ and $F(x) = 0$, then $F(2x) = 0$. To see it, observe that $F \ge 0$ everywhere, so a zero of F is a global minimum of a function differentiable on all of $\mathbb{R}$; Fermat's rule gives $F'(x) = 0$, and (4) turns this into $2F(2x) = 0$.

Suppose now that $F(x) = 0$ for some $x > 0$. Since $2^n x \to \infty$, there is a smallest integer $n \ge 0$ with $2^n x \ge \frac{1}{2}$. For $0 \le m < n$ we have $2^m x < \frac{1}{2}$, so the doubling step applies at $2^m x$ and propagates the vanishing: from $F(x) = 0$ we obtain $F(2^m x) = 0$ for every $m \le n$, and in particular $F(2^n x) = 0$. But F is nondecreasing and $2^n x \ge \frac{1}{2}$, so $F(2^n x) \ge F(\frac{1}{2}) = \frac{1}{2}$, where the midpoint value comes from (3) at $x = \frac{1}{2}$. This contradiction proves that F has no zero on the positive half-line; combined with (1) it proves (i).

Assertion (ii) follows from (i) and (17): $F(x) < 1$ if and only if $F(1 - x) > 0$, that is, if and only if $1 - x > 0$.

For (iii), let $0 < x < 1$. Then $2x - 1 \in (-1, 1)$, so $1 - |2x - 1| \in (0, 1]$, and (i) gives $\operatorname{up}(2x - 1) = F(1 - |2x - 1|) > 0$. Now (16) yields $F'(x) > 0$. A continuous function on $[0, 1]$ whose derivative is positive throughout the interior is strictly increasing on $[0, 1]$.

For (iv), F is continuous with $F(0) = 0$ and $F(1) = 1$, so the intermediate value theorem makes $F([0, 1])$ contain $[0, 1]$; since F takes values in $[0, 1]$, the image is exactly $[0, 1]$. Strict monotonicity from (iii) gives injectivity. □

The fold (8) now transfers the whole picture to Rvachev's bump, and it does so in strict form.

Theorem 2.6 (Strict unimodality of up)**.** *Rvachev's function satisfies the following.*

1. up *is strictly increasing on* $[-1, 0]$ *and strictly decreasing on* $[0, 1]$.
2. $\operatorname{up}(x) = 1$ *if and only if* $x = 0$*; the maximum value* 1 *is attained at the origin and nowhere else.*
3. $\operatorname{up}(x) = 0$ *if and only if* $x \notin (-1, 1)$*. Thus* up *is nonzero exactly on* $(-1, 1)$*, and* $\operatorname{supp} \operatorname{up} = [-1, 1]$.
4. $\operatorname{up}'(x) > 0$ *for* $-1 < x < 0$ *and* $\operatorname{up}'(x) < 0$ *for* $0 < x < 1$.

Proof. By (8), $\operatorname{up}(x) = F(1 - x)$ for $x \ge 0$. As x increases through $[0, 1]$ the argument $1 - x$ decreases through $[0, 1]$, on which F is strictly increasing by theorem 2.5(iii); hence up is strictly decreasing on $[0, 1]$, and evenness gives (i) on $[-1, 0]$.

For (ii), $\operatorname{up}(x) = F(1 - |x|)$ and theorem 2.5(ii) say that $\operatorname{up}(x) < 1$ unless $1 - |x| \ge 1$, that is, unless $x = 0$; and $\operatorname{up}(0) = F(1) = 1$. For (iii), $\operatorname{up}(x) > 0$ if and only if $1 - |x| > 0$ by theorem 2.5(i), and $\operatorname{up} \ge 0$, so the zero set is exactly the complement of $(-1, 1)$; the closed support is the closure $[-1, 1]$.

For (iv), the identity $\operatorname{up}(x) = F(1 - x)$ holds on the whole open half-line $x > 0$, hence may be differentiated there: $\operatorname{up}'(x) = -F'(1-x)$, which is negative for $0 < x < 1$ by theorem 2.5(iii). Evenness gives $\operatorname{up}'(x) = -\operatorname{up}'(-x)$, so $\operatorname{up}' > 0$ on $(-1, 0)$. □

Parts (i) and (iii) are the classical description of the bump [6, 7]; the strict forms, and the equivalences in theorem 2.5, are exactly what the formal development records.

2.5 Convexity and the maximal slope

The derivative formula transfers the monotonicity of up to the derivative of F.

Theorem 2.7 (Convexity, concavity, and the maximal slope)**.** *The derivative* F' *is nondecreasing on* $(-\infty, \frac{1}{2}]$ *and nonincreasing on* $[\frac{1}{2}, \infty)$*. It is strictly increasing on* $[0, \frac{1}{2}]$ *and strictly* *decreasing on* $[\frac{1}{2}, 1]$*. Consequently* F *is convex on* $(-\infty, \frac{1}{2}]$*, concave on* $[\frac{1}{2}, \infty)$*, strictly convex on* $[0, \frac{1}{2}]$*, and strictly concave on* $[\frac{1}{2}, 1]$*. Moreover*

$$F'(\tfrac{1}{2}) = 2 = \max_{x\in\mathbb{R}} F'(x), \tag{18}$$

and equality $F'(x) = 2$ *holds exactly at* $x = \frac{1}{2}$*.*

Formal proof. The derivative monotonicity statements are `monotoneOn_deriv_fabiusReal`, `antitoneOn_deriv_fabiusReal`, `strictMonoOn_deriv_fabiusReal`, and `strictAntiOn_deriv_fabiusReal`. The four convexity statements are `convexOn_fabiusReal_Iic_half`, `concaveOn_fabiusReal_Ici_half`, `strictConvexOn_fabiusReal_firstHalf`, and `strictConcaveOn_fabiusReal_secondHalf`. Finally `deriv_fabiusReal_half`, `deriv_fabiusReal_eq_two_iff`, and `isGreatest_deriv_fabiusReal` prove the exact maximal-slope statement. All are in `Convexity.lean`. □

2.6 The optimal Lipschitz constant

The bound $F' \le 2$ of (18) is a Lipschitz estimate, and because the bound is attained the estimate cannot be improved.

Theorem 2.8 (Two is the least Lipschitz constant)**.** *Both* F *and* up *are* 2*-Lipschitz on* $\mathbb{R}$*:*

$$|F(x) - F(y)| \le 2|x-y|, \qquad |\operatorname{up}(x) - \operatorname{up}(y)| \le 2|x-y| \qquad (x, y \in \mathbb{R}). \tag{19}$$

Moreover 2 *is the least admissible constant for each: if* $K \ge 0$ *is such that* $|F(x) - F(y)| \le K|x-y|$ *for all* x, y*, or such that* $|\operatorname{up}(x) - \operatorname{up}(y)| \le K|x-y|$ *for all* x, y*, then* $K \ge 2$*.*

Proof. By (16) and $0 \le \operatorname{up} \le 1$ we have $0 \le F' \le 2$ on all of $\mathbb{R}$, so the mean value theorem gives the first estimate in (19). The map $x \mapsto 1 - |x|$ is 1-Lipschitz, so the second estimate follows from the first through the fold (8).

For optimality, let K be admissible for F. Dividing $|F(x) - F(\frac{1}{2})| \le K|x - \frac{1}{2}|$ by $|x - \frac{1}{2}|$ and letting $x \to \frac{1}{2}$ gives $K \ge |F'(\frac{1}{2})| = 2$ by (18). For up the same limit at the same point works: the proof of theorem 2.6(iv) gives $\operatorname{up}'(\frac{1}{2}) = -F'(\frac{1}{2}) = -2$, so $K \ge 2$. □

The two extremal points are the same point in the two pictures: $\frac{1}{2}$ is the centre of the sigmoid and the midpoint of the decreasing half of the bump, and the fold sends one to the other.

Corollary 2.9 (Linear majorants at the endpoints)**.** *For all admissible arguments,*

$$F(x) \le 2x \qquad (x \ge 0), \tag{20}$$

$$1 - F(x) \le 2(1-x) \qquad (x \le 1), \tag{21}$$

$$\operatorname{up}(x) \le 2(1 - |x|) \qquad (|x| \le 1). \tag{22}$$

Proof. Applying (19) with $y = 0$ and using $F(0) = 0$ gives (20). Replacing x by $1 - x$ there and using (17) gives (21). Substituting $1 - |x| \ge 0$ into (20) and using (8) gives (22). □

The content of (20) lies on $[0, \frac{1}{2}]$; for $x \ge \frac{1}{2}$ it is weaker than $F \le 1$. Its value is that it holds with no hypothesis at all on the size of x, so it can be used as a linear majorant in any estimate near the left endpoint, and (22) says that the bump vanishes at least linearly at both ends of its support. These majorants and the optimality of the constant 2 are contributions of the formal development; the plain 2-Lipschitz bound is used implicitly throughout [7, 6].

3 Existence and uniqueness by a contraction

The most economical construction of F is a Banach fixed-point argument. It has an additional advantage: the contraction estimate later gives a clean uniqueness theorem, without presupposing Fourier analysis or probability.

3.1 The admissible function space

Let $I = [0, 1]$ and let $C(I)$ carry the supremum norm. Define $\mathcal{A} \subset C(I)$ by

$$\mathcal{A} = \{f \in C(I) : 0 \le f \le 1,\ f(1 - x) = 1 - f(x)\}. \tag{23}$$

The set $\mathcal{A}$ is closed in the Banach space $C(I)$, hence complete. It is nonempty; for example $f(x) = x$ belongs to it.

Extend $f \in \mathcal{A}$ to a continuous function $\widetilde{f}$ on $\mathbb{R}$ by clamping the argument to I:

$$\widetilde{f}(t) = \begin{cases} f(0), & t \le 0, \\ f(t), & 0 \le t \le 1, \\ f(1), & t \ge 1. \end{cases} \tag{24}$$

Symmetry gives

$$\int_0^1 f(t)\,\mathrm{d}t = \frac{1}{2}. \tag{25}$$

Indeed, replacing t by $1 - t$ and adding the two integrals gives $2\int_0^1 f = 1$.

3.2 The integral transform

For $f \in \mathcal{A}$, define Tf by

$$(Tf)(x) = \begin{cases} \displaystyle\int_0^{2x} \widetilde{f}(t)\,\mathrm{d}t, & 0 \le x \le \frac{1}{2}, \\ \displaystyle 1 - \int_0^{2-2x} \widetilde{f}(t)\,\mathrm{d}t, & \frac{1}{2} < x \le 1. \end{cases} \tag{26}$$

The two definitions agree at $x = 1/2$ by (25). Positivity and the upper bound $f \le 1$ show that $0 \le Tf \le 1$, and a direct substitution shows $(Tf)(1 - x) = 1 - (Tf)(x)$. Hence $T : \mathcal{A} \to \mathcal{A}$.

The central estimate is stronger than the naive integral bound.

Lemma 3.1 (Contraction estimate). *For $f, g \in \mathcal{A}$,*

$$\|Tf - Tg\|_\infty \le \frac{1}{2}\|f - g\|_\infty. \tag{27}$$

Proof. Write $h = f - g$. Since both f and g have integral $1/2$, $\int_0^1 h = 0$. For $0 \le y \le 1$ we may therefore estimate the primitive in two ways:

$$\left|\int_0^y h(t)\,\mathrm{d}t\right| \le y\|h\|_\infty,$$
$$\left|\int_0^y h(t)\,\mathrm{d}t\right| = \left|\int_y^1 h(t)\,\mathrm{d}t\right| \le (1 - y)\|h\|_\infty.$$

Consequently

$$\left|\int_0^y h(t)\,\mathrm{d}t\right| \le \min(y, 1 - y)\|h\|_\infty \le \frac{1}{2}\|h\|_\infty.$$

Every value of $Tf - Tg$ is one of these primitives, possibly with a minus sign. Taking the supremum proves (27). □

Theorem 3.2 (Existence of the bounded Fabius function)**.** *The transform T has a unique fixed point $f_* \in \mathcal{A}$. Extending f_* by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$ produces the bounded Fabius function F.*

Proof. The space $\mathcal{A}$ is complete and T is a strict contraction with constant $1/2$, so Banach's fixed-point theorem gives a unique fixed point. On $[0, 1/2]$ its fixed-point equation is

$$F(x) = \int_0^{2x} F(t)\,\mathrm{d}t. \tag{28}$$

Differentiating gives $F'(x) = 2F(2x)$. The second half of (26) supplies symmetry and the matching right-hand formula. The endpoint values are immediate. □

3.3 The differential equation for up and smoothness

Folding (4) and (5) yields the refinement-type differential equation

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big) \qquad (x \in \mathbb{R}). \tag{29}$$

At $x = 0$ the two one-sided derivatives glue because $\mathrm{up}(1) = \mathrm{up}(-1) = 0$. Once up is continuously differentiable, (29) bootstraps regularity: if $\mathrm{up} \in C^r$, then the right side is C^r, so $\mathrm{up} \in C^{r+1}$. Induction gives $\mathrm{up} \in C^\infty$ and then $F \in C^\infty$.

Proposition 3.3 (Basic shape)**.** *The functions F and* up *have the following properties.*

1. *F is strictly increasing on $(0, 1)$ and satisfies $0 < F(x) < 1$ there.*
2. up *is even, strictly positive on $(-1, 1)$, and zero outside $[-1, 1]$.*
3. $\int_{-1}^{1} \mathrm{up}(x)\,\mathrm{d}x = 1$.
4. *Every derivative of* up *vanishes at $x = \pm 1$.*

Proof. For $0 < x \le 1/2$, (4) gives $F'(x) = 2F(2x)$. Positivity follows first near $1/2$ from $F(1/2) = 1/2$, and then propagates toward zero by repeatedly doubling the argument; symmetry handles the right half. This proves strict monotonicity and, by (8), positivity of up in the interior of its support. Finally,

$$\int_{-1}^{1} \mathrm{up}(x)\,\mathrm{d}x = 2\int_0^1 F(t)\,\mathrm{d}t = 1,$$

where the middle identity uses the fold and the last uses symmetry. Flatness at the support endpoints follows from smoothness and the fact that up is identically zero just outside the support. □

3.4 The original Rvachev characterization

The formal predicate `IsOriginalFabius` packages the source characterization exactly. For a function $\varphi : \mathbb{R} \to \mathbb{R}$ and a scale k, it requires C^∞-smoothness, topological support $[-1, 1]$, strict positivity on $(-1, 1)$, the normalization $\varphi(0) = 1$, positivity of k, and

$$\varphi'(x) = k\big(\varphi(2x+1) - \varphi(2x-1)\big) \qquad (x \in \mathbb{R}). \tag{30}$$

The positivity of the scale is redundant: the smart constructor `IsOriginalFabius.mk_of_derivative_law` derives it from the other hypotheses.

Theorem 3.4 (Original characterization)**.** *If* `IsOriginalFabius` φ k*, then*

$$k = 2, \qquad \varphi = \mathrm{up}\,. \tag{31}$$

Conversely, the canonical up *satisfies this predicate at scale* 2*; in particular there is exactly one original solution.*

Formal proof. The scale is fixed by `originalFabius_scale_eq_two`; the canonical witness is `canonical_isOriginalFabius`. Uniqueness and the displayed characterization are `originalFabius_eq_canonical`, `isOriginalFabius_iff_eq_canonical`, and `existsUnique_originalFabius` in `OriginalUniqueness.lean`. □

The fold loses only the unused right tail of a bounded candidate. Precisely,

$$\mathrm{up}_G = \mathrm{up}_H \quad\Longleftrightarrow\quad G(t) = H(t) \text{ for every } t \le 1. \tag{32}$$

Thus a bounded candidate is canonical exactly when its fold satisfies the original characterization at scale 2 and its right tail is constantly 1. These are the exact declarations `rvachevUp_eq_iff_eqOn_Iic_one`, `isFabius_iff_isOriginalFabius_rvachevUp_and_rightTail`, and `isOriginalFabius_iff_existsUnique_isFabius` in `OriginalUniqueness.lean`.

4 The probability law behind the functions

The fixed-point construction is analytic. A probabilistic construction explains the same self-similarity almost without calculation and makes monotonicity and positivity transparent.

4.1 An infinite weighted sum of uniforms

Let $U_0, U_1, \ldots$ be independent random variables, each uniformly distributed on $[0, 1]$, and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \tag{33}$$

The series converges absolutely for every outcome and satisfies $0 \le X \le 1$. Removing the first coordinate gives an independent copy X' of X, and

$$X = \frac{U_0 + X'}{2}. \tag{34}$$

Reflecting every coordinate, $U_n \mapsto 1 - U_n$, preserves the product measure and sends X to $1 - X$.

Let $H(x) = \mathbb{P}(X \le x)$ be the cumulative distribution function. Conditioning on U_0 in (34) gives

$$H(y) = \int_0^1 H(2y - u)\,\mathrm{d}u. \tag{35}$$

For $0 \le y \le 1/2$, the integrand vanishes when $u > 2y$, so

$$H(y) = \int_0^{2y} H(t)\,\mathrm{d}t. \tag{36}$$

Reflection gives $H(1 - y) = 1 - H(y)$. Thus the restriction of H to $[0, 1]$ is a fixed point of the contraction T from section 3.

Theorem 4.1 (Global probabilistic representation). *For every $x, y \in \mathbb{R}$,*

$$F(x) = \mathbb{P}\left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \le x\right), \tag{37}$$

$$\mathrm{up}(y) = \mathbb{P}\left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \le 1 - |y|\right). \tag{38}$$

The second identity includes both tails: if $|y| \ge 1$, both sides vanish.

Formal proof. The product-space construction proves continuity and measurability of the uniformly convergent coordinate sum, identifies its pushforward distribution, establishes the head–tail self-similarity and the coordinate-reflection law, and invokes uniqueness of the bounded Fabius fixed point. The exact canonical specializations are `fabiusReal_eq_weightedSum_probability` and `rvachevUp_eq_weightedSum_probability_global` in `ProbabilityRepresentation.lean`. □

4.2 Finite atomic laws and weak convergence

There is also a completely discrete approximation theorem. For $k \ge 0$, let β_k be the probability measure assigning mass $1/2$ to each of $\pm 2^{-k-1}$. Starting from $\nu_0 = \delta_0$, define

$$\nu_{n+1} = \nu_n * \beta_{n+1}^{*(n+1)}. \tag{39}$$

Thus ν_n is a finite atomic probability measure. Its characteristic function is the finite weighted cosine product

$$\widehat{\nu}_n(t) = \prod_{k=0}^{n-1} \left(\cos \frac{t}{2^{k+2}}\right)^{k+1}. \tag{40}$$

The indexing in (39) is worth noticing: the new scale at stage $n+1$ is 2^{-n-2}, and it occurs with multiplicity $n+1$.

Let μ_{up} denote the formal measure `rvachevMeasure`; by definition it is Lebesgue measure with density ofReal $\circ$ up. The finite laws converge weakly to μ_{up}:

$$\nu_n \Longrightarrow \mu_{\mathrm{up}}. \tag{41}$$

Equivalently, for every bounded continuous real-valued map g,

$$\lim_{n\to\infty} \int_{\mathbb{R}} g(x)\, \mathrm{d}\nu_n(x) = \int_{\mathbb{R}} g(x)\, \mathrm{d}\mu_{\mathrm{up}}(x). \tag{42}$$

The same test-function theorem is formalized for complex-valued, and more generally real-or-complex-like, bounded continuous functions.

Formal proof. The definitions are `centeredBernoulliMeasure` and `finiteConvolutionMeasure`. Their probability-measure instances and (40) are proved in `EarlyApproximants.lean`. Pointwise convergence of the characteristic functions, weak convergence (41), and the test-function formulations (42) are respectively `finiteConvolutionMeasure_charFun_tendsto`, `finiteConvolutionProbability_tendsto_fabius`, and `integral_finiteConvolutionMeasure_tendsto_fabius` in `WeakConvergence.lean`. □

4.3 Combinatorial step functions and the half-weight convention

A second finite model replaces each atom by a narrow interval. Define polynomials with nonnegative integer coefficients by

$$\mathcal{P}_0(X) = 1, \qquad \mathcal{P}_m(X) = \mathcal{P}_{m-1}(X^2)(1+X)^m \quad (m \geq 1). \tag{43}$$

They also have the finite product form

$$\mathcal{P}_m(X) = \prod_{j=1}^{m} (1 + X + \cdots + X^{2^j - 1}). \tag{44}$$

Writing $\mathcal{P}_m(X) = \sum_r a_{m,r} X^r$, the coefficient $a_{m,r}$ counts the tuples

$$(s_1, \ldots, s_m), \qquad 0 \leq s_j \leq 2^j - 1, \qquad s_1 + \cdots + s_m = r. \tag{45}$$

Moreover,

$$g_m := \deg \mathcal{P}_m = 2^{m+1} - m - 2, \qquad \mathcal{P}_m(1) = 2^{\binom{m+1}{2}}. \tag{46}$$

For $\alpha < \beta$, let $\mathbf{1}^*_{[\alpha,\beta]}$ be 1 in the interior, 1/2 at either endpoint, and 0 elsewhere. Put

$$I_{m,r} = \left[\frac{2r - 1 - g_m}{2^{m+1}}, \frac{2r + 1 - g_m}{2^{m+1}} \right] \tag{47}$$

and

$$\mathcal{W}_m(x) = \frac{2^m}{2^{\binom{m+1}{2}}} \sum_{r=0}^{g_m} a_{m,r} \, \mathbf{1}^*_{I_{m,r}}(x). \tag{48}$$

Theorem 4.2 (Corrected step approximation)**.** *For every $m \geq 0$ and $x \in \mathbb{R}$,*

$$0 \leq \mathcal{W}_m(x) \leq 1, \qquad \mathcal{W}_m(0) = 1, \qquad \int_{\mathbb{R}} \mathcal{W}_m(x) \, \mathrm{d}x = 1.$$

The function $\mathcal{W}_m$ is nondecreasing on $(-\infty, 0)$ and nonincreasing on $(0, \infty)$. For every real x,

$$\mathcal{W}_m(x) \longrightarrow \mathrm{up}(x). \tag{49}$$

The proof of pointwise convergence passes through a useful exact comparison. Set $h_m = 2^{-m-1}$, and use the finite atomic law ν_m from (39). Whenever $a \leq b$,

$$\nu_m\big((a + h_m, b - h_m]\big) \;\leq\; \int_a^b \mathcal{W}_m(x) \, \mathrm{d}x \;\leq\; \nu_m\big((a - h_m, b + h_m]\big). \tag{50}$$

Consequently,

$$\lim_{m\to\infty} \int_a^b \mathcal{W}_m(x) \, \mathrm{d}x = \int_a^b \mathrm{up}(x) \, \mathrm{d}x. \tag{51}$$

Monotonicity on the two open half-lines, the exact value at the origin, and continuity of up then upgrade these interval limits to (49).

Formal proof. The polynomial recursion, product, degree, total coefficient mass, and bounded-partition interpretation are respectively proved by `approximationPolynomial_succ`, the definition `approximationPolynomial`, `approximationPolynomial_natDegree`, `approximationDegree_eq_sub`, `approximationPolynomial_eval_one`, and `approximationPolynomial_coeff_eq_card` in `EarlyApproximants.lean`. The finite atomic description is identified with `finiteConvolutionMeasure` by `polynomialMeasure_eq_finiteConvolutionMeasure` in `EarlyMeasureBridge.lean`.

For nonnegativity, see `stepApproximant_nonneg` in `EarlyApproximants.lean`. The remaining analytic assertions are `stepApproximant_le_one`, `stepApproximant_monotoneOn_Iio`, `stepApproximant_antitoneOn_Ioi`, `stepApproximant_apply_zero`, and `stepApproximant_tendsto_fabius` in `StepApproximationLimit.lean`. The normalization, the two sides of (50), and (51) are `integral_stepApproximant`, `finiteConvolutionMeasure_inner_le_intervalIntegral_stepApproximant`, `intervalIntegral_stepApproximant_le_finiteConvolutionMeasure_outer`, and `intervalIntegral_stepApproximant_tendsto_of_le` in `StepMeasureBridge.lean`. □

Warning 4.3 (Why the endpoint convention is explicit). The half-weighted cell representative agrees almost everywhere with the ordinary interval indicator, so it has the same integral. Its point values are nevertheless part of the formal statement: adjacent cells share endpoints, and each contributes exactly half of its coefficient there. In particular, the corrected representative satisfies $\mathcal{W}_m(0) = 1$ at every level. These facts are formalized by `halfEndpointIntervalIndicator_ae_eq_indicator`, `intervalIntegral_halfEndpointIntervalIndicator_of_le`, and `stepApproximant_apply_zero`.

4.4 Centered finite splines and an explicit error bound

The preceding finite random sums also give a completely explicit polynomial approximation to F. Put

$$T_p = \sum_{i=0}^{p-1} \frac{U_i}{2^{i+1}}, \qquad C_p(x) = \mathbb{P}\left(T_p \le x - 2^{-p-1}\right), \tag{52}$$

where $p \ge 1$. The shift by half of the omitted tail is important: it centers the one-sided approximation and improves its error by a factor of two.

Let

$$N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right), \qquad \varepsilon_r = (-1)^{\mathrm{wt}_2(r)}, \tag{53}$$

and interpret a sum with $N_p(x) = 0$ as empty. Define

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)^p. \tag{54}$$

Equivalently,

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r \ge 0} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)_+^p, \tag{55}$$

where only the terms $r < N_p(x)$ can be nonzero and $t_+ = \max(t, 0)$. The cutoff in (53) is a safe natural-number form of the inclusive upper limit

$$0 \le r \le \left\lfloor 2^p x - \frac{1}{2} \right\rfloor.$$

It handles the empty case without assigning a negative upper endpoint to a finite sum.

Theorem 4.4 (Centered spline as a finite distribution function)**.** *For $p \ge 1$ and $0 \le x \le 1$,*

$$S_p(x) = C_p(x). \tag{56}$$

Consequently $0 \le S_p(x) \le 1$ on the unit interval.

Proof. For $p = 1$, $T_1 = U_0/2$, and direct evaluation gives

$$C_1(x) = \max\left(0, \min\left(1, 2x - \frac{1}{2}\right)\right).$$

Formula (55) gives exactly the same broken-linear function: the term $r = 0$ starts at $x = 1/4$, and the term $r = 1$ subtracts the excess after $x = 3/4$.

For the induction step, split off the first uniform coordinate. The finite sums satisfy

$$T_{p+1} \overset{\mathrm{d}}{=} \frac{U_0 + T_p'}{2},$$

where T_p' is an independent copy of T_p. Conditioning on U_0 and accounting for the midpoint shifts gives the exact smoothing recurrence

$$C_{p+1}(x) = \int_0^1 C_p(2x - u)\,\mathrm{d}u. \tag{57}$$

On the spline side, integrate each truncated power in (55). The two binary children $2r$ and $2r + 1$ have opposite Thue–Morse signs, because $\varepsilon_{2r} = \varepsilon_r$ and $\varepsilon_{2r+1} = -\varepsilon_r$. After the change of scale this yields

$$S_{p+1}(x) = \int_0^1 S_p(2x - u)\,\mathrm{d}u. \tag{58}$$

The normalization is exactly the one required by

$$\sum_{r=0}^{2^p-1} \varepsilon_r r^p = (-1)^p 2^{\binom{p}{2}} p!,$$

while every lower power sum vanishes. Thus the two recurrences and their common base case prove (56). □

Theorem 4.5 (Quantitative centered approximation). *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$F(x - 2^{-p-1}) \leq S_p(x) \leq F(x + 2^{-p-1}), \tag{59}$$

and hence

$$|S_p(x) - F(x)| \leq 2^{-p}. \tag{60}$$

Proof. Write $X = T_p + R_p$, where the omitted tail satisfies

$$0 \leq R_p = \sum_{i=p}^{\infty} \frac{U_i}{2^{i+1}} \leq 2^{-p}.$$

If $X \leq x - 2^{-p-1}$, then certainly $T_p \leq x - 2^{-p-1}$. Conversely, if $T_p \leq x - 2^{-p-1}$, then $X \leq x + 2^{-p-1}$. Taking probabilities and using (37) and (56) proves (59).

The bounded Fabius function is globally 2-Lipschitz. Indeed, on the open unit interval its derivative is a rescaled value of up (or of the signed extension), whose absolute value is at most 1, and outside the unit interval F is constant. Thus

$$|F(x \pm 2^{-p-1}) - F(x)| \leq 2 \cdot 2^{-p-1} = 2^{-p}.$$

Combining these two inequalities with the sandwich gives (60). □

The bound is uniform on the closed interval and includes both endpoints. It is stronger than mere weak convergence of the finite distributions, and it is the quantitative input for the computability proof in section 19.

The same finite formula also has the correct signed global behavior. For $x \le 0$ its safe cutoff is empty, so $S_p(x) = \mathcal{F}(x) = 0$. On the nonnegative half-line, binary block translation and midpoint reflection extend the local identification. Consequently, for every $p \ge 1$,

$$\sup_{x\in\mathbb{R}} |S_p(x) - \mathcal{F}(x)| \le 2^{-p}, \qquad S_p \longrightarrow \mathcal{F} \quad \text{uniformly on } \mathbb{R}. \tag{61}$$

Indeed, every nonnegative two-unit block is a common signed copy of the first block, and on the second unit cell both functions use the same complement. Neither operation changes the absolute error in (60); the nonpositive half-line is exact. This global uniform form strengthens and packages the pointwise analytic input used by the complex-shift and discrete-limit results below.

5 Fourier products, inversion, and Poisson summation

Use the convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i} xz}\,\mathrm{d}x. \tag{62}$$

The formal Fourier transform of up is entire and has two exact infinite-product descriptions.

Theorem 5.1 (Sinc and cosine products)**.** *For every $z \in \mathbb{C}$,*

$$\widehat{\mathrm{up}}(z) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi z}{2^n}\right), \tag{63}$$

$$= \prod_{m=1}^{\infty} \left(\cos\frac{\pi z}{2^m}\right)^m. \tag{64}$$

The transform also satisfies

$$\widehat{\mathrm{up}}(z) = \operatorname{sinc}(\pi z)\widehat{\mathrm{up}}(z/2). \tag{65}$$

Formal proof. The finite-product refinement and its infinite limit are `rvachevFourier_eq_finite_product` and `rvachevFourier_eq_product` in `FourierProduct.lean`. The entire-function theorem is `rvachevFourier_differentiable_analytic` in `FourierAnalytic.lean`. The weighted cosine product is `rvachevFourierProduct_eq_cosineProduct` in `OriginalPaperSupplement.lean`; its finite telescoping identity is proved internally in the same module. □

A consequence needed for Poisson summation is the formally proved integer-zero statement

$$\widehat{\mathrm{up}}(m) = 0 \qquad (m \in \mathbb{Z} \setminus \{0\}). \tag{66}$$

This is exactly `rvachevFourier_int_eq_zero`. No converse or zero-multiplicity classification is asserted here.

5.1 Rapid decay and inversion

For every $N \in \mathbb{N}$ there is a constant $C_N > 0$ such that

$$|\widehat{\mathrm{up}}(\xi)| \le C_N(1 + |\xi|)^{-N} \qquad (\xi \in \mathbb{R}), \tag{67}$$

and the stronger formal theorem gives polynomially weighted bounds for every real-frequency derivative. Fourier inversion holds in the exact form

$$\mathrm{up}(x) = \int_{\mathbb{R}} \widehat{\mathrm{up}}(\xi) \mathrm{e}^{2\pi \mathrm{i} x\xi}\, \mathrm{d}\xi. \tag{68}$$

These are `rvachevFourier_real_rapidDecay`, `rvachevFourier_real_iteratedDeriv_rapidDecay`, and `rvachev_fourier_inversion_analytic` in `PoissonSummation.lean` and `FourierAnalytic.lean`.

5.2 Poisson summation and partitions of unity

For $u > 0$ and $t \in \mathbb{R}$, the formal Poisson identity is

$$\sum_{k\in\mathbb{Z}} \mathrm{up}(t + uk) = \frac{1}{u} \sum_{k\in\mathbb{Z}} \widehat{\mathrm{up}}(k/u)\mathrm{e}^{2\pi \mathrm{i} kt/u}. \tag{69}$$

It is `rvachev_poisson_summation`. Combining it with (66) gives, for every positive integer m,

$$\sum_{k\in\mathbb{Z}} \mathrm{up}\left(t + \frac{k}{m}\right) = m, \tag{70}$$

and in particular

$$\sum_{k\in\mathbb{Z}} \mathrm{up}(t + k) = 1. \tag{71}$$

These are `rvachev_partition_one_over_nat` and `rvachev_partition_unity` in `PoissonSummation.lean`.

5.3 Moment expansion of the Fourier transform

With

$$c_n = \int_{-1}^{1} x^{2n} \,\mathrm{up}(x)\, \mathrm{d}x, \tag{72}$$

the exact entire series is

$$\widehat{\mathrm{up}}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n}. \tag{73}$$

This is `rvachevFourier_eq_momentSeries` in `AnalyticMoments.lean`.

6 The global extension, derivatives, flatness, and inverse

6.1 The signed extension and all iterated derivatives

The translate series (13) is locally finite. It agrees with the bounded F on $[0, 1]$, vanishes on $(-\infty, 0]$, and satisfies

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \qquad (x \in \mathbb{R}). \tag{74}$$

These facts are `extendedFabius_eq_fabiusReal`, `extendedFabius_eq_zero_of_nonpos`, and `extendedFabius_hasDerivAt` in `GlobalExtension.lean`.

Iterating the dilation law gives the exact identities

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x), \tag{75}$$

$$\mathrm{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(x + 1)) \qquad (-1 \le x \le 1). \tag{76}$$

They are iteratedDeriv_extendedFabius and iteratedDeriv_rvachev. The global extension obeys

$$|\mathcal{F}(x)| \le 1, \tag{77}$$

by abs_extendedFabius_le_one. Hence, for all $m \in \mathbb{N}$ and $x \in \mathbb{R}$,

$$|F^{(m)}(x)| \le 2^{\binom{m+1}{2}}, \tag{78}$$

and this bound is attained at $x = 2^{-m}$:

$$F^{(m)}(2^{-m}) = 2^{\binom{m+1}{2}}. \tag{79}$$

The exact bound, attainment, and supremum statements are abs_iteratedDeriv_fabiusReal_le, iteratedDeriv_fabiusReal_inv_two_pow_all, and isGreatest_abs_iteratedDeriv_fabiusReal_all in BoundedDerivatives.lean.

6.2 Effective flatness

For every $n \in \mathbb{N}$, $x \ge 0$, and $2^n x \le 1$, the formal development proves both

$$F(x) \le 2^{\binom{n+1}{2}} x^n, \tag{80}$$

$$F(x) \le \frac{2^{\binom{n+1}{2}}}{n!} x^n. \tag{81}$$

These are fabiusReal_le_two_pow_mul_pow in EffectiveFlatness.lean and fabiusReal_le_two_pow_div_factorial_mul_pow in SharpFlatness.lean. Their reflected forms at 1 and their folded forms for up are also formalized in those modules. In particular, for each natural power,

$$F(x) = o(x^n) \qquad (x \to 0),$$

as stated by fabiusReal_isLittleO_pow_at_zero in FabiusFlatness.lean.

6.3 The precise analytic statements

The exact real-analytic locus of the bounded function is

$$F \text{ is analytic at } x \quad \Longleftrightarrow \quad x \notin [0,1]. \tag{82}$$

This is fabius_analyticAt_iff, with canonical specialization canonical_fabius_analyticAt_iff. For the other global forms the current proved statements are deliberately one-sided:

$$\mathrm{up} \text{ is not analytic at any } x \in [-1,1], \tag{83}$$

$$\mathcal{F} \text{ is not analytic at any } x \in [0,2). \tag{84}$$

They are canonical_rvachev_not_analyticAt and extendedFabius_not_analyticAt in NowhereAnalytic.lean. No stronger global locus is inferred here.

6.4 The totalized inverse

Let $I : \mathbb{R} \to [0,1]$ be the inverse order isomorphism on $[0,1]$, extended constantly by 0 on the left and 1 on the right. This is the formal fabiusInv constructed from fabiusOrderIso. It is continuous and nondecreasing; on $[0,1]$ it is a strictly increasing bijection and satisfies

$$F(I(y)) = y, \qquad I(F(x)) = x \qquad (x, y \in [0,1]). \tag{85}$$

For $0 < y < 1$, it is smooth and

$$I'(y) = \frac{1}{F'(I(y))} = \frac{1}{2\,\mathrm{up}(2I(y) - 1)} > 0. \tag{86}$$

Formal proof. The range and inverse identities are `fabiusInv_mem_Icc`, `fabiusReal_fabiusInv`, and `fabiusInv_fabiusReal`. The order, continuity, and bijectivity statements are `strictMonoOn_fabiusInv`, `monotone_fabiusInv`, `continuous_fabiusInv`, and `bijOn_fabiusInv`. Interior smoothness and the derivative chain are `fabiusInv_contDiffOn_Ioo`, `deriv_fabiusInv`, `deriv_fabiusInv_eq_inv_two_mul_rvachevUp`, and `deriv_fabiusInv_pos`, all in `FabiusInverse.lean`. □

The same module gives the exact second derivative on the open interval,

$$I''(y) = -\frac{F''(I(y))}{F'(I(y))^3}, \qquad 0 < y < 1, \tag{87}$$

and proves the sharp closed-half shape statements

$$I \text{ is strictly concave on } [0, \tfrac{1}{2}], \qquad I \text{ is strictly convex on } [\tfrac{1}{2}, 1]. \tag{88}$$

These are `deriv_fabiusInv_hasDerivAt`, `deriv_deriv_fabiusInv`, `strictConcaveOn_fabiusInv_firstHalf`, and `strictConvexOn_fabiusInv_secondHalf`. The closed endpoints in (88) use continuity; no endpoint differentiability is claimed.

The inverse inherits the exact symmetry and midpoint

$$I(1-y) = 1 - I(y), \qquad I(1/2) = 1/2, \tag{89}$$

and for $0 \le a \le 2^n$,

$$I\left(F\left(\frac{a}{2^n}\right)\right) = \frac{a}{2^n}. \tag{90}$$

These are `fabiusInv_one_sub`, `fabiusInv_half`, and `fabiusInv_fabiusDyadicUnit_of_le`. Its exact analytic locus is likewise

$$I \text{ is analytic at } y \quad \Longleftrightarrow \quad y \notin [0,1], \tag{91}$$

by `fabiusInv_analyticAt_iff`; in the open interval its derivative never vanishes by `deriv_fabiusInv_ne_zero`.

At the two clamping endpoints the inverse is steeper than every root: for every $n \in \mathbb{N}$,

$$y = o(I(y)^n) \quad (y \downarrow 0), \qquad 1 - y = o((1 - I(y))^n) \quad (y \uparrow 1). \tag{92}$$

These are `id_isLittleO_fabiusInv_pow_at_zero_right` and `one_sub_isLittleO_one_sub_fabiusInv_pow_at_one_left`. In particular the one-sided difference quotients tend to $+\infty$, as stated by `tendsto_fabiusInv_div_atTop` and its reflected companion.

6.5 Precise non-elementarity

The inductive predicate `IsElementary` contains constants and the identity and is closed under arithmetic operations, composition, real powers on positive inputs, exponential, logarithm, and trigonometric operations. Every such expression has a dense real-analytic locus:

$$\texttt{IsElementary}\ g \quad \Longrightarrow \quad \overline{\{x : g \text{ is analytic at } x\}} = \mathbb{R}.$$

This is `IsElementary.dense_analyticLocus`. It yields the proved global obstructions

$$F, \quad \mathrm{up}, \quad \mathcal{F}, \quad I \quad \text{are not elementary.} \tag{93}$$

The exact declarations are `canonical_fabius_not_isElementary`, `rvachev_not_isElementary`, `extendedFabius_not_isElementary`, and `fabiusInv_not_isElementary`; local versions for F and I on $(0,1)$ are `canonical_fabius_not_isElementary_on_Ioo` and `canonical_fabiusInv_not_isElementary_on_Ioo`.

The larger formal class `IsElementaryOrInverse` admits an inverse-branch constructor only with an open branch domain, continuity and a right-inverse identity there, and analyticity of the chosen total extension on the interior of the complement. Even that class cannot represent F or I on $(0,1)$, by `canonical_fabius_not_isElementaryOrInverse_on_Ioo` and `canonical_fabiusInv_not_isElementaryOrInverse_on_Ioo` in `InverseNotElementary.lean`.

7 Moments, generating functions, and reciprocal powers

7.1 Even and half moments

Define

$$c_n = \int_{-1}^{1} x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x, \qquad c_0 = 1, \tag{94}$$

and

$$d_0 = 1, \qquad d_n = n\int_0^1 t^{n-1}\,\mathrm{up}(t)\,\mathrm{d}t \quad (n \geq 1). \tag{95}$$

Both are positive rational sequences. Their exact triangular recurrences, for $n \geq 1$, are

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k, \tag{96}$$

$$(n+1)(2^n-1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \tag{97}$$

These are the rational declarations `moment_succ` and `halfMoment_succ` in `Arithmetic.lean`; positivity is `moment_pos` and `halfMoment_pos`.

For denominator arguments, write the exact natural normalization as

$$c_n = \frac{C_n}{(2n+1)!!\prod_{j=1}^{n}(2^{2j}-1)}, \tag{98}$$

where C_n is the positive integer `momentNumerator` at index n.

The entire generating function

$$G(z) = 1 + z\int_0^1 \mathrm{up}(t)\mathrm{e}^{zt}\,\mathrm{d}t = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!} \tag{99}$$

satisfies

$$G(2z) = \frac{\mathrm{e}^z - 1}{z}\, G(z), \tag{100}$$

with the removable value at $z = 0$. The exact formal statements are `proposition_two_formula` and `complexGeneratingFunction_eq_series_formula` in `AnalyticMoments.lean`.

The two sequences are linked by

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k, \tag{101}$$

and, at odd indices,

$$d_{2k+1} = \frac{2k+1}{2}\, c_k. \tag{102}$$

These are `halfMoment_eq_evenMomentSum` and `halfMoment_odd_eq_moment` in `MomentPowerSeries.lean`. Division-free natural numerators for c_n and d_n, together with their exact recurrences and odd-index bridge, are formalized by `momentNumerator`, `halfMomentNumerator`, `halfMomentNumerator_succ`, `halfMomentNumerator_odd_index_eq`, and `halfMomentNumerator_odd_index_formula`. No unproved numerical table is needed.

7.2 The probability–Laplace bridge

Let μ_X be the distribution of the weighted sum X from (33). Its survival function on the nonnegative half-line is exactly

$$\mu_X((t,\infty)) = \mathrm{up}(t) \qquad (t \geq 0). \tag{103}$$

If $g, g' : \mathbb{R} \to \mathbb{R}$ are continuous and g has derivative $g'(t)$ at every $t \in [0,1]$, then

$$\int_{[0,1]} g(x)\ \mathrm{d}\mu_X(x) = g(0) + \int_0^1 g'(t)\,\mathrm{up}(t)\,\mathrm{d}t. \tag{104}$$

For $s \in \mathbb{R}$ and $k \in \mathbb{N}$, put

$$L_k(s) = \int_{[0,1]} \mathrm{e}^{-sx} x^k\ \mathrm{d}\mu_X(x). \tag{105}$$

These raw tilted moments agree with the formal analytic Fabius Laplace moments, and at zero tilt

$$L_n(0) = \int_{[0,1]} (1-x)^n\ \mathrm{d}\mu_X(x) = d_n. \tag{106}$$

Formal proof. The survival identity and integration-by-parts formula are `weightedSumDistribution_real_Ioi_eq_rvachevUp_of_nonneg` and `integral_unit_eq_zero_add_integral_deriv_mul_rvachevUp`. The tilted and zero-tilt identities are `unitLaplaceMoment_weightedSumDistribution_eq_fabiusLaplaceMoment`, `unitLaplaceMoment_weightedSumDistribution_zero_eq_halfMoment`, and `unitEndpointMoment_weightedSumDistribution_eq_halfMoment` in `ProbabilityLaplaceMoments.lean`. □

7.3 Exact values at reciprocal powers of two

For every $n \in \mathbb{N}$,

$$d_n = n!\,2^{\binom{n}{2}} F(2^{-n}), \qquad F(2^{-n}) = \frac{d_n}{n!\,2^{\binom{n}{2}}}. \tag{107}$$

This is `halfMoment_eq_fabius_formula` in `AnalyticMoments.lean`. The numerator-cleared version is `fabiusAtInverseTwoPow_eq_halfMomentNumerator_formula` in `ExactInversePower.lean`.

Writing $V_n = F(2^{-n})$, one has $V_0 = 1$, and for $n \geq 1$,

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!}\, V_k. \tag{108}$$

This is `fabiusAtInverseTwoPow_recurrence`. The same sequence has exact finite path and ordered-composition descriptions, proved by `fabiusAtInverseTwoPow_eq_pathSum`, `fabiusAtInverseTwoPow_eq_composition_formula`, `pathWeight_eq_boundary_product`, and `fabiusRecurrenceSequence_eq_sum_compositions` in `FabiusInverseDyadicClosedForm.lean`. Their positivity and recurrence uniqueness are `fabiusCompositionSum_pos` and `triangularRecurrence_unique`.

7.4 Parity and the exact two-adic valuation

Lucas parity is packaged as `odd_binomial_coefficient_counts` and `card_odd_inner_binomial_coefficients`. It implies the exact numerator theorem `momentNumerator_odd`. For every $n \geq 1$, the reduced numerator and denominator of $2d_n$ are both odd, equivalently

$$\nu_2(d_n) = -1. \tag{109}$$

This is `two_mul_halfMoment_num_den_odd` together with `halfMoment_padicVal_two`.

Consequently, for $n \geq 1$,

$$\nu_2\big(F(2^{-n})\big) = -\binom{n}{2} - 1 - \nu_2(n!), \tag{110}$$

by `fabiusAtInverseTwoPow_padicVal_two`. The binary form uses the proved Legendre identity

$$\nu_2(n!) = n - \mathrm{wt}_2(n), \tag{111}$$

namely `factorial_padicVal_two`; the combined declaration is `fabiusAtInverseTwoPow_padicVal_two_closed` in `TwoAdic.lean`.

7.5 Bernoulli recurrences

The formal development also proves Bernoulli convolution and recurrence forms for both sequences. For $n \geq 1$,

$$c_n = \frac{1}{2^{2n}-1}\sum_{k=1}^{n} 2^{2n-2k}(2^{2k}-2)\binom{2n}{2k}B_{2k}c_{n-k}, \tag{112}$$

$$d_n = \frac{n\,2^n}{4(2^n-1)}d_{n-1} - \frac{1}{2^n-1}\sum_{k=1}^{\lfloor n/2\rfloor}\binom{n}{2k}2^{n-2k}B_{2k}d_{n-2k}. \tag{113}$$

Here $B_1 = -1/2$. These are `moment_bernoulli_recurrence` and `halfMoment_bernoulli_recurrence`; their convolution forms and combined source-paper statement are `moment_bernoulli_convolution`, `halfMoment_bernoulli_convolution`, and `proposition_twenty_two_formula` in `BernoulliRecurrences.lean`.

8 Exact evaluation at arbitrary dyadic rationals

The rationality theorem for dyadic arguments comes with two complementary exact presentations:

1. a closed finite formula involving Thue–Morse signs and the rational moments c_k;
2. an executable bit-recursive Taylor evaluator.

8.1 The closed Thue–Morse formula

Theorem 8.1 (Exact dyadic formula)**.** *Let $n, a \in \mathbb{N}$ with $0 \leq a \leq 2^n$. Then*

$$\boxed{F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!}\sum_{h=0}^{a-1}(-1)^{\mathrm{wt}_2(h)}\sum_{k=0}^{\lfloor n/2\rfloor}\binom{n}{2k}c_k(2a-2h-1)^{n-2k}.} \tag{114}$$

An empty outer sum has value 0*.*

Proof. Write $x = a/2^n$ and use $F(x) = \mathrm{up}(x-1)$. Since up and all its derivatives vanish at -1, Taylor's theorem with integral remainder gives

$$\mathrm{up}(x-1) = \frac{1}{n!}\int_{-1}^{x-1}(x-1-t)^n\,\mathrm{up}^{(n+1)}(t)\,\mathrm{d}t. \tag{115}$$

Insert the exact derivative identity (76):

$$\mathrm{up}^{(n+1)}(t) = 2^{\binom{n+2}{2}}\mathcal{F}(2^{n+1}(t+1)).$$

Put $s = 2^{n+1}(t+1)$. As t runs from -1 to $x-1$, s runs from 0 to $2a$. Split this interval into the blocks $[2h, 2h+2]$, $0 \le h < a$. On the hth block, (14) gives

$$\mathcal{F}(s) = (-1)^{\mathrm{wt}_2(h)}\,\mathrm{up}(s-2h-1).$$

The substitution $u = s-2h-1$ maps the block to $[-1,1]$. Gathering the powers of two gives

$$F(a/2^n) = \frac{2^{-\binom{n+1}{2}}}{n!}\sum_{h=0}^{a-1}(-1)^{\mathrm{wt}_2(h)}\int_{-1}^{1}(2a-2h-1-u)^n\,\mathrm{up}(u)\,\mathrm{d}u. \tag{116}$$

Expand the nth power. Odd powers of u integrate to zero because up is even, while

$$\int_{-1}^{1}u^{2k}\,\mathrm{up}(u)\,\mathrm{d}u = c_k.$$

The surviving even terms are exactly the inner sum in (114). □

Every quantity on the right of (114) is rational, so dyadic rationality is immediate. More importantly, the formula lives entirely in $\mathbb{Q}$ and can be evaluated without roundoff.

Theorem 8.2 (Global dyadic formula)**.** *For all $n, a \in \mathbb{N}$ the signed global extension satisfies*

$$\boxed{\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!}\sum_{h=0}^{a-1}\varepsilon_h\sum_{k=0}^{\lfloor n/2\rfloor}\binom{n}{2k}c_k(2a-2h-1)^{n-2k}.} \tag{117}$$

No upper bound on a is imposed. An empty outer sum has value 0.

Proof. By (75) the mth derivative of $\mathcal{F}$ is $2^{\binom{m+1}{2}}\mathcal{F}(2^m x)$, and $\mathcal{F}$ vanishes on $(-\infty, 0]$. Hence $\mathcal{F}^{(m)}(0) = 0$ for every $m \ge 0$, and Taylor's theorem with integral remainder at the base point 0, applied through order n, gives for $x \ge 0$

$$\mathcal{F}(x) = \frac{1}{n!}\int_0^x (x-t)^n\mathcal{F}^{(n+1)}(t)\,\mathrm{d}t. \tag{118}$$

Put $x = a/2^n$ and insert $\mathcal{F}^{(n+1)}(t) = 2^{\binom{n+2}{2}}\mathcal{F}(2^{n+1}t)$. The substitution $u = 2^{n+1}t$ carries $[0,x]$ onto $[0,2a]$ and produces $(x-t)^n = 2^{-n(n+1)}(2a-u)^n$ together with $\mathrm{d}t = 2^{-(n+1)}\,\mathrm{d}u$. Since

$$\binom{n+2}{2} - n(n+1) - (n+1) = -\binom{n+1}{2},$$

this is exactly

$$\mathcal{F}\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!}\int_0^{2a}(2a-u)^n\mathcal{F}(u)\,\mathrm{d}u. \tag{119}$$

Split $[0, 2a]$ into the a blocks $[2h, 2h+2]$ with $0 \le h < a$. On the hth block (14) gives $\mathcal{F}(u) = \varepsilon_h \operatorname{up}(u - 2h - 1)$, and the substitution $v = u - 2h - 1$ maps that block onto $[-1, 1]$:

$$\int_{2h}^{2h+2} (2a-u)^n \mathcal{F}(u)\,\mathrm{d}u = \varepsilon_h \int_{-1}^{1} (2a-2h-1-v)^n \operatorname{up}(v)\,\mathrm{d}v.$$

Expand the nth power by the binomial theorem. The function up is even, so every odd moment vanishes, while $\int_{-1}^{1} v^{2k} \operatorname{up}(v)\,\mathrm{d}v = c_k$ and the surviving sign $(-1)^{2k}$ is $+1$. Summing over h gives (117). □

Two special cases are the ones used in practice.

Corollary 8.3 (Bounded case)**.** *If $0 \le a \le 2^n$ then $\mathcal{F}(a/2^n) = F(a/2^n)$, so (117) reduces to (114).*

Proof. The two functions agree on $[0, 1]$ and $a/2^n \in [0, 1]$. □

Corollary 8.4 (Bump case)**.** *Let $n \in \mathbb{N}$ and $q \in \mathbb{Z}$. If $|q| < 2^n$, then*

$$\operatorname{up}\left(\frac{q}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{q+2^n-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k \left(2q + 2^{n+1} - 2h - 1\right)^{n-2k}, \tag{120}$$

that is, (117) evaluated at $a = q + 2^n$. If $|q| \ge 2^n$, then $\operatorname{up}(q/2^n) = 0$.

Proof. Taking $b = 0$ in (14) gives $\mathcal{F}(x) = \operatorname{up}(x-1)$ for $0 \le x \le 2$. With $x = 1 + q/2^n = (q+2^n)/2^n$ and $|q| < 2^n$ we have $0 < x < 2$, so $\operatorname{up}(q/2^n) = \mathcal{F}\big((q+2^n)/2^n\big)$, and (117) with $a = q + 2^n$ applies. The last statement is the support condition $\operatorname{supp} \operatorname{up} = [-1, 1]$. □

Corollary 8.4 is the global form invoked in the proof of the uniform denominator multiplier theorem below. The point is that it genuinely requires Theorem 8.2 and not (114): for $0 < q < 2^n$ the numerator $a = q + 2^n$ exceeds 2^n, so the bounded statement does not apply. At the endpoints $|q| = 2^n$ the two clauses agree, both giving 0; for $q = -2^n$ the outer sum is empty and for $q = 2^n$ the value is $\mathcal{F}(2) = 0$.

Remark 8.5 (Why the sum is finite)*.* Finiteness has two independent sources, and neither involves a limit. Horizontally, only the a complete translated bumps indexed by $0 \le h < a$ meet the integration range in (119); the locally finite sum (13) therefore contributes finitely many terms. Vertically, expanding $(2a - 2h - 1 - v)^n$ produces only powers v^j with $j \le n$, and all odd j integrate to zero, leaving the $\lfloor n/2 \rfloor + 1$ even moments $c_0, \dots, c_{\lfloor n/2 \rfloor}$. Every one of these is an explicit rational number produced by (96), so (117) is an exact algorithm over $\mathbb{Q}$ with no roundoff and no truncation. It is not, however, an efficient one: the outer range has length a, whereas the bit recursion of Algorithm 8.11 needs only $\operatorname{wt}_2(a)$ steps.

Example 8.6 (The closed formula evaluated)**.** The formula is stated, globalized, used to bound denominators, and proved finite above without ever being evaluated. Three small cases, using $c_0 = 1$ and $c_1 = 1/9$.

For $n = 2$, $a = 1$ the outer sum has the single term $h = 0$ with $\varepsilon_0 = +1$ and $2a - 2h - 1 = 1$, so

$$F\left(\frac{1}{4}\right) = \frac{2^{-3}}{2!}\left[\binom{2}{0} c_0 \cdot 1^2 + \binom{2}{2} c_1 \cdot 1^0\right] = \frac{1}{16}\left(1 + \frac{1}{9}\right) = \frac{5}{72}.$$

For $n = 3$, $a = 1$ the same single term gives

$$F\left(\frac{1}{8}\right) = \frac{2^{-6}}{3!}\left[\binom{3}{0} c_0 \cdot 1^3 + \binom{3}{2} c_1 \cdot 1^1\right] = \frac{1}{384}\left(1 + \frac{3}{9}\right) = \frac{1}{288}.$$

For $n = 3$, $a = 3$ the Thue–Morse signs finally do some work: $h = 0, 1, 2$ carry $\varepsilon_h = +1, -1, -1$ and $2a - 2h - 1 = 5, 3, 1$, so

$$F\left(\frac{3}{8}\right) = \frac{2^{-6}}{3!}\left[\left(5^3 + 3\cdot\frac{5}{9}\right) - \left(3^3 + 3\cdot\frac{3}{9}\right) - \left(1^3 + 3\cdot\frac{1}{9}\right)\right]$$
$$= \frac{1}{384}\left[\left(125 + \frac{5}{3}\right) - 28 - \left(1 + \frac{1}{3}\right)\right] = \frac{1}{384}\cdot\frac{292}{3} = \frac{73}{288},$$

in agreement with the level-4 table below. The third case is the smallest one in which the signs are not all positive, and it shows the characteristic feature of the formula: the individual bracketed terms are much larger than the answer, and the cancellation among them is what produces a value in $[0, 1]$.

8.2 Thue–Morse cancellation

The mechanism that makes the dyadic formulas simplify is the Prouhet cancellation carried by the Thue–Morse signs.

Lemma 8.7 (Affine power cancellation)**.** *If $d < b$, then for all $X, Y \in \mathbb{Q}$,*

$$\sum_{h=0}^{2^b-1}(-1)^{\mathrm{wt}_2(h)}(X + Yh)^d = 0. \tag{121}$$

Proof. It is enough to prove the claim for h^j, $0 \le j \le d$, and then expand the affine power. Let

$$S_{b,j} = \sum_{h=0}^{2^b-1}(-1)^{\mathrm{wt}_2(h)}h^j.$$

Split the sum for $S_{b+1,j}$ at 2^b and use $\varepsilon_{2^b+h} = -\varepsilon_h$:

$$S_{b+1,j} = \sum_{h<2^b}\varepsilon_h\left(h^j - (2^b + h)^j\right)$$
$$= -\sum_{r=0}^{j-1}\binom{j}{r}2^{b(j-r)}S_{b,r}.$$

If $j < b + 1$, every index r on the right satisfies $r < b$, so the induction hypothesis makes all terms zero. The base case is immediate. ☐

A compact generating-function version is

$$\sum_{h<2^b}(-1)^{\mathrm{wt}_2(h)}\mathrm{e}^{ht} = \prod_{j=0}^{b-1}(1 - \mathrm{e}^{2^j t}), \tag{122}$$

whose right side has a zero of order b at $t = 0$. Differentiating fewer than b times at zero gives the same cancellation.

This lemma explains why apparently large translated sums in (114) collapse to short Taylor blocks when the numerator crosses a binary boundary.

The cancellation lemma states that the first b moments of the signed block vanish. The first surviving moment has an equally clean closed value, and that value is the constant that governs every leading coefficient in the q-binomial analysis of Section 8.7.

Proposition 8.8 (The first surviving Thue–Morse moment)**.** *For every $k \ge 0$,*

$$\boxed{\sum_{r=0}^{2^k-1}\varepsilon_r\, r^k = (-1)^k 2^{\binom{k}{2}}k!.} \tag{123}$$

Proof. By (122),

$$\sum_{r<2^k} \varepsilon_r \mathrm{e}^{rt} = \prod_{j=0}^{k-1} (1-\mathrm{e}^{2^j t}). \tag{124}$$

Each factor is an entire function with a simple zero at $t=0$, and

$$1-\mathrm{e}^{2^j t} = -2^j t + \mathcal{O}(t^2) \qquad (t \to 0),$$

so the product has a zero of order exactly k at the origin, with

$$\prod_{j=0}^{k-1} (1-\mathrm{e}^{2^j t}) = \Big(\prod_{j=0}^{k-1}(-2^j)\Big) t^k + \mathcal{O}(t^{k+1}) = (-1)^k 2^{\binom{k}{2}} t^k + \mathcal{O}(t^{k+1}),$$

because $\sum_{j=0}^{k-1} j = \binom{k}{2}$. The left side of (124) is the exponential generating function

$$\sum_{d\ge 0} \Big(\sum_{r<2^k} \varepsilon_r r^d\Big) \frac{t^d}{d!}.$$

Comparing coefficients of t^d for $d<k$ recovers (121); comparing coefficients of t^k gives

$$\frac{1}{k!} \sum_{r<2^k} \varepsilon_r r^k = (-1)^k 2^{\binom{k}{2}}.$$

The case $k=0$ is the empty product, both sides being 1. □

Corollary 8.9 (Translation invariance)**.** *For every $k \ge 0$ and every Q in a field of characteristic zero,*

$$\sum_{r=0}^{2^k-1} \varepsilon_r\,(r+Q)^k = (-1)^k 2^{\binom{k}{2}} k!. \tag{125}$$

Proof. Expand $(r+Q)^k = \sum_{d=0}^k \binom{k}{d} Q^{k-d} r^d$ and interchange the two finite sums. For $d<k$ the inner sum $\sum_{r<2^k} \varepsilon_r r^d$ vanishes by (121), so only $d=k$ survives, with coefficient 1. Apply (123). □

Thus the value is unchanged if r is replaced by $r+Q$ for any constant Q, precisely because all the lower moments vanish: a shift can only alter the coefficients of degrees below the top, and those coefficients are already zero. The centered choice $Q=-2^k$ gives

$$\sum_{r=0}^{2^k-1} \varepsilon_r\,(r-2^k)^k = (-1)^k 2^{\binom{k}{2}} k!, \tag{126}$$

which is the formally proved centered version of the same exact moment identity. Together with the lower-moment annihilation, it proves that the zero in (148) has order exactly k.

8.3 A Taylor block that removes the leading bit

Suppose $0<x<1$, and let r be the unique integer such that

$$2^{-r} \le x < 2^{-r+1}. \tag{127}$$

Put $y = x - 2^{-r}$. Then $0 \le y < 2^{-r}$.

Theorem 8.10 (Exact Taylor-block recursion)**.** *With the notation above,*

$$F(x) = -F(y) + \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} y^j. \tag{128}$$

Proof. Apply Taylor's theorem at the point 2^{-r} through degree r:

$$F(2^{-r} + y) = \sum_{j=0}^{r} \frac{F^{(j)}(2^{-r})}{j!} y^j + R_r(y). \tag{129}$$

The derivative scaling formula gives

$$F^{(j)}(2^{-r}) = 2^{\binom{j+1}{2}} F(2^{j-r}), \qquad 0 \le j \le r. \tag{130}$$

It remains to identify the integral remainder. The scaling equation carries the interval $[2^{-r}, 2^{-r} + y]$ into the adjacent signed translate of the interval $[0, y]$; on that translate the Thue–Morse sign is negative. Reversing the substitutions in the derivative identity shows

$$R_r(y) = -\frac{1}{r!} \int_0^y (y-t)^r F^{(r+1)}(t)\, \mathrm{d}t = -F(y),$$

where the last equality is Taylor's integral formula at the flat point 0. Substitution in (129) proves (128). ☐

Using (107), the coefficients may be written entirely in terms of the half moments:

$$\frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} = \frac{2^{\binom{j+1}{2} - \binom{r-j}{2}} d_{r-j}}{j!(r-j)!}. \tag{131}$$

This is the form that appears in the original arithmetic recurrence.

8.4 The bit-recursive evaluator

Let $x = a/2^e$ with $0 < a < 2^e$. Write

$$b = \lfloor \log_2 a \rfloor, \qquad r = e - b, \qquad q = a - 2^b. \tag{132}$$

Then 2^{-r} is the leading binary term of x and

$$y = \frac{q}{2^e}. \tag{133}$$

Applying (128) reduces $F(a/2^e)$ to $F(q/2^e)$; the highest set bit of the numerator has disappeared.

Algorithm 8.11 (Exact dyadic evaluator)**.** Precompute $V_j = F(2^{-j})$ for $0 \le j \le e$ from (107). Define $\mathrm{Eval}(e, a)$ recursively by

$$\mathrm{Eval}(e, a) = \begin{cases} 0, & a \le 0, \\ 1, & a \ge 2^e, \\ T_r(q/2^e) - \mathrm{Eval}(e, q), & 0 < a < 2^e, \end{cases}$$

where b, r, q are given by (132) and

$$T_r(z) = \sum_{j=0}^{r} \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} z^j. \tag{134}$$

Evaluate T_r by Horner's rule.

A literal pseudocode version is:

```
Eval(e, a, V):
    if a <= 0:      return 0
    if a >= 2^e:    return 1

    b = floor_log2(a)
    r = e - b
    q = a - 2^b
    y = q / 2^e

    p = V[0]                        // highest Taylor coefficient, j = r
    for m = 1 .. r:
        j = r - m
        p = V[m] + (2^(j+1) * y / (j+1)) * p

    return p - Eval(e, q, V)
```

The displayed loop is one convenient Horner ordering; any exact evaluation of (134) is equivalent.

Theorem 8.12 (Correctness and termination). *For every $e \in \mathbb{N}$ and $a \in \mathbb{Z}$,*

$$\mathrm{Eval}(e, a) = F(a/2^e) \tag{135}$$

under the bounded convention $F = 0$ to the left and $F = 1$ to the right.

Proof. The boundary clauses agree with (1) and (2). In the interior, (128) is precisely the recursive clause. Since $0 \leq q < a$, strong induction proves correctness and termination. ☐

8.5 Reduction of an arbitrary dyadic argument

Algorithm 8.11 evaluates the bounded function F on $[0, 1]$. A total evaluator for the signed global extension first folds the argument back into the unit interval and records the associated sign. The fold is a reduction modulo 2 followed by a reflection.

Theorem 8.13 (Triangular reduction). *Let $n \in \mathbb{N}$ and put $s = 2^n$. Let $a \in \mathbb{N}$ and write*

$$a = 2sB + R, \qquad B = \left\lfloor \frac{a}{2s} \right\rfloor, \qquad 0 \leq R < 2s, \tag{136}$$

and set

$$C = \begin{cases} R, & R \leq s, \\ 2s - R, & R > s. \end{cases} \tag{137}$$

Then $0 \leq C \leq s$ and

$$\boxed{\mathcal{F}\left(\frac{a}{2^n}\right) = \varepsilon_B \, F\left(\frac{C}{2^n}\right).} \tag{138}$$

For $a \in \mathbb{Z}$ with $a < 0$ one has $\mathcal{F}(a/2^n) = 0$.

Proof. Write $x = a/2^n = 2B + R/s$. Since $0 \leq R < 2s$ we have $2B \leq x < 2B + 2$, so x lies in the Bth block and (14) gives

$$\mathcal{F}(x) = \varepsilon_B \, \mathrm{up}(x - 2B - 1) = \varepsilon_B \, \mathrm{up}\left(\frac{R}{s} - 1\right).$$

Put $t = R/s - 1$, so that $-1 \leq t < 1$. The fold relation (7) says $\mathrm{up}(t) = F(t + 1)$ for $t \leq 0$ and $\mathrm{up}(t) = F(1 - t)$ for $t > 0$; on $[-1, 1]$ both clauses read $\mathrm{up}(t) = F(1 - |t|)$. If

$R \le s$ then $|t| = 1 - R/s$ and $F(1-|t|) = F(R/s)$. If $R > s$ then $|t| = R/s - 1$ and $F(1-|t|) = F(2-R/s) = F\big((2s-R)/2^n\big)$. In both cases the argument is $C/2^n$ with C as in (137), and $0 \le C \le s$ is immediate from $0 \le R < 2s$. The final statement is the vanishing of $\mathcal{F}$ on the negative half-line. □

Combining (138) with Algorithm 8.11 gives a total exact evaluator for $\mathcal{F}$ at every dyadic rational: reduce with (136)–(137), compute the block sign $\varepsilon_B = (-1)^{\mathrm{wt}_2(B)}$ from the binary digits of $B = \lfloor a/2^{n+1} \rfloor$, and call $\mathrm{Eval}(n, C)$.

The reduction is compatible with unreduced presentations. Replacing (n, a) by $(n+1, 2a)$ doubles s and R, leaves B unchanged, and doubles C; the right side of (138) is therefore unchanged, as it must be.

Remark 8.14 (Recognizing dyadic arguments exactly). A reduced rational p/q with $q > 0$ is dyadic exactly when q is a power of two, and in that case $n = \log_2 q$ and $a = p$ are read off directly. An exact rational interface to F, up, and $\mathcal{F}$ therefore tests a single integer condition on the reduced denominator, returns "not dyadic" when it fails, and otherwise invokes the signed integer evaluator above. No floating-point recognition of powers of two is involved anywhere, so the interface is exact on every input and cannot be defeated by rounding in the recognition step.

8.6 Denominator bounds

Let

$$M_m = \prod_{j=1}^{m} (2^{2j} - 1), \qquad (2m+1)!! = 1 \cdot 3 \cdots (2m+1). \tag{139}$$

Theorem 8.15 (Uniform denominator multiplier)**.** *Let $n \ge 1$ and $a \in \mathbb{Z}$ with $|a| < 2^n$. Then*

$$\mathrm{up}(a/2^n)\, n!\, 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!!\, M_{\lfloor n/2 \rfloor} \tag{140}$$

is an integer.

Proof. Apply the global dyadic form of (114). Replace each c_k by (98). For $k \le \lfloor n/2 \rfloor$, both the odd double factorial in the denominator of c_k and its even-Mersenne product divide the two common products in (140). The remaining factors are integers. Thus every summand becomes integral after multiplication by the displayed common factor. □

Let Δ_n be the least common multiple of the reduced denominators of the values $\mathrm{up}(a/2^n)$ with odd a in the support. The theorem gives the explicit divisibility bound

$$\Delta_n \mid n!\, 2^{\binom{n+1}{2}} (2\lfloor n/2 \rfloor + 1)!! M_{\lfloor n/2 \rfloor}. \tag{141}$$

A complementary lower statement is available at no cost, and it is what makes the normalization of the conjecture below meaningful.

Proposition 8.16 (The reciprocal power is one of the samples)**.** *For every $n \ge 1$ the reduced denominator of $F(2^{-n})$ divides Δ_n. Consequently*

$$v_2(\Delta_n) \ge \binom{n}{2} + 1 + v_2(n!). \tag{142}$$

Proof. By the fold (8), $F(2^{-n}) = \mathrm{up}(1 - 2^{-n}) = \mathrm{up}\big((2^n - 1)/2^n\big)$, and the numerator $2^n - 1$ is odd. So $F(2^{-n})$ is itself one of the samples whose denominators Δ_n is the least common multiple of, and its reduced denominator therefore divides Δ_n. The valuation statement follows from (110), which gives $v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!)$, so the reduced denominator contains exactly that power of two. □

Together, (141) and (142) give the formally proved denominator information used in this article. Proposed exact factorizations of the least common denominator, together with the associated numerical tables, are kept in the dyadic research-frontier notebooks until their individual statements are formalized.

8.7 The exact half-base q-binomial layer

The dyadic dilation forces the specialization $q = 1/2$. For $a, q \in \mathbb{Q}$, define

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \tag{143}$$

and write $\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}$ for the Gaussian coefficient at that fixed base. For $k \le n$, its exact Mersenne normalization is

$$\begin{bmatrix} n \\ k \end{bmatrix}_{1/2} = \frac{\mathfrak{m}_n}{\mathfrak{m}_k \mathfrak{m}_{n-k}\, 2^{k(n-k)}}, \qquad \mathfrak{m}_j = \prod_{\ell=1}^{j} (2^\ell - 1). \tag{144}$$

In particular the row is symmetric under $k \mapsto n - k$.

Theorem 8.17 (Finite half-base q-binomial theorem)**.** *For every $n \in \mathbb{N}$ and $z \in \mathbb{Q}$,*

$$(z; 1/2)_n = \sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} z^k. \tag{145}$$

At the dyadic nodes this gives

$$\begin{aligned} &\sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} (2^m)^k = 0 && (m < n), \\ &\sum_{k=0}^{n} (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} (2^n)^k = (-1)^n 2^{\binom{n+1}{2}} (1/2; 1/2)_n. \end{aligned} \tag{146}$$

Moreover,

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{m}_n. \tag{147}$$

Formal proof. These are `halfQBinomial_theorem`, `halfQBinomial_two_pow_sum_eq_zero`, `halfQBinomial_two_pow_sum_eq_self`, `halfQPochhammer_eq_mersenne_div`, `halfQBinomial_eq_mersenne`, and `halfQBinomial_symm` in `HalfQBinomial.lean`. □

The inner cancellation in the dyadic formula is the exact Thue–Morse block identity

$$\sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)} z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}), \tag{148}$$

formalized as `thueMorseBlockPolynomial_eq_product`. Its affine power-sum form annihilates every degree below k, while its first surviving moment is $(-1)^k 2^{\binom{k}{2}} k!$; see `thueMorse_affine_power_sum_eq_zero` and `thueMorse_affine_power_sum_self`.

For comparison with the moment recurrence, put

$$a_n = \frac{d_n}{n!}, \qquad \mathscr{A}(X) = \sum_{n \ge 0} a_n X^n, \qquad E(X) = \frac{\mathrm{e}^X - 1}{X}, \tag{149}$$

with the removable value $E(0) = 1$. The refinement is

$$\mathscr{A}(2X) = E(X)\mathscr{A}(X), \tag{150}$$

and its coefficient recurrence is

$$(2^n - 1)a_n = \sum_{j=0}^{n-1} \frac{a_j}{(n-j+1)!} \quad (n \geq 1), \qquad a_0 = 1. \tag{151}$$

These identities are the formal-series bridge used by the specialized dyadic theorem.

For $m, k, d \in \mathbb{N}$ and $Q \in \mathbb{C}$, write

$$\sigma_{m;k,d}(Q) = \sum_{r=0}^{m2^k-1} (-1)^{\mathrm{wt}_2(r)}(r - m2^k + Q)^d. \tag{152}$$

The exact result is the following finite identity.

Theorem 8.18 (Translated global dyadic formula)**.** *For all $m, n \in \mathbb{N}$ and $Q \in \mathbb{C}$,*

$$\begin{aligned}\mathcal{F}(m/2^n) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ \times \sum_{r=0}^{m2^k-1} (-1)^{\mathrm{wt}_2(r)}(r - m2^k + Q)^{n+k}.\end{aligned} \tag{153}$$

The real number on the left is embedded in $\mathbb{C}$. The entire right-hand side is independent of the common translation Q. When $m \leq 2^n$, the left side is the bounded value $F(m/2^n)$.

Formal proof. The numerator is proved to be a constant polynomial in the translation by `qBinomialThueMorseDyadicTranslatedFormulaPolynomial_eq_const`. Its identification with the signed dyadic value over $\mathbb{C}$ is `globalFabius_dyadic_eq_qBinomialThueMorseDyadic_translated_sum_complex`; the rational core and real, complex, and Gaussian-rational specializations are in `FabiusQBinomialFormula.lean` and `FabiusDyadicQBinomialScalar.lean`. □

Binary complementation gives the raw inverse-dyadic form

$$\boxed{\mathcal{F}(2^{-n}) = \frac{1}{(-2)^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)}(r+Q)^{n+k}.} \tag{154}$$

Its translated numerator is constant by `qBinomialThueMorseRawTranslatedNumeratorPolynomial_eq_const`, and the value identity is `fabiusAtInverseTwoPow_cast_eq_qBinomialThueMorse_rawTranslated_sum` in `FabiusRawQBinomialScalar.lean`. Generic-q interpolation formulas, barycentric identities, numerical cancellation studies, and complexity comparisons are maintained in the dyadic research-frontier notebooks rather than used as theorems here.

9 The global binary-reduction telescope

The Taylor-block recursion also applies to the signed global extension $\mathcal{F}$. It can therefore be iterated along the binary expansion of an arbitrary nonnegative real number, rather than only along the finite expansion of a dyadic rational. The reductions need not terminate, but their residuals tend geometrically to zero.

Define the finite reduction polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^{m} 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \tag{155}$$

All powers of 2 here are integer powers. In particular, negative exponents are allowed, and $\mathcal{T}_0(y) = 1$. The signed Taylor-block identity is

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \qquad 0 \le y < 2^{-m}. \tag{156}$$

Fix $x \ge 0$. At scale m, put

$$p_m = \lfloor 2^m x \rfloor, \qquad y_m = x - \frac{p_m}{2^m}, \qquad 0 \le y_m < 2^{-m}. \tag{157}$$

The previous prefix requires a genuine half-scale value at $m = 0$:

$$p^*_{m-1} = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}, & m \ge 1. \end{cases} \tag{158}$$

Uniformly, $p^*_{m-1} = \lfloor p_m/2 \rfloor$. Thus $p_m = 2p^*_{m-1} + \delta_m$ with $\delta_m \in \{0, 1\}$. Define

$$\alpha_m(x) = (-1)^{\mathrm{wt}_2(p_m)}(2p^*_{m-1} - p_m), \qquad R_m(x) = (-1)^{\mathrm{wt}_2(p_m)}\mathcal{F}(y_m). \tag{159}$$

The coefficient α_m is zero or a sign, and it is nonzero precisely when the newly exposed binary digit is 1.

Lemma 9.1 (One-step residual identity)**.** *For $m \ge 1$,*

$$\alpha_m(x)\mathcal{T}_m(y_m) = R_{m-1}(x) - R_m(x). \tag{160}$$

Proof. If $p_m = 2p_{m-1}$, then $y_m = y_{m-1}$, the Thue–Morse signs agree, and $\alpha_m = 0$, so both sides vanish. If $p_m = 2p_{m-1} + 1$, then

$$y_{m-1} = 2^{-m} + y_m, \qquad (-1)^{\mathrm{wt}_2(p_m)} = -(-1)^{\mathrm{wt}_2(p_{m-1})}, \qquad \alpha_m = (-1)^{\mathrm{wt}_2(p_{m-1})}.$$

Apply (156) to y_{m-1} and multiply by the old prefix sign; the two copies of $\mathcal{F}(y_m)$ combine exactly into $R_{m-1} - R_m$. ☐

The scale-zero identity is separate and retains the parity of the integer block:

$$\mathcal{F}(x) = \alpha_0(x)\mathcal{T}_0(y_0) + R_0(x). \tag{161}$$

Indeed, write $p_0 = 2p^*_{-1} + \delta_0$ and use the two-unit translation law (14). When $\delta_0 = 0$, the polynomial term vanishes. When $\delta_0 = 1$, the identity $\mathcal{F}(1 + y_0) = -\mathcal{F}(y_0) + 1$ supplies it.

Theorem 9.2 (Exact finite binary telescope)**.** *For $x \ge 0$ and $N \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{N} \alpha_m(x)\mathcal{T}_m(y_m(x)) + R_N(x).} \tag{162}$$

Proof. Start with (161). Summing (160) for $m = 1, \dots, N$ telescopes all intermediate residuals. The case $N = 0$ is exactly the scale-zero identity. ☐

For $N \ge 1$, $0 \le y_N < 2^{-N} \le 1/2$. On this interval $0 \le \mathcal{F}(y) = F(y) \le 2y$, so

$$|R_N(x)| \le 2^{1-N}. \tag{163}$$

The estimate is independent of x. Therefore, if $B_N(x)$ denotes the finite sum in (162), then

$$\sup_{x \ge 0} |B_N(x) - \mathcal{F}(x)| \le 2^{1-N}. \tag{164}$$

Lean's natural-floor convention extends every B_N by zero on $x \le 0$, where $\mathcal{F}$ also vanishes. In that convention (164) holds with the supremum over all of $\mathbb{R}$, so the finite telescopes converge uniformly on the whole line. The one-step identity then gives, for $m \ge 2$,

$$|\alpha_m(x)\mathcal{T}_m(y_m)| \le |R_{m-1}| + |R_m| \le 4\,2^{-(m-1)}. \tag{165}$$

Thus the limiting series is absolutely, not merely conditionally, convergent.

Theorem 9.3 (Absolutely convergent global binary series)**.** *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x)\mathcal{T}_m(y_m(x)).} \tag{166}$$

On $0 \leq x \leq 1$, the sum is equivalently $F(x)$.

The index $m = 0$ is indispensable globally. For $0 \leq x < 1$ it vanishes, so a series starting at $m = 1$ happens to be valid there. At $x = 1$, however, the zeroth term is 1 and all later coefficients vanish. More generally it is the zeroth term that distinguishes even and odd integer blocks of the signed extension.

9.1 The corrected parity-power presentation

The same summands have a more literal presentation in terms of prefix parity and inverse dyadic values. Define, using integer exponents,

$$\kappa_j = \frac{((j : \mathbb{Z}) + 2)((j : \mathbb{Z}) - 1)}{2} = \binom{j+1}{2} - 1. \tag{167}$$

The integer interpretation matters: $\kappa_0 = -1$, hence $2^{\kappa_0} = 1/2$; natural subtraction would silently give the wrong term. Put

$$\mathcal{I}_m(y) = \sum_{j=0}^{m} 2^{\kappa_j} \mathcal{F}\big(2^{(j:\mathbb{Z})-(m:\mathbb{Z})}\big) \frac{y^j}{j!}. \tag{168}$$

Lemma 9.4. *For every m and y,*

$$\mathcal{I}_m(y) = \frac{1}{2}\mathcal{T}_m(y). \tag{169}$$

Proof. For $j \leq m$, set $r = m - j$. The inverse-dyadic identity (107) gives

$$\mathcal{F}(2^{j-m}) = \frac{d_{m-j}}{(m-j)!\, 2^{\binom{m-j}{2}}}.$$

Together with $\kappa_j = \binom{j+1}{2} - 1$, this identifies each summand with one half of the corresponding summand in (155). ☐

Let $\tau(a) = \mathrm{wt}_2(a) \bmod 2$. Since

$$\frac{(-1)^a - 1}{2} = 2\left\lfloor \frac{a}{2} \right\rfloor - a,$$

the literal summand

$$\begin{aligned} \Lambda_m(x) := &\big((-1)^{\lfloor 2^m x \rfloor} - 1\big)(-1)^{\tau(\lfloor 2^m x \rfloor)} \\ &\times \sum_{j=0}^{m} 2^{((j+2)(j-1))/2} \mathcal{F}\big(2^{j-m}\big) \frac{(x - \lfloor 2^m x \rfloor / 2^m)^j}{j!} \end{aligned} \tag{170}$$

is exactly $\alpha_m(x)\mathcal{T}_m(y_m)$. Both exponents $((j+2)(j-1))/2$ and $j - m$ in this display are integer exponents, not truncated natural subtractions.

Warning 9.5. In (152) and (153), the summation indices are cast into the ambient characteristic-zero field before the subtraction in $r - m2^k + Q$ is performed. This is the typing used by the formal theorem; the displayed subtraction is therefore field subtraction, never truncated subtraction on natural numbers.

Theorem 9.6 (Corrected parity-power series)**.** *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \Lambda_m(x),} \tag{171}$$

and the series converges absolutely. On the half-open unit interval $0 \leq x < 1$, the zeroth summand vanishes and one may start at $m = 1$.

Proof. The elementary parity identity and (169) prove $\Lambda_m = \alpha_m \mathcal{T}_m(y_m)$ term by term. The claim follows from (166). The half-open corollary follows from $\lfloor x \rfloor = 0$; the closed endpoint $x = 1$ requires the restored zeroth term. □

9.2 The global complex q-binomial series

The finite constant-polynomial identity from section 8.7 can be inserted at every scale of the binary telescope. For $Q \in \mathbb{C}$, define

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)} (r - 2^k + Q)^{n+k}. \tag{172}$$

Translation invariance and the inverse-dyadic normalization give

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n} = \frac{2^n d_n}{n!}. \tag{173}$$

The normalization differs from the 2^{n^2} in (153): here it produces a Taylor coefficient rather than a function value.

For $m \in \mathbb{N}$, put

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^{m} \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}}(1/2;1/2)_n}. \tag{174}$$

Theorem 9.7 (q-binomial Taylor identity)**.** *For every m, every $y \in \mathbb{R}$, and every $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y), \tag{175}$$

where the real polynomial on the right is embedded in $\mathbb{C}$. In particular, each finite summand is pointwise independent of Q before any infinite sum is taken.

Proof. Insert (173) into (174) and set $j = m - n$. Collecting the powers of 2 gives

$$2^{-\binom{m+1}{2}}\, 2^{(m+1)j}\, 2^{m-j} = 2^{\binom{j+1}{2} - \binom{m-j}{2}}.$$

The remaining factorials and moments are exactly those in (155). □

Theorem 9.8 (Global complex q-binomial series)**.** *For fixed $Q \in \mathbb{C}$ and every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x)),} \tag{176}$$

with the left side embedded in $\mathbb{C}$. The series converges absolutely.

This is not a new infinite basic-hypergeometric identity. The outer sum is the analytic binary telescope, while at each scale a finite q-binomial identity presents the same Taylor polynomial. Expanding one summand makes the architecture explicit:

$$\alpha_m(x)\, 2^{-\binom{m+1}{2}} \sum_{n=0}^{m} \frac{(2^{m+1}x - 2p_m(x))^{m-n}}{(m-n)!} \frac{1}{2^{\binom{n}{2}}(1/2;1/2)_n} \times \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\mathrm{wt}_2(r)}(r - 2^k + Q)^{n+k}. \tag{177}$$

Convergence comes entirely from the binary-tail majorant (165); the special-function manipulations are finite and algebraic.

10 A complex q-binomial discrete limit

There is a second way in which the same Gaussian row produces $\mathcal{F}$. It is a genuine limit of finite shifted splines rather than a sequence of partial sums of (176). Reuse the safe midpoint cutoff $N_p(x)$ from (53). For $x \geq 0$ it is equivalent to the inclusive source range

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor, \tag{178}$$

including the empty case where the upper endpoint is -1.

For fixed $Q \in \mathbb{C}$ and $x \in \mathbb{R}$, define

$$\mathcal{D}_n^{(Q)}(x) = \frac{1}{2^{n^2}(1/2;1/2)_n} \sum_{k=0}^{n} \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \times \sum_{r=0}^{N_{n+k}(x)-1} (-1)^{\mathrm{wt}_2(r)} \left(r - 2^{n+k}x + Q\right)^{n+k}. \tag{179}$$

Theorem 10.1 (Complex discrete-limit formula)**.** *For every fixed $Q \in \mathbb{C}$ and every $x \in \mathbb{R}$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x)} \tag{180}$$

in $\mathbb{C}$. If $0 \leq x \leq 1$, the limit is equivalently $F(x)$.

Proof. If $x \leq 0$, every safe cutoff $N_p(x)$ is zero. Hence every shifted spline and every approximant (179) is zero, as is $\mathcal{F}(x)$. It remains to treat $x \geq 0$.

For a degree p and complex shift Q, introduce the finite shifted spline

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\mathrm{wt}_2(r)}(r - 2^p x + Q)^p. \tag{181}$$

At $Q = 1/2$ this is exactly the centered spline (54), so by (61),

$$S_p^{(1/2)}(x) \longrightarrow \mathcal{F}(x).$$

Lemma 10.2 (Fixed-cutoff Taylor expansion of a complex shift)**.** *Read (54) at $p = 0$ as*

$$S_0(x) = \sum_{r=0}^{N_0(x)-1} \varepsilon_r, \tag{182}$$

so that a centered branch S_d is defined for every integer $d \ge 0$. Fix $p \ge 1$ and $x \in \mathbb{R}$, and abbreviate $M = N_p(x)$. For every d with $0 \le d \le p$, the degree-d branch carried by that one fixed cutoff satisfies

$$\frac{1}{2^{\binom{d}{2}} d!} \sum_{r=0}^{M-1} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)^d = S_d\left(2^{p-d} x\right), \tag{183}$$

and in particular $\left|S_d\left(2^{p-d}x\right)\right| \le 1$. Moreover, for every $Q \in \mathbb{C}$,

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}}(p-d)!} \left(\frac{1}{2} - Q\right)^{p-d} S_d\left(2^{p-d}x\right). \tag{184}$$

Proof. Write the shifted spline in its descending orientation. Since $(-1)^p(r - 2^p x + Q)^p = (2^p x - Q - r)^p$, definition (181) reads

$$S_p^{(Q)}(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r=0}^{M-1} \varepsilon_r \left(2^p x - Q - r\right)^p, \qquad M = N_p(x), \tag{185}$$

and at $Q = 1/2$ this is (54). The cutoff M depends on p and x only. It is unaffected by Q, and it is held fixed in what follows while the degree of the summand varies; that is the whole point of (183).

We prove (183). The dilation $x \mapsto 2^{p-d}x$ compensates exactly for the drop in degree, in the sense that

$$N_d\left(2^{p-d}x\right) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right) = N_p(x) = M \tag{186}$$

by (53). If $M = 0$, both sides of (183) are empty sums and there is nothing to prove. Assume $M \ge 1$. Every index in the range satisfies

$$r \le M - 1 = \left\lfloor 2^p x + \frac{1}{2} \right\rfloor - 1 \le 2^p x - \frac{1}{2},$$

so every base $2^p x - \frac{1}{2} - r$ occurring in (183) is nonnegative and the truncation in (55) is inactive. For $d \ge 1$, formula (55) evaluated at the point $2^{p-d}x$, whose own cutoff is M by (186), is therefore literally the left side of (183). For $d = 0$ both sides equal $\sum_{r=0}^{M-1} \varepsilon_r$, by (182) and (186).

For the uniform bound, the case $d \ge 1$ is the global estimate in (61). For $d = 0$ pair the indices: since $\varepsilon_{2r} = \varepsilon_r$ and $\varepsilon_{2r+1} = -\varepsilon_r$, consecutive pairs cancel, so $\sum_{r=0}^{2m-1} \varepsilon_r = 0$ and $\sum_{r=0}^{2m} \varepsilon_r = \varepsilon_m$. Thus S_0 takes only the values 0 and ± 1.

Now expand. Put $w = 2^p x - \frac{1}{2}$ for the descending argument at $Q = 1/2$ and $t = \frac{1}{2} - Q$ for its increment, so that $2^p x - Q = w + t$. The binomial theorem gives

$$(w + t - r)^p = \sum_{d=0}^{p} \binom{p}{d} t^{p-d} (w - r)^d.$$

Multiply by $\varepsilon_r / (2^{\binom{p}{2}} p!)$, sum over $0 \le r \le M-1$, and interchange the two finite sums. Using (185) on the left and (183) on each inner sum,

$$\begin{aligned} S_p^{(Q)}(x) &= \sum_{d=0}^{p} \frac{t^{p-d}}{2^{\binom{p}{2}} p!} \binom{p}{d} \sum_{r=0}^{M-1} \varepsilon_r (w - r)^d \\ &= \sum_{d=0}^{p} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}} (p-d)!} t^{p-d} S_d\left(2^{p-d}x\right), \end{aligned} \tag{187}$$

because $\binom{p}{d}/p! = 1/(d!\,(p-d)!)$ and the inner sum is $2^{\binom{d}{2}} d!\, S_d(2^{p-d}x)$. The term $d = p$ is $S_p(x) = S_p^{(1/2)}(x)$; subtracting it leaves (184). $\square$

Remark 10.3 (Which convention makes the sign disappear). The expansion (184) is free of alternating signs only because the increment is measured in the *descending* variable $2^p x - Q$ of (185), that is, because the coefficient carries $\left(\frac{1}{2} - Q\right)^{p-d}$ and not $\left(Q - \frac{1}{2}\right)^{p-d}$. If one instead measures the shift by the ascending increment $q = Q - \frac{1}{2}$, which is the natural reading when the branch is written as a function of the ascending argument $r - 2^p x + Q$ of (181), then the identity acquires exactly the factor $(-1)^{p+d}$:

$$S_p^{(Q)}(x) - S_p^{(1/2)}(x) = \sum_{d=0}^{p-1} (-1)^{p+d} \frac{2^{\binom{d}{2}}}{2^{\binom{p}{2}}(p-d)!}\, q^{p-d}\, S_d\left(2^{p-d}x\right), \qquad q = Q - \frac{1}{2}. \tag{188}$$

The two displays are the same statement, since $(-1)^{p+d} = (-1)^{p-d}$ and therefore $(-1)^{p+d} q^{p-d} = \left(\frac{1}{2} - Q\right)^{p-d}$. It is worth being explicit about where the sign does *not* live: it is not a matter of renormalizing the lower branches. The alternating prefactor $(-1)^d$ of (54) is cancelled exactly by the $(-1)^d$ produced when the ascending argument is reversed, so the branch values $S_d(2^{p-d}x)$ appearing in (184) and in (188) are the same numbers. The sign belongs to the increment alone, and the convention that removes it is the descending one.

Remark 10.4 (Where the two constants of the bound come from). Taking absolute values in (184) and using $\left|S_d(2^{p-d}x)\right| \le 1$ gives

$$\left|S_p^{(Q)}(x) - S_p^{(1/2)}(x)\right| \le \sum_{d=0}^{p-1} 2^{\binom{d}{2} - \binom{p}{2}} \frac{\left|Q - \frac{1}{2}\right|^{p-d}}{(p-d)!}. \tag{189}$$

Both constants of (191) are visible in (189). Write $e = p - d$, so that $1 \le e \le p$ and

$$\binom{p}{2} - \binom{d}{2} = \frac{(p-d)(p+d-1)}{2} = \frac{e\,(2p - e - 1)}{2}.$$

As a function of e this is a downward parabola, so on $1 \le e \le p$ it attains its minimum at an endpoint; the value at $e = 1$ is $p - 1$ and the value at $e = p$ is $\binom{p}{2} \ge p - 1$. Hence every power of two in (189) is at most $2^{-(p-1)}$, with equality exactly in the top term $d = p - 1$, and

$$\left|S_p^{(Q)}(x) - S_p^{(1/2)}(x)\right| \le 2^{-(p-1)} \sum_{e=1}^{p} \frac{\left|Q - \frac{1}{2}\right|^{e}}{e!} \le 2^{-(p-1)} \left(\exp\left(\left|Q - \tfrac{1}{2}\right|\right) - 1\right). \tag{190}$$

This is (191), marginally sharpened. The factor $2^{-(p-1)}$ is the extremal case $d = p - 1$ of the triangular exponent; the exponential is the truncated exponential series of the increment; and the only input from the normalized Thue–Morse branches is the uniform bound $|S_d| \le 1$, which is why no further constant appears.

Expanding a fixed cutoff branch about $Q = 1/2$ gives an exact finite Taylor combination of lower-degree centered branches. The normalized Thue–Morse bounds then give, for $p \ge 1$,

$$\left|S_p^{(Q)}(x) - S_p^{(1/2)}(x)\right| \le 2^{-(p-1)} \exp\left(|Q - \tfrac{1}{2}|\right). \tag{191}$$

Thus every fixed complex shift has the same spline limit.

Now reindex the outer sum in (179) by $j = n - k$. Direct collection of signs, powers, and factorials gives the exact finite identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^{n} \omega_{n,j} S_{2n-j}^{(Q)}(x), \tag{192}$$

where

$$\omega_{n,j} = \frac{(-1)^j \left[{n \atop j}\right]_{1/2} 2^{-\binom{j+1}{2}}}{(1/2;1/2)_n}. \tag{193}$$

The finite q-binomial theorem gives the row normalization

$$\sum_{j=0}^{n} \omega_{n,j} = 1. \tag{194}$$

Applying it with the opposite sign also gives the uniform variation bound

$$\sum_{j=0}^{n} |\omega_{n,j}| = \frac{(-1/2;1/2)_n}{(1/2;1/2)_n} \le 16. \tag{195}$$

Finally, every sampled degree satisfies $2n - j \ge n$. A finite-row Toeplitz lemma therefore carries the convergence of $S_p^{(Q)}(x)$ to the weighted rows (192). Explicitly, given $\epsilon > 0$, choose P so that $|S_p^{(Q)}(x) - \mathcal{F}(x)| < \epsilon/16$ for $p \ge P$. If $n \ge P$, then every $2n - j \ge n \ge P$, and the row-sum identity together with (195) gives

$$\left| \sum_{j=0}^{n} \omega_{n,j} S_{2n-j}^{(Q)}(x) - \mathcal{F}(x) \right| \le \sum_{j=0}^{n} |\omega_{n,j}|\, |S_{2n-j}^{(Q)}(x) - \mathcal{F}(x)| < \epsilon.$$

This proves (180). □

Two distinctions prevent common misreadings of this theorem. First, unlike the complete finite dyadic numerator and each summand of the global q-series, a finite value $\mathcal{D}_n^{(Q)}(x)$ can depend on Q; it is the limit that is independent of a fixed shift. Second, the Toeplitz row (192) is not a finite binary telescope or a partial sum of (176). These finite expressions share a limit but are not equal term by term.

The two properties of the reindexed row (193) used above are both evaluations of the finite q-binomial theorem (145), at $z = 1/2$ and at $z = -1/2$ respectively. Both use the exponent identity $\binom{j}{2} + j = \binom{j+1}{2}$.

For the row sum, evaluate (145) at $z = 1/2$. The left side is $(1/2;1/2)_n$ by definition, and the right side is

$$(1/2;1/2)_n = \sum_{j=0}^{n} (-1)^j 2^{-\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \left(\frac{1}{2}\right)^j = \sum_{j=0}^{n} (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}. \tag{196}$$

Dividing by $(1/2;1/2)_n$ gives exactly $\sum_{j=0}^{n} \omega_{n,j} = 1$, which is (194).

For the total variation, evaluate (145) at $z = -1/2$ instead. Each factor $(-1)^j(-1/2)^j = 2^{-j}$ is now positive, so the alternating signs disappear:

$$(-1/2;1/2)_n = \sum_{j=0}^{n} (-1)^j 2^{-\binom{j}{2}} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} \left(-\frac{1}{2}\right)^j = \sum_{j=0}^{n} \begin{bmatrix} n \\ j \end{bmatrix}_{1/2} 2^{-\binom{j+1}{2}}. \tag{197}$$

Since $\left[{n \atop j}\right]_{1/2} > 0$, the summands of (197) are the absolute values of those of (196). Dividing by $(1/2;1/2)_n$ gives the exact evaluation

$$\sum_{j=0}^{n} |\omega_{n,j}| = \frac{(-1/2;1/2)_n}{(1/2;1/2)_n}, \tag{198}$$

which is the first equality in (195). The formally proved finite-product estimate supplies the bound by 16 used in the Toeplitz argument above.

11 Iterated prefix sums of the signed Thue–Morse sequence

The signed Thue–Morse sequence $\varepsilon_n = (-1)^{\mathrm{wt}_2(n)}$ has already appeared twice in this article: once as the coefficient sequence of the locally finite global expansion (13), and once as the sign pattern that makes the finite dyadic sums of section 8 collapse. There is a third, entirely discrete, way in which the same signs produce the Fabius function. Sum the sequence, sum the result, and iterate. After k summations the sequence has been convolved with an explicit binomial kernel, and after a dyadic normalization the resulting integer grid reproduces, in the limit, the Rvachev bump and hence F.

This section develops that construction: the convolution law and its explicit binomial kernel, the exact zero runs forced by Prouhet cancellation, the two generating-series identities, the discrete functional equation satisfied by the normalized grid, and the precise sense in which the normalized grid converges. The convergence statement requires one shift of the prefix order and the centered reading of the grid cells; the literal normalization does not converge, while the corrected centered formulation below does. Two further pieces of foundational material are attached at the end: the lower branch of Lambert's W-function, which the asymptotic sections of this article use but never construct, and an exact binomial–logarithm formula for the Thue–Morse bit.

11.1 Iterated prefix sums and the binomial convolution law

Definition 11.1 (Iterated prefix sums)**.** Set $\sigma_0(n) = \varepsilon_n$ and, for $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{j=0}^{n} \sigma_k(j). \tag{199}$$

All sums are *inclusive*: the index n itself is summed.

Every $\sigma_k(n)$ is an integer, and $\sigma_k(0) = 1$ for all k. The inclusive convention has one consequence that must be kept in view throughout, because it is the source of an index shift later:

$$\sigma_{k+1}(n+1) - \sigma_{k+1}(n) = \sigma_k(n+1). \tag{200}$$

The increment of a prefix sum is the lower-order value at the *new* index, not at the old one.

Proposition 11.2 (Odd zeros of the first prefix sum)**.** *For every $m \geq 0$, $\sigma_1(2m+1) = 0$.*

Proof. By (12), $\varepsilon_{2j} + \varepsilon_{2j+1} = 0$ for every j. The range $0 \leq n \leq 2m+1$ is the disjoint union of the pairs $\{2j, 2j+1\}$ for $0 \leq j \leq m$, so the sum of ε_n over that range vanishes. □

The general structure is a discrete convolution. Write

$$\mathcal{B}_k(i) = \binom{i+k-1}{k-1} \qquad (k \geq 1,\ i \geq 0) \tag{201}$$

for the number of ways of writing i as an ordered sum of k nonnegative integers. The family $\mathcal{B}_k$ is the k-fold discrete convolution power of the constant kernel $\mathcal{B}_1 \equiv 1$. Because the argument $i = n - m$ is automatically nonnegative in the sums below, no indicator function is needed.

Theorem 11.3 (Binomial convolution law)**.** *For every $k \geq 1$ and every $n \geq 0$,*

$$\sigma_k(n) = \sum_{m=0}^{n} \mathcal{B}_k(n-m)\,\varepsilon_m = \sum_{m=0}^{n} \binom{n-m+k-1}{k-1} \varepsilon_m. \tag{202}$$

Equivalently, in a form valid for every $k \geq 0$,

$$\sigma_{k+1}(n) = \sum_{m=0}^{n} \binom{n-m+k}{k} \varepsilon_m. \tag{203}$$

Proof. Induct on k in the form (203). For $k = 0$ the binomial coefficient is 1 and the assertion is the definition of σ_1. Assume (203) for k. Then

$$\begin{aligned}\sigma_{k+2}(n) &= \sum_{j=0}^{n} \sigma_{k+1}(j) = \sum_{j=0}^{n}\sum_{m=0}^{j} \binom{j-m+k}{k}\varepsilon_m \\ &= \sum_{m=0}^{n} \varepsilon_m \sum_{i=0}^{n-m} \binom{i+k}{k} = \sum_{m=0}^{n} \binom{n-m+k+1}{k+1}\varepsilon_m,\end{aligned}$$

where the interchange is the triangular reindexing $i = j - m$ and the inner evaluation is the hockey-stick identity $\sum_{i=0}^{N} \binom{i+k}{k} = \binom{N+k+1}{k+1}$. □

Remark 11.4. The kernel in (202) is exactly the coefficient sequence of $(1-z)^{-k}$. Theorem 11.3 is therefore the coefficient form of the series identity proved in theorem 11.10 below, and either statement may be taken as the definition of the other. Both are recorded here because the coefficient form is what one actually evaluates, while the series form is what one manipulates.

11.2 Prouhet annihilation and exact dyadic zero runs

Equation (121) already records that a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below the block exponent. For the prefix sums we also need the sharp boundary case, which identifies the first power that survives.

Lemma 11.5 (Sharp Prouhet extraction)**.** *Write $S_{r,d} = \sum_{h<2^r} \varepsilon_h h^d$. Then $S_{r,d} = 0$ for $0 \le d < r$, and*

$$S_{r,r} = (-1)^r 2^{\binom{r}{2}} r!. \tag{204}$$

Consequently, for all rational numbers $a_0, \dots, a_r$,

$$\sum_{h<2^r} \varepsilon_h \sum_{d=0}^{r} a_d h^d = a_r\,(-1)^r 2^{\binom{r}{2}} r!, \tag{205}$$

so summation against a complete dyadic block of Thue–Morse signs annihilates every polynomial of degree below r and, on polynomials of degree at most r, extracts the leading coefficient up to the explicit nonzero factor $(-1)^r 2^{\binom{r}{2}} r!$.

Proof. The vanishing statement is (121) with $X = 0$, $Y = 1$. For the boundary case, split the block by parity rather than by halves. Since $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$ by (12),

$$S_{r+1,d} = \sum_{j<2^r} \varepsilon_j \left[(2j)^d - (2j+1)^d\right] = -\sum_{c=0}^{d-1} \binom{d}{c} 2^c S_{r,c},$$

because $(2j+1)^d - (2j)^d = \sum_{c<d} \binom{d}{c}(2j)^c$. Take $d = r+1$. All terms with $c < r$ vanish by the first assertion, leaving only $c = r$:

$$S_{r+1,r+1} = -\binom{r+1}{r} 2^r S_{r,r} = -(r+1)2^r S_{r,r}.$$

Since $S_{0,0} = \varepsilon_0 = 1$, induction gives $S_{r,r} = (-1)^r r!\, 2^{0+1+\cdots+(r-1)} = (-1)^r r!\, 2^{\binom{r}{2}}$, which is (204). Formula (205) follows by exchanging the two sums: every index $d < r$ contributes $a_d S_{r,d} = 0$, and the index $d = r$ contributes $a_r S_{r,r}$. □

Corollary 11.6 (Affine sharp case)**.** *For all $X, Y \in \mathbb{Q}$,*

$$\sum_{h<2^r} \varepsilon_h (X+Yh)^r = (-1)^r Y^r 2^{\binom{r}{2}} r!.$$

In particular the sum does not depend on the translation X.

Proof. Expand $(X+Yh)^r$ by the binomial theorem and apply (205): the coefficient of h^r is Y^r. □

Equivalently, (205) is the statement that the block polynomial of (148) has a zero of exact order r at $z=1$: the mth derivative of $\prod_{j<r}(1-z^{2^j})$ at $z=1$ vanishes for $m<r$, while the rth derivative equals $(-1)^r r! \prod_{j<r} 2^j = (-1)^r r!\, 2^{\binom{r}{2}}$. We use the polynomial form because it applies directly to the binomial kernel (201), which is a polynomial in the summation index.

Theorem 11.7 (Exact dyadic zero runs)**.** *Let $1 \le k \le r$. Then*

$$\sigma_k(n) = 0 \qquad \text{for } 2^r - k \le n \le 2^r - 1, \tag{206}$$

$$\sigma_k(2^r - k - 1) = (-1)^{r-k}, \tag{207}$$

$$\sigma_k(2^r) = -1. \tag{208}$$

Formula (207) also holds for $k=0$, where it reduces to $\varepsilon_{2^r-1} = (-1)^r$, and (208) also holds for $k=0$. Hence the zero run terminating at 2^r-1 has length exactly k: it is bounded on the left by a value of modulus one and on the right by the value -1.

Proof. Endpoint. Put $q = k-1$, so $0 \le q < r$. By (202),

$$\sigma_k(2^r-1) = \sum_{m=0}^{2^r-1} \binom{2^r-1-m+q}{q} \varepsilon_m = \frac{1}{q!} \sum_{m<2^r} \varepsilon_m \prod_{i=1}^{q} (2^r-1-m+i).$$

The product is a polynomial in m of degree $q<r$, so the sum vanishes by (205).

Rest of the run. We prove, by induction on d, that $\sigma_k(2^r-1-d) = 0$ whenever $1 \le k \le r$ and $0 \le d < k$. The case $d=0$ is the endpoint. Suppose the claim holds for d and that $d+1<k$, so $k \ge 2$ and $1 \le k-1 \le r$. The induction hypothesis applied at order k gives $\sigma_k(2^r-1-d)=0$, and applied at order $k-1$, which is legitimate because $d<k-1$, gives $\sigma_{k-1}(2^r-1-d)=0$. Now use (200) with $n = 2^r-1-(d+1)$, so that $n+1 = 2^r-1-d$:

$$0 - \sigma_k(n) = \sigma_k(n+1) - \sigma_k(n) = \sigma_{k-1}(n+1) = 0.$$

This proves (206), since the indices $2^r-k, \dots, 2^r-1$ are exactly the 2^r-1-d with $0 \le d < k$.

Left boundary. Induct on k. For $k=0$ the number 2^r-1 has r binary digits, all equal to one, so $\varepsilon_{2^r-1} = (-1)^r$. Assume (207) at order k and let $k+1 \le r$. Put $n = 2^r-(k+1)-1$, so $n+1 = 2^r-k-1$. By the induction hypothesis $\sigma_k(n+1) = (-1)^{r-k}$, and by (206) at order $k+1$ we have $\sigma_{k+1}(n+1) = \sigma_{k+1}(2^r-(k+1)) = 0$. Then (200) gives $-\sigma_{k+1}(n) = (-1)^{r-k}$, that is, $\sigma_{k+1}(n) = (-1)^{r-(k+1)}$.

Right boundary. Induct on k. For $k=0$, $\mathrm{wt}_2(2^r)=1$, so $\varepsilon_{2^r} = -1$. If (208) holds at order k and $k+1 \le r$, then by (200) and (206), $\sigma_{k+1}(2^r) = \sigma_{k+1}(2^r-1) + \sigma_k(2^r) = 0-1 = -1$. □

Remark 11.8. The right boundary value is not $+1$. Every iterated prefix sum, at every order $k \le r$, takes the value -1 at the index 2^r. This single sign is what makes the literal normalization of section 11.4 fail at the right endpoint.

11.3 Two generating-series identities

All series in this subsection are formal power series over $\mathbb{Z}$; no convergence hypothesis on a complex variable is used or needed.

Let

$$\mathcal{K}_r(z) = \sum_{n<2^r} \varepsilon_n z^n, \qquad \mathcal{K}(z) = \sum_{n\geq 0} \varepsilon_n z^n, \qquad \mathcal{S}_k(z) = \sum_{n\geq 0} \sigma_k(n) z^n. \tag{209}$$

Proposition 11.9 (Block product and coefficient stabilization)**.** *For every $r \geq 0$,*

$$\mathcal{K}_r(z) = \prod_{j=0}^{r-1} (1 - z^{2^j}), \tag{210}$$

which is (148). Moreover, for every $n < 2^r$ the coefficient of z^n in the finite product (210) equals ε_n and is therefore independent of r. Consequently the infinite product $\prod_{j\geq 0}(1 - z^{2^j})$ is a well-defined formal power series, and it equals $\mathcal{K}(z)$.

Proof. Induct on r. For $r = 0$ both sides are 1. For the step, split the block of length 2^{r+1} at 2^r and use $\varepsilon_{2^r+n} = -\varepsilon_n$ for $n < 2^r$:

$$\mathcal{K}_{r+1}(z) = \sum_{n<2^r} \varepsilon_n z^n - z^{2^r} \sum_{n<2^r} \varepsilon_n z^n = \mathcal{K}_r(z)(1 - z^{2^r}),$$

and the product formula follows. For the stabilization, $\mathcal{K}_r$ is a polynomial of degree $2^r - 1$ whose coefficient of z^n is ε_n for every $n < 2^r$; adjoining the factor $1 - z^{2^r}$ changes only coefficients of index $\geq 2^r$. Thus for fixed n the coefficient of z^n in $\prod_{j<r}(1 - z^{2^j})$ is constant once $2^r > n$, which is the definition of convergence of the infinite product in the formal topology, and the stable value is ε_n. □

Theorem 11.10 (Prefix generating series)**.** *For every $k \geq 0$,*

$$\mathcal{S}_k(z) = \frac{\mathcal{K}(z)}{(1 - z)^k}, \tag{211}$$

equivalently, in denominator-cleared form,

$$(1 - z)^k \, \mathcal{S}_k(z) = \mathcal{K}(z) = \prod_{j\geq 0} (1 - z^{2^j}). \tag{212}$$

Both identities have a finite verification: for every r and every $n < 2^r$, the coefficient of z^n on either side of (212) may be computed from the polynomial $\prod_{j<r}(1 - z^{2^j})$ alone.

Proof. For $k = 0$ both sides of (211) are $\mathcal{K}(z)$. For $k \geq 1$, expand $(1-z)^{-k} = \sum_{i\geq 0} \binom{i+k-1}{k-1} z^i = \sum_{i\geq 0} \mathcal{B}_k(i) z^i$. The coefficient of z^n in $\mathcal{K}(z)(1 - z)^{-k}$ is then the Cauchy product $\sum_{m=0}^{n} \mathcal{B}_k(n - m)\varepsilon_m$, which is $\sigma_k(n)$ by (202). Since $(1-z)^k$ is a unit multiple of the invertible series $(1-z)^{-k}$, the cleared form (212) is equivalent, and the product expression is proposition 11.9. The finite verification is the stabilization statement of that proposition: no coefficient of index below 2^r sees any factor $1 - z^{2^j}$ with $j \geq r$. □

The two identities of theorem 11.10 connect the prefix sums to the polynomial approximants used in section 11.5. These are the combinatorial step polynomials already introduced in (44); their coefficient interpretation and exact degree were proved there. In the index convention of this section,

$$\mathcal{P}_n(z) = \prod_{i=1}^{n} (1 + z + \cdots + z^{2^i - 1}), \qquad g_n = \deg \mathcal{P}_n = \sum_{i=1}^{n} (2^i - 1) = 2^{n+1} - n - 2. \tag{213}$$

Since $1 - z^{2^j} = (1-z)(1+z+\cdots+z^{2^j-1})$ and the factor with $j=0$ is trivial, (210) factors as

$$\mathcal{K}_{n+1}(z) = (1-z)^{n+1}\mathcal{P}_n(z). \tag{214}$$

Comparing (214) with (212) at order $k = n+1$ and using stabilization below 2^{n+1} gives the exact coefficient identification

$$\sigma_{n+1}(m) = [z^m]\mathcal{P}_n(z) \qquad (0 \le m < 2^{n+1}). \tag{215}$$

Note that (215) must fail at $m = 2^{n+1}$, because $g_n < 2^{n+1}$ makes the right side zero while the left side is -1 by (208). That single index is the exceptional right endpoint of theorem 11.16.

11.4 The functional equation of the normalized prefix grid

The natural dyadic normalization of the prefix sums is

$$\mathcal{V}_k(j) = \frac{\sigma_k(j)}{2^{\binom{k}{2}}}, \qquad x_{k,j} = \frac{j}{2^k}, \tag{216}$$

so that $\mathcal{V}_k$ is a rational-valued function on the level-k dyadic grid. The exponent $\binom{k}{2}$ is forced: it is the exponent that appears in the sharp Prouhet value (204), in the centered spline (54), and in the derivative scaling (75).

Theorem 11.11 (Discrete functional equation)**.** *Let $q \ge 0$ and $0 \le j < 2^q$, and put $h = 2^{-(q+1)}$ and $x = x_{q+1,j}$. Then*

$$\frac{\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j)}{h} = 2\,\mathcal{V}_q(j+1), \tag{217}$$

and the abscissa attached to the right-hand value is

$$x_{q,j+1} = 2(x+h) = 2x_{q+1,j+1} \in [0,1]. \tag{218}$$

The same identity holds verbatim for the shifted grid $\widetilde{\mathcal{V}}_k(j) = \sigma_{k+1}(j)/2^{\binom{k}{2}}$ of section 11.5.

Proof. Because $\binom{q+1}{2} = \binom{q}{2} + q$, the forward difference (200) gives

$$\mathcal{V}_{q+1}(j+1) - \mathcal{V}_{q+1}(j) = \frac{\sigma_{q+1}(j+1) - \sigma_{q+1}(j)}{2^{\binom{q}{2}+q}} = \frac{\sigma_q(j+1)}{2^{\binom{q}{2}}} \cdot \frac{1}{2^q} = \frac{\mathcal{V}_q(j+1)}{2^q}.$$

Dividing by $h = 2^{-(q+1)}$ multiplies by 2^{q+1} and gives (217). For (218), $2(x+h) = 2(j+1)/2^{q+1} = (j+1)/2^q = x_{q,j+1}$, and the hypothesis $j < 2^q$ is exactly the condition $j+1 \le 2^q$ that keeps this abscissa in the unit interval. The shifted grid satisfies the same computation with σ replaced by $\sigma_{\bullet+1}$ throughout. □

Equation (217) is the discrete counterpart of the global differential equation (74): a difference quotient at level $q+1$ equals twice a level-q value at the doubled abscissa. Two features of the discrete statement have no continuous analogue and are easy to lose. First, the right-hand side is evaluated at the *forward* node $2(x+h)$ rather than at $2x$; this is the inclusive-prefix convention (200). Second, the identity is an assertion about a level-q value, so it is meaningful as a statement about F or up only when the doubled abscissa lies in the relevant interval, that is, only when $j < 2^q$. Both features are used in section 11.5.

11.5 Corrected normalization and pointwise convergence

We now identify the normalized prefix grid with the histogram approximants of the probabilistic construction, and deduce convergence. Two adjustments to the naive reading are required, and both are forced by identities already proved: the prefix order must be shifted by one, because (215) matches σ_{n+1} with $\mathcal{P}_n$; and the grid index m must be read as labelling the *center* of a cell of width 2^{-n}, not the left endpoint of a cell of width $2^{-(n+1)}$.

For the polynomial $\mathcal{P}_n$ of (213), place the coefficient of the monomial z^m at

$$\xi_{n,m} = \frac{2m - g_n}{2^{n+1}}, \tag{219}$$

and spread it uniformly over the cell $I_{n,m}$ of width 2^{-n} centered at $\xi_{n,m}$. The consecutive centers are spaced 2^{-n} apart, so these finitely many cells tile their support interval

$$\left[-1 + \frac{n+1}{2^{n+1}},\ 1 - \frac{n+1}{2^{n+1}}\right].$$

The resulting step approximant is

$$\mathcal{W}_n(x) = \frac{2^n}{2^{\binom{n+1}{2}}} \sum_{m=0}^{g_n} [z^m]\mathcal{P}_n(z) \cdot \mathbf{1}_{I_{n,m}}(x), \tag{220}$$

where a point shared by two adjacent cells is given the weight $\frac{1}{2}$ in each, so that the two contributions add to the single value the formula is intended to have. This endpoint convention is not cosmetic: with ordinary closed-interval indicators the value at $x = 0$ would be counted twice already for $n = 1$.

Theorem 11.12 (Histogram approximants)**.** *For every $n \geq 0$ the function $\mathcal{W}_n$ satisfies $0 \leq \mathcal{W}_n \leq 1$, is nondecreasing on $(-\infty, 0)$, is nonincreasing on $(0, \infty)$, and satisfies $\mathcal{W}_n(0) = 1$. Moreover $\mathcal{W}_n(x) \to \operatorname{up}(x)$ for every real x.*

Proof. For $n \geq 1$, this is exactly theorem 4.2, since (213) and (44) define the same polynomial, while (219) and (220) give the same cells and normalization as (47) and (48). At $n = 0$ the empty product is $\mathcal{P}_0 = 1$, $g_0 = 0$, and $\mathcal{W}_0$ is the half-weighted indicator of $[-1/2, 1/2]$, so the stated order and normalization properties are immediate. The pointwise convergence of the sequence is already the final assertion of theorem 4.2. □

Theorem 11.13 (Exact identification of the corrected grid)**.** *Put $\widetilde{\mathcal{V}}_n(m) = \sigma_{n+1}(m)/2^{\binom{n}{2}}$. For every $n \geq 0$ and every m with $0 \leq m < 2^{n+1}$,*

$$\widetilde{\mathcal{V}}_n(m) = \mathcal{W}_n(\xi_{n,m}). \tag{221}$$

Proof. If $m \leq g_n$, the point $\xi_{n,m}$ is the center of $I_{n,m}$ and lies in no other cell, so (220) evaluates there to

$$\frac{2^n}{2^{\binom{n+1}{2}}} [z^m]\mathcal{P}_n(z).$$

Since $\binom{n+1}{2} = \binom{n}{2} + n$, this is $[z^m]\mathcal{P}_n/2^{\binom{n}{2}} = \widetilde{\mathcal{V}}_n(m)$ by (215). If $g_n < m < 2^{n+1}$, then $\xi_{n,m}$ lies strictly to the right of the finite support interval above, so $\mathcal{W}_n(\xi_{n,m}) = 0$; meanwhile $[z^m]\mathcal{P}_n = \sigma_{n+1}(m) = 0$ because $\deg \mathcal{P}_n = g_n$. Thus both sides again agree. □

Sampling at the dyadic floor now gives an approximation scheme. For $x \geq 0$ set

$$j_n(x) = \lfloor 2^{n+1}x \rfloor, \qquad \mathcal{J}_n(x) = \widetilde{\mathcal{V}}_n(j_n(x)) = \frac{\sigma_{n+1}(j_n(x))}{2^{\binom{n}{2}}}. \tag{222}$$

Lemma 11.14 (Moving cell centers)**.** *For every $x \in [0,1)$, $\xi_{n,j_n(x)} \to 2x-1$ as $n \to \infty$.*

Proof. By (219), $\xi_{n,j_n(x)} = 2 \cdot j_n(x)2^{-(n+1)} - g_n 2^{-(n+1)}$. The first term tends to $2x$ because $0 \le 2^{n+1}x - j_n(x) < 1$, and $g_n 2^{-(n+1)} = 1-(n+2)2^{-(n+1)} \to 1$. □

Lemma 11.15 (Stability under moving arguments)**.** *If $u_n \to y$ in $\mathbb{R}$, then $\mathcal{W}_n(u_n) \to \mathrm{up}(y)$.*

Proof. Let $\varepsilon > 0$. Suppose first $y > 0$. By continuity of up choose $d \in (0, y/2]$ with $|\mathrm{up}(y \pm d) - \mathrm{up}(y)| < \varepsilon/2$. For all large n we have $u_n \in [y-d, y+d] \subset (0,\infty)$, so the monotonicity of theorem 11.12 on $(0,\infty)$ gives

$$\mathcal{W}_n(y+d) \le \mathcal{W}_n(u_n) \le \mathcal{W}_n(y-d).$$

By pointwise convergence at the two fixed points $y \pm d$, for all large n both outer terms lie within $\varepsilon/2$ of $\mathrm{up}(y \pm d)$, hence within ε of $\mathrm{up}(y)$. The case $y < 0$ is identical using monotonicity on $(-\infty, 0)$. For $y = 0$ the upper bound is $\mathcal{W}_n(u_n) \le 1 = \mathrm{up}(0)$, and the lower bound is $\mathcal{W}_n(-d)$, $\mathcal{W}_n(0) = 1$, or $\mathcal{W}_n(d)$ according to the sign of u_n; each of the three converges to a value within $\varepsilon/2$ of $\mathrm{up}(0)$. □

Theorem 11.16 (Corrected prefix approximation)**.** *For every $x \in [0,1]$,*

$$\mathcal{J}_n(x) \longrightarrow \mathrm{up}(2x-1), \tag{223}$$

and consequently, for every $x \in [0,1]$,

$$\mathcal{J}_n\left(\frac{x}{2}\right) \longrightarrow F(x). \tag{224}$$

Proof. Let $0 \le x < 1$. Then $2^{n+1}x < 2^{n+1}$, so $j_n(x) < 2^{n+1}$ and theorem 11.13 applies: $\mathcal{J}_n(x) = \mathcal{W}_n(\xi_{n,j_n(x)})$. Combining lemma 11.14 with lemma 11.15 gives $\mathcal{J}_n(x) \to \mathrm{up}(2x-1)$.

At $x = 1$ the identification fails, because $j_n(1) = 2^{n+1}$ is exactly the first index not covered by (215). Here the value is computed instead from (208) with $k = r = n+1$:

$$\mathcal{J}_n(1) = \frac{\sigma_{n+1}(2^{n+1})}{2^{\binom{n}{2}}} = -\frac{1}{2^{\binom{n}{2}}} \longrightarrow 0 = \mathrm{up}(1),$$

since $\binom{n}{2} \to \infty$ and $\mathrm{up}(1) = F(0) = 0$. This proves (223) on all of $[0,1]$. For (224), apply it at $x/2 \in [0,1]$ and use $2(x/2) - 1 = x - 1 \le 0$, so that $\mathrm{up}(x-1) = F(x)$ by (7). □

Warning 11.17. Theorem 11.16 is a pointwise statement and nothing more. No uniform rate is asserted, and none is proved. The proof passes through lemma 11.15, whose only quantitative inputs are the shared monotonicity of the $\mathcal{W}_n$ and the modulus of continuity of the limit up at the single point $2x-1$; the sandwich it uses degrades as that modulus degrades. A uniform bound of the kind supplied for the centered splines by (60) is available from section 4.4, but it is a different approximation scheme, and the two should not be conflated.

Remark 11.18. The two corrections are independent of one another and both are necessary. Shifting the prefix order from k to $k+1$ is what makes (215) available; reading the index m as the center (219) of a cell of width 2^{-n} rather than as an abscissa $m2^{-k}$ is what supplies the affine change of variable $x \mapsto 2x-1$ between the unit interval and the support of up. In particular the corrected grid is a sample of the compactly supported bump, not of the distribution function; this is why its right-endpoint limit is 0 and not 1.

11.6 The lower Lambert branch

The asymptotic analysis of section 13 is organized around the coordinate (272), where W_{-1} denotes the lower real branch of Lambert's W-function [11]. That section, and the elementary expansions derived from it in section 15, use three facts about W_{-1} without proving any of them: that the branch exists on $(-\mathrm{e}^{-1}, 0)$, that it is characterized by being the solution below -1, and that it has the two-term expansion recorded below. We supply them here, since they are the foundational input for (274) and for every subsequent estimate in the Lambert variable.

Proposition 11.19 (Existence, defining equation, and uniqueness)**.** *The map $u \mapsto u \log u$ is a strictly decreasing bijection from $(0, \mathrm{e}^{-1})$ onto $(-\mathrm{e}^{-1}, 0)$. For $z \in (-\mathrm{e}^{-1}, 0)$ let $u(z)$ be its unique preimage and set $W_{-1}(z) = \log u(z)$. Then*

$$W_{-1}(z) < -1, \qquad W_{-1}(z)\,\mathrm{e}^{W_{-1}(z)} = z, \tag{225}$$

and $W_{-1}(z)$ is the unique real $w < -1$ with $w\mathrm{e}^w = z$.

Proof. On $(0, \mathrm{e}^{-1})$ the derivative of $u \log u$ is $1 + \log u < 0$, so the map is strictly decreasing there. Its limit at $u \downarrow 0$ is 0 and its value at $u = \mathrm{e}^{-1}$ is $-\mathrm{e}^{-1}$; by the intermediate value theorem and strict monotonicity it is a bijection onto $(-\mathrm{e}^{-1}, 0)$. Write $u = u(z)$ and $w = \log u$. Since $0 < u < \mathrm{e}^{-1}$ we get $w < -1$, and $\mathrm{e}^w = u$, so $w\mathrm{e}^w = u \log u = z$, which is (225). Conversely, if $w < -1$ and $w\mathrm{e}^w = z$, put $v = \mathrm{e}^w$; then $v \in (0, \mathrm{e}^{-1})$ and $v \log v = w\mathrm{e}^w = z$, so $v = u(z)$ by injectivity, hence $w = \log u(z) = W_{-1}(z)$. □

Corollary 11.20 (Solution of the saddle equation)**.** *Let $x > 0$ with $Lx < \mathrm{e}^{-1}$, where $L = \log 2$ as in (231), and put $\lambda = -W_{-1}(-Lx)/L$ as in (273). Then $\lambda > 1/L$ and $\lambda 2^{-\lambda} = x$, which is (274).*

Proof. The number $z = -Lx$ lies in $(-\mathrm{e}^{-1}, 0)$, so proposition 11.19 applies. Write $W = W_{-1}(z)$, so $\lambda = -W/L$ and $2^{-\lambda} = \mathrm{e}^{-L\lambda} = \mathrm{e}^W$. Hence $\lambda 2^{-\lambda} = (-W/L)\mathrm{e}^W = -z/L = x$. Finally $W < -1$ gives $\lambda > 1/L$. □

Proposition 11.21 (Two-term expansion)**.** *As $\epsilon \downarrow 0$,*

$$W_{-1}(-\epsilon) - \big(\log \epsilon - \log|\log \epsilon|\big) \longrightarrow 0. \tag{226}$$

Proof. For $0 < \epsilon < \mathrm{e}^{-1}$ put $y = y(\epsilon) = -W_{-1}(-\epsilon) > 1$. Negating (225) gives the defining equation in the convenient form

$$y\,\mathrm{e}^{-y} = \epsilon.$$

The map $t \mapsto t\mathrm{e}^{-t}$ is strictly decreasing on $[1, \infty)$ with limit 0 at infinity, so $y(\epsilon) \to \infty$ as $\epsilon \downarrow 0$. Taking logarithms in the displayed equation gives $\log \epsilon = \log y - y$, hence

$$\frac{-\log \epsilon}{y} = 1 - \frac{\log y}{y} \longrightarrow 1,$$

because $\log y/y \to 0$. Finally, $\epsilon < 1$ makes $\log \epsilon < 0$, so $|\log \epsilon| = -\log \epsilon$, and

$$\begin{aligned} W_{-1}(-\epsilon) - \big(\log \epsilon - \log|\log \epsilon|\big) &= -y - (\log y - y) + \log(-\log \epsilon) \\ &= \log \frac{-\log \epsilon}{y} \longrightarrow \log 1 = 0. \end{aligned}$$ □

11.7 A binomial–logarithm formula for the Thue–Morse bit

The zero-one Thue–Morse bit is $\tau(n) = \mathrm{wt}_2(n) \bmod 2$, so that $\varepsilon_n = 1 - 2\tau(n)$. Its primitive-recursive evaluation computes one binary digit at a time:

$$b(0) = 0, \qquad b(m) = \big(b(\lfloor m/2 \rfloor) + (m \bmod 2)\big) \bmod 2 \quad (m > 0). \tag{227}$$

It also admits a closed formula in which the entire binary structure is hidden inside a single row of Pascal's triangle. Set

$$\mathcal{X}(n) = \sum_{k=0}^{n} (-1)^{\binom{n}{k}} \in \mathbb{Z}. \tag{228}$$

Lemma 11.22 (Glaisher count)**.** *The number of odd entries in the nth row of Pascal's triangle is* $2^{\mathrm{wt}_2(n)}$.

Proof. Let $\nu(n)$ denote that number, so that $\nu(n)$ is the number of nonzero coefficients of $(1+z)^n$ in $\mathbb{F}_2[z]$. There $(1+z)^2 = 1 + z^2$, hence

$$(1+z)^{2n} = \big(1+z^2\big)^n, \qquad (1+z)^{2n+1} = \big(1+z^2\big)^n(1+z).$$

The substitution $z \mapsto z^2$ does not change the number of nonzero coefficients, so $\nu(2n) = \nu(n)$; and multiplying a polynomial in z^2 by $1+z$ produces two summands with disjoint supports, one even and one odd, so $\nu(2n+1) = 2\nu(n)$. Since $\mathrm{wt}_2(2n) = \mathrm{wt}_2(n)$ and $\mathrm{wt}_2(2n+1) = \mathrm{wt}_2(n)+1$ by (11), and $\nu(0) = 1 = 2^{\mathrm{wt}_2(0)}$, induction on the binary representation gives $\nu(n) = 2^{\mathrm{wt}_2(n)}$. □

Proposition 11.23 (The logarithm is exact)**.** *For every $n \ge 0$,*

$$n + 1 - \mathcal{X}(n) = 2^{\mathrm{wt}_2(n)+1}. \tag{229}$$

In particular $n + 1 - \mathcal{X}(n)$ is a positive power of two, and $\log_2\big(n+1-\mathcal{X}(n)\big) = \mathrm{wt}_2(n)+1$ is an integer.

Proof. Each summand of (228) is -1 at an odd entry and $+1$ at an even entry, and the row has $n+1$ entries. Hence $\mathcal{X}(n) = (n+1) - 2\nu(n)$ with ν as in lemma 11.22, so $n+1-\mathcal{X}(n) = 2\nu(n) = 2^{\mathrm{wt}_2(n)+1}$. □

Theorem 11.24 (Binomial–logarithm formula)**.** *For every $n \ge 0$,*

$$\tau(n) = \frac{1 + (-1)^{\log_2\left(n+1-\sum_{k=0}^{n}(-1)^{\binom{n}{k}}\right)}}{2}, \qquad \varepsilon_n = 1 - 2\tau(n). \tag{230}$$

Proof. By proposition 11.23 the exponent equals $\mathrm{wt}_2(n)+1$. If $\mathrm{wt}_2(n)$ is even then $\tau(n) = 0$ and $(-1)^{\mathrm{wt}_2(n)+1} = -1$, so the right side is 0; if $\mathrm{wt}_2(n)$ is odd then $\tau(n) = 1$ and $(-1)^{\mathrm{wt}_2(n)+1} = 1$, so the right side is 1. The second identity is $\varepsilon_n = (-1)^{\mathrm{wt}_2(n)} = 1 - 2\big(\mathrm{wt}_2(n) \bmod 2\big)$. □

Remark 11.25. The order of the two steps matters. As printed, (230) raises -1 to a logarithm, an operation that is meaningful only because the argument of that logarithm has first been proved to be exactly a power of two. The formal development therefore proves (229) before it interprets the outer expression at all, evaluates the logarithm on the natural numbers, and reads the final division in $\mathbb{Q}$. A floating-point evaluation of the same display, or an interpretation of the logarithm as a real-valued function whose output is then rounded, is an approximation to a formula rather than the formula.

12 Endpoint asymptotics: exact logarithmic decomposition

We now turn to the behavior of $F(x)$ as $x \downarrow 0$. This is the technically deepest part of the theory and the place where a very small, but mathematically unavoidable, log-periodic fluctuation appears.

Throughout the asymptotic sections, put

$$L = \log 2. \tag{231}$$

We use the standard Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + \mathcal{O}(s^2), \tag{232}$$

where γ is Euler's constant and γ_1 is the first Stieltjes constant.

12.1 An exact delay identity and the coarse quadratic scale

Before introducing the saddle point, it is useful to isolate what follows directly from the differential equation. Put

$$\phi(t) = F(2^{-t}), \qquad \Gamma(t) = -\log \phi(t). \tag{233}$$

For $t \geq 1$, the argument 2^{-t} lies in the left half of the unit interval, and the chain rule with $F'(x) = 2F(2x)$ gives

$$\phi'(t) = -L\, 2^{1-t}\phi(t-1), \qquad \Gamma'(t) = L\, 2^{1-t}\frac{\phi(t-1)}{\phi(t)} > 0. \tag{234}$$

Taking logarithms of the second identity yields the exact delay relation

$$\boxed{\Gamma(t) - \Gamma(t-1) = \log \Gamma'(t) - \log L - (1-t)L, \qquad t \geq 1.} \tag{235}$$

The restriction $t \geq 1$ is essential: it is precisely what licenses the left-half bounded differential law.

The recurrence and monotonicity already prove, independently of the sharper periodic saddle decomposition below,

$$\frac{\Gamma(t)}{t^2} \longrightarrow \frac{L}{2}, \tag{236}$$

and, for all sufficiently large t,

$$\left|\Gamma(t) - \frac{L}{2}t^2\right| \leq 16t\log t. \tag{237}$$

At integral scales the elementary estimate is sharper:

$$\left|\Gamma(n) - \frac{L}{2}n^2\right| \leq 3n\log(n+1) \qquad (n \geq 1). \tag{238}$$

Monotonicity of Γ and $\lfloor t \rfloor \leq t < \lfloor t \rfloor + 1$ transfer the integer estimate to real t. These facts are formalized separately in the log-scale modules: they establish the quadratic law before the analysis below resolves the linear and logarithmic drift, the exact phase, and its unavoidable periodic fluctuation.

12.2 The negative Laplace product

The function G from (99) is the moment generating function of the random variable X in (33):

$$G(z) = \mathbb{E}\mathrm{e}^{zX}. \tag{239}$$

Therefore, for $s > 0$, $G(-s)$ is the ordinary Laplace transform of the probability law.

Iterating the functional equation (100) in the negative direction gives an exact product.

Proposition 12.1 (Negative Laplace product)**.** *For $s > 0$,*

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - \mathrm{e}^{-s/2^j}}{s/2^j}. \tag{240}$$

Consequently

$$\Lambda(s) := \log G(-s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \qquad \kappa(u) = \log \frac{1 - \mathrm{e}^{-u}}{u}. \tag{241}$$

The series converges absolutely and locally uniformly on $(0, \infty)$*.*

Proof. Putting $z = -s/2$ in (100) gives

$$G(-s) = \frac{1 - \mathrm{e}^{-s/2}}{s/2} G(-s/2).$$

After N iterations, the remaining factor is $G(-s/2^N)$, which tends to $G(0) = 1$. For small u,

$$\kappa(u) = -\frac{u}{2} + \mathcal{O}(u^2),$$

so the logarithmic series is dominated on compact s-intervals by a geometric series. □

The product has the exact dilation law

$$\Lambda(2s) = \kappa(s) + \Lambda(s). \tag{242}$$

This equation resembles a renewal equation on the logarithmic scale. Its homogeneous solutions are periodic functions of $\log_2 s$, which is the conceptual origin of the fluctuation in the final asymptotic.

12.3 Removing the quadratic drift

Define the forward tail

$$T(s) = \sum_{m=0}^{\infty} \log(1 - \mathrm{e}^{-s2^m}) < 0 \tag{243}$$

and the positive tail error

$$E(s) = -T(s) \geq 0. \tag{244}$$

The series is exponentially small as $s \to \infty$.

Lemma 12.2 (Forward-tail bound)**.** *For $s > 0$,*

$$0 \leq E(s) \leq \frac{\mathrm{e}^{-s}}{(1 - \mathrm{e}^{-s})^2}. \tag{245}$$

In particular $E(s) \leq 4\mathrm{e}^{-s}$ for $s \geq \log 2$.

Proof. For $0 < q < 1$, $-\log(1-q) \le q/(1-q)$. Hence

$$E(s) \le \frac{1}{1-\mathrm{e}^{-s}} \sum_{m\ge 0} \mathrm{e}^{-s2^m}.$$

Since $2^m \ge m+1$, the last sum is at most $\sum_{m\ge 0} \mathrm{e}^{-s(m+1)} = \mathrm{e}^{-s}/(1-\mathrm{e}^{-s})$. □

For $t \in \mathbb{R}$, set

$$P(t) = \Lambda(2^t) + \frac{L}{2}(t^2 - t) + T(2^t). \tag{246}$$

Theorem 12.3 (Exact periodic decomposition)**.** *The function P is 1-periodic and smooth. For every $s > 0$, with $t = \log_2 s$, one has the exact identity*

$$\boxed{\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s).} \tag{247}$$

Proof. Write $s = 2^t$. Using (242),

$$\Lambda(2s) = \Lambda(s) + \log(1-\mathrm{e}^{-s}) - \log s.$$

Also

$$T(2s) = T(s) - \log(1-\mathrm{e}^{-s}).$$

The change in the quadratic correction is

$$\frac{L}{2}\left((t+1)^2 - (t+1) - (t^2 - t)\right) = Lt = \log s.$$

The three changes cancel, proving $P(t+1) = P(t)$. Rearranging (246) and using $E = -T$ yields (247). Smoothness follows by differentiating locally uniformly convergent series; periodicity then makes every derivative bounded. □

Let

$$\overline{P} = \int_0^1 P(t)\,\mathrm{d}t, \qquad \Psi(t) = P(t) - \overline{P}. \tag{248}$$

Then Ψ is smooth, real-valued, 1-periodic, and has mean zero.

12.4 Fourier coefficients of the periodic correction

The periodic term is not an unspecified bounded nuisance: all of its Fourier coefficients are explicit. Put

$$\chi_k = \frac{2\pi \mathrm{i} k}{L} \qquad (k \in \mathbb{Z}), \tag{249}$$

and use the convention

$$\widehat{\Psi}(k) = \int_0^1 \Psi(t)\mathrm{e}^{-2\pi \mathrm{i} k t}\,\mathrm{d}t. \tag{250}$$

Theorem 12.4 (Gamma–zeta Fourier coefficients)**.** *One has $\widehat{\Psi}(0) = 0$, and for $k \neq 0$,*

$$\boxed{\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}.} \tag{251}$$

Every nonzero Fourier coefficient is nonzero. In particular, Ψ is genuinely nonconstant.

Proof. The zero coefficient is zero by definition. For $k \neq 0$, change variables $s = 2^t$ in (250). Dyadic periodization turns the integral over one logarithmic period into a Mellin transform; this is the standard harmonic-sum framework for dyadic periodicities [12]. The basic Mellin identity is

$$\int_0^\infty s^{z-1} \log(1 - \mathrm{e}^{-s})\, \mathrm{d}s = -\Gamma(z)\zeta(1+z), \qquad \Re z > 0. \tag{252}$$

It follows by expanding $\log(1 - \mathrm{e}^{-s}) = -\sum_{m \geq 1} \mathrm{e}^{-ms}/m$ and integrating term by term. Near $s = 0$ the kernel contains logarithmic and linear divergences; the quadratic correction in (246) is exactly the finite-part subtraction that removes them. Analytic continuation of the regularized Mellin transform to the imaginary axis gives (251) at $z = -\chi_k$.

The Gamma function has no zeros. Moreover $1 - \chi_k$ lies on the line $\Re z = 1$ away from the pole at 1, and the zeta function has no zeros there. Hence the product in (251) is nonzero. ☐

Conjugate symmetry $\widehat{\Psi}(-k) = \overline{\widehat{\Psi}(k)}$ reconstructs a real-valued Fourier series. Stirling's estimate for Γ on vertical lines implies exponential decay of the coefficients, so the Fourier series and all of its differentiated series converge rapidly.

The first coefficient is tiny:

$$\widehat{\Psi}(1) \approx (7.5586475410 - 7.4701035590\,\mathrm{i}) \times 10^{-7}, \tag{253}$$

with magnitude approximately $1.0627116252 \times 10^{-6}$. The second is smaller by six orders of magnitude,

$$\widehat{\Psi}(2) \approx (3.2481234772 + 5.8517589871\,\mathrm{i}) \times 10^{-13}, \tag{254}$$

of magnitude $6.6927863679 \times 10^{-13}$, so the first harmonic already describes Ψ to better than one part in a million of its own size.

Warning 12.5 (A numerical hazard specific to these points). Evaluating (251) numerically is delicate for a reason that has nothing to do with rounding. Many implementations of the zeta function compute it from the alternating Dirichlet series through

$$\zeta(s) = \frac{1}{1 - 2^{1-s}} \sum_{m \geq 1} \frac{(-1)^{m-1}}{m^s}. \tag{255}$$

At the points that (251) needs, $s = 1 - \chi_k$, one has $2^{1-s} = 2^{\chi_k} = \mathrm{e}^{\chi_k L} = \mathrm{e}^{2\pi \mathrm{i} k} = 1$ exactly, so the prefactor in (255) has a pole precisely there. The singularity is removable in ζ but not in that formula, and a routine built on it returns an overflow or, worse, a plausible wrong number. This is not a rounding story; it is a defect of the algorithm at exactly the arguments this problem cares about, and it explains how two independently published decimal values for $\widehat{\Psi}(1)$ can agree with each other to eight figures and both be wrong.

The symptom that identifies the defect is that raising the working precision does not shrink the error. It moves it:

working digits	value returned for $10^7\widehat{\Psi}(1)$	
30	$7.55865005 - 7.47011001\,\mathrm{i}$	
50	$7.55864674 - 7.47009636\,\mathrm{i}$	
80	$7.55865060 - 7.47008702\,\mathrm{i}$	
any	$7.55864754 - 7.47010356\,\mathrm{i}$	(Euler–Maclaurin)

The first three rows come from a widely used arbitrary-precision library whose complex zeta routes through (255); the last is a hand-written Euler–Maclaurin evaluation, which is stable at all three precisions and agrees with (253).[1] Digits that wander as the precision rises are the signature of a removable singularity being divided by, and they are exactly what would persuade a careful reader to trust a wrong value. Use an Euler–Maclaurin or Hurwitz evaluation instead, and check the answer at two working precisions.

Thus the visible oscillation is on the scale of a few parts in a million in the logarithm. Tiny is not the same as zero: its nonvanishing changes the logical status of an asymptotic equivalent.

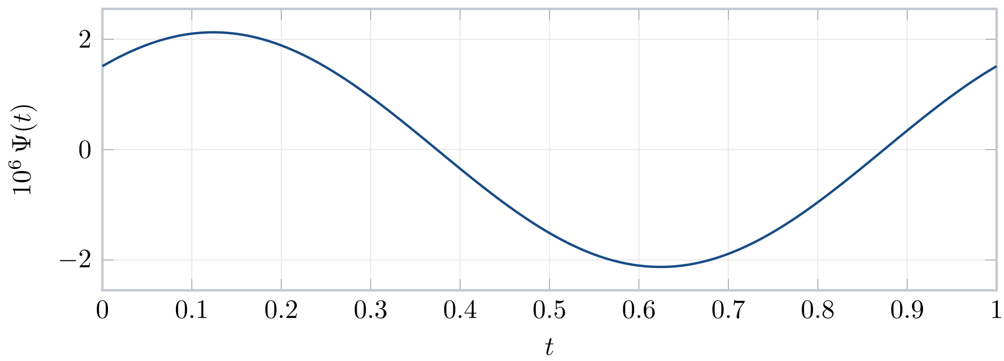

Figure 1: The periodic correction Ψ, magnified by 10^6. The curve is indistinguishable from its first Fourier harmonic at this resolution, and the half-period antisymmetry $\Psi(t+\frac{1}{2}) \approx -\Psi(t)$ visible here is the statement that the even harmonics are negligible. The vertical scale is the point of the picture: the oscillation is a few parts in a million in the logarithm, but it does not tend to zero.

12.5 The finite part at the Mellin origin

The constant that enters the saddle asymptotic comes from the finite part of $-\Gamma(z)\zeta(1+z)$ at $z=0$. Using

$$\Gamma(z) = \frac{1}{z} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12}\right) z + \mathcal{O}(z^2),$$
$$\zeta(1+z) = \frac{1}{z} + \gamma - \gamma_1 z + \mathcal{O}(z^2),$$

one finds

$$-\Gamma(z)\zeta(1+z) + \frac{1}{z^2} = A_0 + \mathcal{O}(z), \tag{256}$$

where

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}. \tag{257}$$

12.5.1 A convergent real finite-part integral

The formal development also realizes this Laurent coefficient as an ordinary sum of two absolutely convergent real integrals. For $x > 0$, write

$$\kappa_0(x) = \frac{1}{x}\log\frac{1-\mathrm{e}^{-x}}{x}, \qquad \kappa_\infty(x) = \frac{1}{x}\log(1-\mathrm{e}^{-x}). \tag{258}$$

[1] The precision table is due to an independent recomputation carried out alongside this article; the Euler–Maclaurin values were checked here at two truncations, 60 terms with 12 Bernoulli corrections and 100 with 16, which agree with each other to 37 digits.

The first kernel stays uniformly bounded as $x \downarrow 0$: the formal estimate is

$$|\kappa_0(x)| \le 1 \qquad (x > 0). \tag{259}$$

The second is integrable at infinity, by an explicit exponential majorant. Thus no analytic-continuation convention is hidden in the following identity.

Theorem 12.6 (Convergent Bose finite part)**.** *For every cutoff $c > 0$, one has the cutoff-independent identity*

$$\boxed{\int_{(0,c]} \kappa_0(x)\,\mathrm{d}x + \int_{(c,\infty)} \kappa_\infty(x)\,\mathrm{d}x + \frac{(\log c)^2}{2} = A_0.} \tag{260}$$

In particular, the unit cutoff gives

$$\int_{(0,1]} \kappa_0(x)\,\mathrm{d}x + \int_{(1,\infty)} \kappa_\infty(x)\,\mathrm{d}x = A_0. \tag{261}$$

Moreover, if $0 < a \le b$, then moving between the two regularizations over the transition interval incurs exactly

$$\int_{(a,b]} (\kappa_\infty(x) - \kappa_0(x))\,\mathrm{d}x = \frac{(\log b)^2 - (\log a)^2}{2}. \tag{262}$$

In Lean, the kernels in (258) are `boseFinitePartSmallKernel` and `boseFinitePartLargeKernel`. Their arbitrary-cutoff integrability is `integrableOn_boseFinitePartSmallKernel_Ioc` and `integrableOn_boseFinitePartLargeKernel_Ioi_of_pos`; the pointwise estimate (259) is `abs_boseFinitePartSmallKernel_le_one_of_pos`. Equation (262) is `integral_boseFinitePartLargeKernel_sub_small_Ioc`, and (260) is the expanded theorem `boseFinitePartIntegral_eq_gammaZetaConstant_of_pos`. Equivalently, Lean packages its left side as `boseFinitePartIntegralAtCutoff`; its cutoff independence and evaluation are `boseFinitePartIntegralAtCutoff_eq` and `boseFinitePartIntegralAtCutoff_eq_gammaZetaConstant`. Before adding the logarithmic correction, the raw quantity `boseFinitePartSplitIntegral` is evaluated by `boseFinitePartSplitIntegral_eq_gammaZetaConstant_sub`, which fixes the minus sign in $A_0 - (\log c)^2/2$. The unit-cutoff specialization (261) is `boseFinitePartIntegral_eq_gammaZetaConstant`. All these declarations are in `BoseFinitePartIntegral.lean`. The proof first establishes the positive-real Mellin identity `bose_mellin_integral_zeta` in `MellinBose.lean`, splits off the elementary integral of $x^{r-1}\log x$, and then uses `tendsto_realGamma_mul_zeta_finitePart` from `MellinFinitePart.lean` as $r \downarrow 0$. In particular, the displayed equality is a theorem about convergent real integrals, not a notation for a divergent integral assigned a regularized value.

The final constant is

$$C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2}\log\pi \approx -2.02798389863885942397. \tag{263}$$

The terms $-7L/12$ and $-(1/2)\log\pi$ arise from the dyadic Euler–Maclaurin finite part and the leading Gaussian saddle mass, respectively.

12.5.2 The Mellin transform in the relevant strip

The identity (252) is stated for the kernel $\log(1-\mathrm{e}^{-s})$ in the half-plane $\Re z > 0$. The kernel that actually occurs in (241) is not that one but

$$\kappa(u) = \log\frac{1-\mathrm{e}^{-u}}{u} = \log(1-\mathrm{e}^{-u}) - \log u,$$

and its Mellin transform lives in a different strip. Both facts are needed, and it is worth saying at once that they do not compete.

Lemma 12.7 (Mellin transform of the dyadic kernel)**.** *The integral*

$$\widetilde{\kappa}(z) = \int_0^\infty \kappa(u) u^{z-1}\, \mathrm{d}u \tag{264}$$

converges absolutely precisely in the strip $-1 < \Re z < 0$*, and there*

$$\widetilde{\kappa}(z) = -\Gamma(z)\zeta(1+z). \tag{265}$$

Proof. As $u \downarrow 0$ one has $\kappa(u) = -u/2 + \mathcal{O}(u^2)$, so $\kappa(u)u^{z-1} = \mathcal{O}(u^{\Re z})$ and the integral converges at the origin exactly when $\Re z > -1$. As $u \to \infty$ one has $\kappa(u) = -\log u + \mathcal{O}(\mathrm{e}^{-u})$, so the integrand is of size $u^{\Re z - 1}\log u$ and the integral converges at infinity exactly when $\Re z < 0$.

Fix z in the strip. Differentiating,

$$\kappa'(u) = \frac{1}{\mathrm{e}^u - 1} - \frac{1}{u}. \tag{266}$$

Integration by parts is legitimate because the boundary term $\kappa(u)u^z/z$ vanishes at both ends: near 0 it is $\mathcal{O}(u^{1+\Re z})$ and near ∞ it is $\mathcal{O}(u^{\Re z}\log u)$, and $-1 < \Re z < 0$. Hence

$$\widetilde{\kappa}(z) = -\frac{1}{z}\int_0^\infty \kappa'(u)u^z\, \mathrm{d}u = -\frac{1}{z}\int_0^\infty \left(\frac{1}{\mathrm{e}^u - 1} - \frac{1}{u}\right) u^z\, \mathrm{d}u.$$

The subtracted Bose integral is the classical analytic continuation of $\int_0^\infty u^{w-1}(\mathrm{e}^u - 1)^{-1}\, \mathrm{d}u = \Gamma(w)\zeta(w)$ from $\Re w > 1$ to the critical strip: for $0 < \Re w < 1$,

$$\int_0^\infty u^{w-1}\left(\frac{1}{\mathrm{e}^u - 1} - \frac{1}{u}\right) \mathrm{d}u = \Gamma(w)\zeta(w). \tag{267}$$

Applying (267) with $w = z+1$, which lies in $0 < \Re w < 1$ exactly when $-1 < \Re z < 0$, and using $\Gamma(z+1) = z\Gamma(z)$ gives

$$\widetilde{\kappa}(z) = -\frac{\Gamma(z+1)\zeta(z+1)}{z} = -\Gamma(z)\zeta(1+z). \qquad \square$$

Remark 12.8 (The two Mellin identities continue each other)*.* A reader who has just met (252) will notice that (265) has the same right-hand side in a disjoint region, and may suspect a clash. There is none. The kernel $\log(1 - \mathrm{e}^{-s})$ decays exponentially at infinity but is logarithmically singular at the origin, so its transform converges exactly in $\Re z > 0$. Subtracting $\log s$ replaces that logarithmic singularity by a zero of order one, which moves the left edge of the strip from 0 to -1; the price is that the subtracted term destroys convergence at infinity, which caps the strip at $\Re z < 0$. The two strips therefore abut along the line $\Re z = 0$, and the two integrals are the restrictions to the two sides of that line of the single meromorphic function $-\Gamma(z)\zeta(1+z)$, which has a double pole at $z = 0$. Neither identity is more fundamental; (265) is simply the one adapted to the harmonic sum (241).

Because Λ is the dyadic harmonic sum of κ, its Mellin transform acquires a geometric factor. For $-1 < \Re z < 0$ the substitution $s \mapsto 2^j s$ gives $\int_0^\infty \kappa(s/2^j)s^{z-1}\, \mathrm{d}s = 2^{jz}\widetilde{\kappa}(z)$, and $\sum_{j \ge 1} 2^{jz}$ converges in that strip, so

$$\widetilde{\Lambda}(z) = \int_0^\infty \Lambda(s)s^{z-1}\, \mathrm{d}s = \frac{2^z}{1 - 2^z}\big(-\Gamma(z)\zeta(1+z)\big), \qquad -1 < \Re z < 0. \tag{268}$$

The factor $2^z/(1-2^z)$ is where the discrete dyadic scale enters, and it is the source of both the periodicity and the constant $\overline{P}$. Writing $w = Lz$ and using $(\mathrm{e}^w - 1)^{-1} = w^{-1} - \frac{1}{2} + w/12 + \mathcal{O}(w^3)$,

$$\frac{2^z}{1-2^z} = -1 - \frac{1}{\mathrm{e}^w - 1} = -\frac{1}{Lz} - \frac{1}{2} - \frac{Lz}{12} + \mathcal{O}(z^3). \tag{269}$$

Its poles are simple and lie exactly at the points $z = \chi_k = 2\pi \mathrm{i}k/L$ of (249), since $1 - 2^z$ vanishes there with derivative $-L$.

Proposition 12.9 (Derivation of the mean of the periodic function)**.** *The mean $\overline{P}$ of (248) is*

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \tag{270}$$

with A_0 as in (257).

Proof. Combining (269) with the finite-part expansion (256), that is $-\Gamma(z)\zeta(1+z) = -z^{-2} + A_0 + \mathcal{O}(z)$, the product (268) has a triple pole at the origin with Laurent expansion

$$\widetilde{\Lambda}(z) = \frac{1}{Lz^3} + \frac{1}{2z^2} + \frac{1}{z}\left(\frac{L}{12} - \frac{A_0}{L}\right) + \mathcal{O}(1). \tag{271}$$

Mellin inversion writes $\Lambda(s)$ as an integral of $\widetilde{\Lambda}(z)s^{-z}$ along a vertical line inside the strip; displacing that line to the right past $\Re z = 0$ contributes minus the residues at the poles crossed. Expanding $s^{-z} = 1 - z\log s + \frac{1}{2}z^2\log^2 s - \cdots$ and reading off the coefficient of z^{-1} in $\widetilde{\Lambda}(z)s^{-z}$ gives

$$\operatorname*{Res}_{z=0} \widetilde{\Lambda}(z)s^{-z} = \frac{\log^2 s}{2L} - \frac{\log s}{2} + \frac{L}{12} - \frac{A_0}{L},$$

so the contribution of the triple pole is

$$-\frac{(\log s)^2}{2L} + \frac{\log s}{2} + \frac{A_0}{L} - \frac{L}{12}.$$

Comparison with the exact decomposition (247), which is proved independently and elementarily, identifies the first two terms with the explicit quadratic drift and the constant with the mean of P; the remaining displaced integral is smaller than every power of $1/\log s$ and is the forward tail $E(s)$. This proves (270). $\square$

The same residue computation reproduces (251) and thereby confirms the normalizations. Near $z = \chi_k$ with $k \neq 0$ one has $1 - 2^z \sim -L(z - \chi_k)$ and $2^z \to 1$, so

$$\operatorname*{Res}_{z=\chi_k} \widetilde{\Lambda}(z)s^{-z} = \frac{\Gamma(\chi_k)\zeta(1+\chi_k)}{L}\, s^{-\chi_k}.$$

With $s = 2^t$ one has $s^{-\chi_k} = \mathrm{e}^{-2\pi \mathrm{i}kt}$. After replacing k by $-k$, minus the sum of these residues is

$$\sum_{k\neq 0} -L^{-1}\Gamma(-\chi_k)\zeta(1-\chi_k)\mathrm{e}^{2\pi \mathrm{i}kt},$$

which is precisely the Fourier series of Ψ with the coefficients (251). Thus the mean and every oscillating mode come from the single factorization (268): the triple pole at the origin produces the nonperiodic part together with $\overline{P}$, and the simple poles on the imaginary axis produce Ψ.

13 The corrected sharp asymptotic

13.1 The lower Lambert coordinate

For sufficiently small $x > 0$, define

$$W = W_{-1}(-Lx), \tag{272}$$

where W_{-1} is the lower real branch of Lambert's W-function [11], and put

$$\lambda = \lambda(x) = -\frac{W}{L}. \tag{273}$$

Then $\lambda \to \infty$ as $x \downarrow 0$ and

$$\lambda 2^{-\lambda} = x. \tag{274}$$

Equivalently,

$$r = 2^{\lambda} = \frac{\lambda}{x}. \tag{275}$$

The real number r is the explicit Lambert saddle radius used in the contour analysis.

13.2 The main term

Define

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda), \tag{276}$$

where C_F is given by (263) and $\lambda = \lambda(x)$. The same expression can be written compactly in the Lambert variable W:

$$\mathcal{M}(x) = -\frac{7L}{12} - \frac{1}{2}\log(\pi x) + \frac{A_0 - W - \frac{1}{2}W^2}{L} + \Psi\left(-\frac{W}{L}\right). \tag{277}$$

The equivalence follows from $W = -L\lambda$ and $\log x = \log\lambda - L\lambda$.

Theorem 13.1 (Corrected sharp endpoint asymptotic)**.** *As $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{\lambda(x)}\right) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{-\log x}\right).} \tag{278}$$

Equivalently,

$$F(x) \sim \exp\mathcal{M}(x). \tag{279}$$

The periodic term is evaluated at the *exact* phase $\lambda(x)$, not at a truncated log–log approximation to it. This matters at the stated error scale.

13.3 Bromwich inversion

The contour representation used below rests on one elementary identity, which is worth recording separately because it is what converts a statement about the probability law into a statement about a Laplace transform with an explicit pole at the origin.

Proposition 13.2 (Laplace transform of the distribution function)**.** *For every $s > 0$,*

$$\int_0^\infty e^{-sx} F(x)\,dx = \frac{G(-s)}{s}, \tag{280}$$

where G is the moment generating function (239).

Proof. By (37) the function F is the distribution function of the random variable X of (33), so $F(0) = 0$, $F(x) = 1$ for $x \geq 1$, and F is continuous and nondecreasing. Split the integral at $x = 1$.

On $[0, 1]$, Stieltjes integration by parts gives

$$G(-s) = \int_0^1 \mathrm{e}^{-sx}\,\mathrm{d}F(x) = \left[\mathrm{e}^{-sx}F(x)\right]_0^1 + s\int_0^1 \mathrm{e}^{-sx}F(x)\,\mathrm{d}x = \mathrm{e}^{-s} + s\int_0^1 \mathrm{e}^{-sx}F(x)\,\mathrm{d}x, \tag{281}$$

because $F(1) = 1$ and $F(0) = 0$. Hence

$$\int_0^1 \mathrm{e}^{-sx}F(x)\,\mathrm{d}x = \frac{G(-s)}{s} - \frac{\mathrm{e}^{-s}}{s}.$$

On $[1, \infty)$ the integrand is the constant tail e^{-sx}, and

$$\int_1^\infty \mathrm{e}^{-sx}\,\mathrm{d}x = \frac{\mathrm{e}^{-s}}{s}.$$

The boundary term e^{-s}/s produced by the integration by parts and the contribution of the constant tail are therefore equal and opposite, and adding the two pieces gives (280). □

Two features of (280) decide the geometry of the inversion. First, since X is bounded, $G(-z) = \mathbb{E}\mathrm{e}^{-zX}$ is entire, so the only singularity of the transform $G(-z)/z$ is the simple pole at $z = 0$ coming from the constant tail of F; the integral in (280) converges absolutely for every z with $\Re z > 0$ and diverges at $z = 0$. The Bromwich representation

$$F(x) = \frac{1}{2\pi\mathrm{i}}\int_{r-\mathrm{i}\infty}^{r+\mathrm{i}\infty} \frac{\mathrm{e}^{zx}G(-z)}{z}\,\mathrm{d}z$$

of (282) is consequently valid on *every* vertical line $\Re z = r$ with $r > 0$, and on no vertical line with $r \leq 0$. The freedom in r is exactly what the saddle choice $r = 2^\lambda$ of (275) exploits. Second, absolute convergence of the contour integral on each such line follows from the rapid decay of $\tau \mapsto G(-r - \mathrm{i}\tau)$, which is the Fourier transform of a smooth compactly supported density up to an exponential tilt and therefore decays faster than every power (section 5). No contour shifting hypothesis is needed: the line $\Re z = r$ is admissible for all $r > 0$ simultaneously.

Because $G(-z) = \mathbb{E}\mathrm{e}^{-zX}$ is entire and F is continuous, Laplace–Stieltjes inversion gives, for every $r > 0$ and $0 < x < 1$,

$$F(x) = \frac{1}{2\pi\mathrm{i}}\int_{r-\mathrm{i}\infty}^{r+\mathrm{i}\infty} \frac{\mathrm{e}^{zx}G(-z)}{z}\,\mathrm{d}z. \tag{282}$$

Set $z = r(1 + \mathrm{i}\theta)$ and choose $r = 2^\lambda = \lambda/x$. Then

$$F(x) = \frac{\mathrm{e}^{\lambda+\Lambda(r)}}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\lambda\theta}\frac{G(-r(1+\mathrm{i}\theta))}{G(-r)}\frac{\mathrm{d}\theta}{1+\mathrm{i}\theta}. \tag{283}$$

The exact real decomposition (247) at $r = 2^\lambda$ gives

$$\lambda + \Lambda(r) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + P(\lambda) + E(r). \tag{284}$$

Since

$$\overline{P} = \frac{A_0}{L} - \frac{L}{12}, \qquad \overline{P} - \frac{1}{2}\log(2\pi) = C_F, \tag{285}$$

the missing factors in (276) are exactly the leading Gaussian mass $1/\sqrt{2\pi\lambda}$ and the centered fluctuation $\Psi = P - \overline{P}$.

13.4 The normalized saddle mass

It is convenient to give the entire remaining error a name. Everything in (283) except the scaled oscillatory integral is explicit, and that integral is a positive real number divided by the Gaussian mass it is supposed to reproduce.

Definition 13.3. For small $x > 0$, with $\lambda = \lambda(x)$ from (273) and $r = 2^\lambda$ from (275), put

$$\mathcal{G}(x) = \frac{F(x)\sqrt{2\pi\lambda}}{\exp(rx + \Lambda(r))}, \tag{286}$$

where $\Lambda = \log G(-\,\cdot\,)$ is the log-transform of (241).

The symbol $\mathcal{G}$ is used nowhere else in this article. Because $F(x) > 0$ for $x > 0$, the quantity $\mathcal{G}(x)$ is a positive real number, and by (283) together with $rx = \lambda$ it has the exact integral representation

$$\mathcal{G}(x) = \sqrt{\frac{\lambda}{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\lambda\theta} \frac{G(-r(1+\mathrm{i}\theta))}{G(-r)} \frac{\mathrm{d}\theta}{1+\mathrm{i}\theta} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp \Phi_\lambda\!\left(\frac{v}{\sqrt{\lambda}}\right) \mathrm{d}v, \tag{287}$$

the second form arising from the substitution (298) and the local phase (295) of the next subsection. Taking logarithms in (286) and using $rx = \lambda$ gives the exact identity

$$\boxed{\log F(x) = rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) + \log \mathcal{G}(x).} \tag{288}$$

No approximation whatever has been made in (288): it is a rearrangement of the definition of $\mathcal{G}$, valid for every x small enough that $\lambda(x)$ is defined. All of the analysis still to be done is the analysis of the single positive number $\mathcal{G}(x)$.

Proposition 13.4 (Exact separation of the error)**.** *For small $x > 0$, with $r = 2^{\lambda(x)}$,*

$$rx + \Lambda(r) - \frac{1}{2}\log(2\pi\lambda) = \mathcal{M}(x) + E(r), \tag{289}$$

where $\mathcal{M}$ is the main term (276) and E is the forward-tail error (244). Consequently

$$\boxed{\log F(x) - \mathcal{M}(x) = \log \mathcal{G}(x) + E(r).} \tag{290}$$

Proof. By (284) and $rx = \lambda$,

$$rx + \Lambda(r) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + P(\lambda) + E(r).$$

Subtract $\frac{1}{2}\log(2\pi\lambda) = \frac{1}{2}\log\lambda + \frac{1}{2}\log(2\pi)$ and split $P = \overline{P} + \Psi$ as in (248). By (285) one has $\overline{P} - \frac{1}{2}\log(2\pi) = C_F$, so the right-hand side becomes

$$-\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + E(r),$$

which is $\mathcal{M}(x) + E(r)$ by (276). Subtracting (289) from (288) gives (290). □

Equation (290) splits the total error into two parts of completely different nature. The term $E(r)$ is beyond all orders: by (245),

$$0 \le E(2^\lambda) \le 4\exp(-2^\lambda), \tag{291}$$

which is smaller than every power of $1/\lambda$, and indeed smaller than $\exp(-\lambda^K)$ for every K. It can never be the obstruction to any inverse-power expansion, and it may be carried along inertly through every order. Everything else is $\log \mathcal{G}(x)$, an algebraic quantity computed from a single explicit integral.

This bookkeeping is preferable to working with the phase integral directly. The phase integral in (283) is a complex oscillatory object whose normalization changes at every stage of the argument, so error statements about it must always be qualified by which prefactors have already been extracted. By contrast, $\mathcal{G}$ is dimensionless and pinned: it is a single positive real number whose distance from 1 *is* the entire error, up to the beyond-all-orders term. The leading theorem (278) is exactly the statement

$$\mathcal{G}(x) = 1 + \mathcal{O}\left(\frac{1}{\lambda(x)}\right), \tag{292}$$

the first correction (311) is the statement that $\log \mathcal{G}(x) = A_1(\lambda)/\lambda + \mathcal{O}(\lambda^{-2})$, and the all-orders theorem (312) is the statement that

$$\log \mathcal{G}(x) = \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}\left(\lambda^{-N}\right) \tag{293}$$

for every N. In each case the periodic coefficients are those of the relative saddle mass (313), and no further normalization has to be tracked.

13.5 The local phase

Let

$$u = \log(1 + \mathrm{i}\theta), \tag{294}$$

with the principal logarithm near the saddle. Subtracting the value at $\theta = 0$ from the complexified version of (247), and including the factor $(1+\mathrm{i}\theta)^{-1}$ in (283), gives the local phase

$$\Phi_\lambda(\theta) = \lambda(\mathrm{i}\theta - u) - \frac{u}{2} - \frac{u^2}{2L} + \Psi\left(\lambda + \frac{u}{L}\right) - \Psi(\lambda) + \mathcal{E}_r(\theta), \tag{295}$$

where $\mathcal{E}_r(0) = 0$ and, on a fixed neighborhood of zero, every derivative of $\mathcal{E}_r$ is exponentially small in r. This last term comes from the forward tail E and is far smaller than every power of $1/\lambda$.

The elementary expansion

$$\log(1 + \mathrm{i}\theta) = \mathrm{i}\theta + \frac{\theta^2}{2} - \frac{\mathrm{i}\theta^3}{3} - \frac{\theta^4}{4} + \mathcal{O}(\theta^5) \tag{296}$$

shows that

$$\lambda(\mathrm{i}\theta - u) = -\frac{\lambda\theta^2}{2} + \frac{\mathrm{i}\lambda\theta^3}{3} + \frac{\lambda\theta^4}{4} + \mathcal{O}(\lambda|\theta|^5). \tag{297}$$

Thus the natural central scale is

$$\theta = \frac{v}{\sqrt{\lambda}}. \tag{298}$$

Uniformly for $|v|$ growing sufficiently slowly with λ,

$$\Phi_\lambda(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \lambda^{-1/2} P_1(v;\lambda) + \lambda^{-1} P_2(v;\lambda) + \mathcal{O}\left(\lambda^{-3/2}(1 + |v|^9)\right), \tag{299}$$

where

$$A(u) = \frac{\Psi'(u)}{L} - \frac{1}{2}, \tag{300}$$

$$B(u) = -\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}, \tag{301}$$

$$P_1(v;u) = \mathrm{i}\left(A(u)v + \frac{v^3}{3}\right), \tag{302}$$

$$P_2(v;u) = \frac{v^4}{4} + B(u)v^2. \tag{303}$$

In (299), the second argument is $u = \lambda$.

13.6 Central and tail estimates

For completeness, the quantitative saddle proof is organized by the following estimates. They are stated here in the form needed for the argument; the formal development proves them with explicit domains and constants.

Lemma 13.5 (Central expansion)**.** *For every fixed truncation order N, there is $\delta > 0$ such that on $|\theta| \le \delta$ the exponent in (283) admits a Taylor expansion through order N after the scaling (298). Its coefficients are smooth 1-periodic functions of λ, and the remainder is dominated by a polynomial in v times the next power of $\lambda^{-1/2}$.*

Proof. The logarithm in (294) is analytic for $|\theta| < 1$. The periodic function Ψ is smooth with bounded derivatives of every order. Taylor's theorem applied to each term of (295), together with the exponentially small bounds on $\mathcal{E}_r$, gives a uniform expansion. The dependence on λ occurs only through $\Psi^{(j)}(\lambda)$, hence is periodic. ☐

Lemma 13.6 (Gaussian domination)**.** *There are positive constants c, δ and λ_0 such that, for $\lambda \ge \lambda_0$ and $|\theta| \le \delta$,*

$$\Re\Phi_\lambda(\theta) \le -c\lambda\theta^2. \tag{304}$$

Proof architecture. The exact function $\Re(\mathrm{i}\theta - \log(1 + \mathrm{i}\theta)) = -(1/2)\log(1 + \theta^2)$ supplies the negative quadratic term. All remaining terms in (295) have bounded first and second derivatives on a fixed neighborhood. For large λ the $-\lambda\theta^2/2$ contribution dominates them. ☐

Lemma 13.7 (Off-saddle decay)**.** *For each N one may choose the central window so that the contribution to (283) outside that window is $O(\lambda^{-N})$ relative to the leading saddle mass.*

Proof architecture. Split the negative-Laplace product at the dyadic index nearest λ. In the intermediate range, a positive proportion of the factors have arguments of order one and their moduli are strictly smaller off the real axis, yielding exponential decay in $\lambda\theta^2$. In the far range, repeated use of the sinc/Laplace product bounds gives arbitrarily high polynomial decay. The two bounds overlap and cover the whole complement of the central window. The factor $(1 + \mathrm{i}\theta)^{-1}$ is harmless. ☐

These lemmas justify extending the scaled central integral to the entire real v-axis and integrating its expansion term by term against $\mathrm{e}^{-v^2/2}$.

13.7 Completion of the leading proof

Keeping only the Gaussian term in (299) gives

$$\int_{\mathbb{R}} \mathrm{e}^{\Phi_\lambda(\theta)}\,\mathrm{d}\theta = \frac{1}{\sqrt{\lambda}}\int_{\mathbb{R}} \mathrm{e}^{-v^2/2}\,\mathrm{d}v\,\Big(1+\mathcal{O}(\lambda^{-1})\Big) = \sqrt{\frac{2\pi}{\lambda}}\,\Big(1+\mathcal{O}(\lambda^{-1})\Big). \tag{305}$$

The term of order $\lambda^{-1/2}$ integrates to zero because P_1 is odd in v and purely imaginary. Combining (284), (285), and (305) yields

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1+\frac{L}{2}\right)\lambda - \frac{1}{2}\log\lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}),$$

which is (278). Finally, $\lambda \asymp -\log x$ follows from (274), so the two displayed error scales are equivalent.

14 First correction and the all-orders saddle expansion

The leading theorem leaves an error of order $1/\lambda$. The same contour analysis can be continued to arbitrary order. The coefficient functions are periodic because every local coefficient is built from derivatives of Ψ evaluated at the exact phase λ.

14.1 The first correction

Expand the exponential in (299):

$$\mathrm{e}^{\Phi_\lambda(v/\sqrt{\lambda})} = \mathrm{e}^{-v^2/2}\Big[1+\lambda^{-1/2}P_1(v;\lambda) \tag{306}$$

$$+\,\lambda^{-1}\big(P_2(v;\lambda)+\tfrac{1}{2}P_1(v;\lambda)^2\big) + \mathcal{O}(\lambda^{-3/2}(1+|v|^{12}))\Big]. \tag{307}$$

Let Z be a standard normal random variable. Since

$$\mathbb{E}Z^2 = 1, \qquad \mathbb{E}Z^4 = 3, \qquad \mathbb{E}Z^6 = 15, \tag{308}$$

the normalized coefficient of λ^{-1} in the saddle mass is

$$\begin{aligned} b_1(u) &= \mathbb{E}\left[P_2(Z;u)+\tfrac{1}{2}P_1(Z;u)^2\right] \\ &= \frac{3}{4} + B(u) - \frac{1}{2}\left(A(u)^2 + 2A(u) + \frac{5}{3}\right). \end{aligned} \tag{309}$$

Substituting (300) and (301) simplifies the expression considerably.

Theorem 14.1 (First saddle correction)**.** *Define*

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u)+\Psi'(u)^2}{2L^2}. \tag{310}$$

Then

$$\boxed{\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \mathcal{O}\Big(\lambda(x)^{-2}\Big).} \tag{311}$$

Proof. The order-$\lambda^{-1/2}$ term in (307) is odd and integrates to zero. Equation (309) gives the first nonzero relative correction to the integral. A direct simplification yields (310). If the normalized saddle mass is $1+b_1/\lambda+O(\lambda^{-2})$, then its logarithm is $b_1/\lambda+O(\lambda^{-2})$, so the same coefficient appears in the logarithmic expansion. The off-saddle contribution is smaller than the stated remainder by the tail estimates of the preceding section. □

The constant part $1/24+1/(2L)$ is the usual type of Gaussian/curvature correction. The terms involving Ψ' and Ψ'' record the fact that the log-periodic environment is not constant across the saddle width.

14.2 All orders

At order N, Taylor expansion of (295) produces finitely many polynomials in v whose coefficients are differential polynomials in

$$\Psi'(u), \Psi''(u), \ldots, \Psi^{(2N)}(u).$$

After exponentiation, all odd powers of $\lambda^{-1/2}$ integrate to zero by parity. The remaining Gaussian moments are integers, so one obtains integer powers of $1/\lambda$.

Theorem 14.2 (Full logarithmic expansion)**.** *There are smooth* 1*-periodic functions $A_j : \mathbb{R} \to \mathbb{R}$, $j \geq 0$, with $A_0 = 0$ and A_1 given by* (310)*, such that for every $N \geq 1$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}\Big(\lambda(x)^{-N}\Big)} \tag{312}$$

as $x \downarrow 0$.

Proof architecture. Fix N. Expand the analytic local phase through sufficiently high order in $\lambda^{-1/2}$, with a Gaussian-dominated remainder. Exponentiate using complete Bell polynomials, integrate term by term, and discard odd terms. The quantitative tail theorem makes the complement of the central region $O(\lambda^{-N})$. This gives a relative saddle mass expansion

$$1 + \sum_{j=1}^{N-1} \frac{b_j(\lambda)}{\lambda^j} + O(\lambda^{-N}), \tag{313}$$

where every b_j is smooth and periodic. Taking the formal logarithm defines the A_j. If

$$\log\left(1 + \sum_{j\geq 1} b_j z^j\right) = \sum_{j\geq 1} A_j z^j,$$

then the coefficients may be generated recursively by

$$A_n = b_n - \frac{1}{n}\sum_{k=1}^{n-1} k A_k b_{n-k}. \tag{314}$$

All operations preserve smooth periodic dependence on the phase. □

This form of the expansion is preferable to a very long series in nested logarithms. The exact Lambert coordinate absorbs the dominant implicit inversion, while the exact periodic phase prevents phase errors from being magnified into false remainder claims.

14.3 The saddle jets in closed form

The description just given defines the A_j only as the output of a chain: a differential recursion for the periodic quantities that enter the local phase, then exponentiation, then Gaussian contraction, then a formal logarithm. Every step after the first is algebraic. The first is not, and it can be solved.

Write $\mathcal{H}_n = \sum_{i\leq n} 1/i$ for the harmonic numbers, and let $\mathcal{Z}_n$ be the periodic *jets* generated by

$$\mathcal{Z}_0 = \frac{1}{2} + \frac{\Psi'}{L}, \qquad \mathcal{Z}_{n+1} = \left(\frac{\mathrm{D}}{L} - (n+1)\right)\mathcal{Z}_n + \frac{(-1)^{n+1} n!}{L}, \tag{315}$$

where D is differentiation in the phase variable. The operators $\mathrm{D}/L - k$ commute, since $(\mathrm{D}/L - j)(\mathrm{D}/L - k)$ is symmetric in j and k, so the homogeneous part of the solution is the falling-factorial operator $\prod_{k=1}^{n}(\mathrm{D}/L - k)$ applied to $\mathcal{Z}_0$, and each inhomogeneous constant is an eigenvector of the remaining factors. Carrying this out gives a closed form.

Theorem 14.3 (Closed form of the jets)**.** *Let $c(n,m)$ be defined by*

$$\prod_{k=1}^{n}(z-k)=\sum_{m=0}^{n}c(n,m)z^m. \tag{316}$$

Then for every $n\geq 0$,

$$\mathcal{Z}_n(u)=(-1)^n n!\left(\frac{1}{2}+\frac{\mathcal{H}_n}{L}\right)+\sum_{m=0}^{n}c(n,m)\,\frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \tag{317}$$

$$\widetilde{\mathcal{Z}}_n(u)=(-1)^n n!\left(\frac{\mathcal{H}_n}{L}-\frac{1}{2}\right)+\sum_{m=0}^{n}c(n,m)\,\frac{\Psi^{(m+1)}(u)}{L^{m+1}}, \tag{318}$$

where $\widetilde{\mathcal{Z}}_n=\mathcal{Z}_n+(-1)^{n+1}n!$ are the bounded exponent jets. The polynomial in (316) is monic of degree exactly n.

In terms of these jets the exponent polynomials collapse: the mth one is

$$\Xi_m(u,v)=\mathrm{i}^m\left(\frac{\widetilde{\mathcal{Z}}_{m-1}(u)\,v^m}{m!}+\frac{(-1)^{m+1}v^{m+2}}{m+2}\right), \tag{319}$$

with no factorial surviving in the second term. Substituting (318) therefore exhibits every A_j as a finite polynomial in $\Psi',\dots,\Psi^{(2j)}$ with coefficients in $\mathbb{Q}[1/L]$, and hence, through (251), as an explicit Gamma–zeta Fourier object. This is what makes Theorem 14.13 below a statement about a formula rather than about a recursion.

Warning 14.4 (Which Stirling numbers)*.* The weights (316) are signed Stirling numbers of the first kind *shifted by one in each index.* With the standard convention $x(x-1)\cdots(x-n+1)=\sum_k s(n,k)x^k$, multiplying (316) by z gives $\prod_{k=0}^{n}(z-k)$, whence

$$c(n,m)=s(n+1,m+1)=(-1)^{n-m}\mathfrak{s}(n+1,m+1), \tag{320}$$

with $\mathfrak{s}$ the unsigned numbers already used in (331). The shift is not cosmetic. One has $s(n,0)=0$ for every $n\geq 1$, whereas $c(n,0)=\prod_{k=1}^{n}(-k)=(-1)^n n!$, and that constant column is precisely what carries the harmonic-number term in (317). Reading $c(n,m)$ as $s(n,m)$ therefore deletes the entire non-oscillatory part of the jet. Explicitly,

$$c(2,\cdot)=(2,-3,1),\quad s(2,\cdot)=(0,-1,1);\qquad c(3,\cdot)=(-6,11,-6,1),\quad s(3,\cdot)=(0,2,-3,1).$$

Remark 14.5. Theorem 14.3 and (319), together with the monicity and degree statement, are machine-checked in the formal development. The identification (320) is elementary and is verified here; note that it was the one item in this circle of results that neither a compiler nor a numerical check could have caught, since every number in sight is computed from the product (316) and never from the name.

Written in the jets, the first two coefficients are far shorter than in the derivatives of Ψ:

$$A_1=-\frac{\widetilde{\mathcal{Z}}_0^2}{2}-\widetilde{\mathcal{Z}}_0-\frac{\widetilde{\mathcal{Z}}_1}{2}-\frac{1}{12}, \tag{321}$$

$$\begin{aligned}A_2=\frac{1}{24}\Big(&8\widetilde{\mathcal{Z}}_0^3+12\widetilde{\mathcal{Z}}_0^2\widetilde{\mathcal{Z}}_1+12\widetilde{\mathcal{Z}}_0^2+48\widetilde{\mathcal{Z}}_0\widetilde{\mathcal{Z}}_1+12\widetilde{\mathcal{Z}}_0\widetilde{\mathcal{Z}}_2\\&+6\widetilde{\mathcal{Z}}_1^2+24\widetilde{\mathcal{Z}}_1+20\widetilde{\mathcal{Z}}_2+3\widetilde{\mathcal{Z}}_3\Big).\end{aligned} \tag{322}$$

Both are machine-checked. Expanding (321) through (318) reproduces (310) exactly: the terms in Ψ' cancel, and the constants collapse to $1/24+1/(2L)$. That is a useful check in its own right, because (310) predates the jet parametrization and so is an independent anchor for it.

These forms also make Theorem 14.13 below legible: the top jet enters (321) with coefficient $-\frac{1}{2}$ and (322) with coefficient $\frac{3}{24}=\frac{1}{8}$, which are $(-1)^j/(2^j j!)$ at $j=1$ and $j=2$.

14.4 Every coefficient in closed form

With the jets solved, the remaining steps are pure algebra and can be carried out once and for all, replacing the recursion (314) by a formula. Write $(m_1, \dots, m_r) \vDash N$ for a composition of N into r positive parts, as in Section 7.3.

Theorem 14.6 (Explicit saddle mass coefficients). *Let $1 + \sum_{j \ge 1} a_j(u)\lambda^{-j}$ be the normalized saddle mass. Then for $j \ge 1$,*

$$a_j(u) = (-1)^j \sum_{r=1}^{2j} \frac{1}{r!} \sum_{(m_1,\dots,m_r)\vDash 2j} \sum_{S\subseteq\{1,\dots,r\}} (2j + 2|S| - 1)!! \times \prod_{i\in S} \frac{(-1)^{m_i+1}}{m_i + 2} \prod_{i\notin S} \frac{\widetilde{\mathcal{Z}}_{m_i-1}(u)}{m_i!}. \tag{323}$$

The set S records which of the two monomials of (319) is taken from each factor: $i \in S$ selects the universal tail $v^{m_i+2}/(m_i + 2)$ and $i \notin S$ selects the jet term. The total degree is then $2j + 2|S|$, always even, and the double factorial is its Gaussian moment. The prefactor is i^{2j}.

Theorem 14.7 (Explicit logarithmic coefficients). *With $a_0 = 1$, for every $j \ge 1$,*

$$A_j(u) = \sum_{r=1}^{j} \frac{(-1)^{r-1}}{r} \sum_{(q_1,\dots,q_r)\vDash j} a_{q_1}(u) \cdots a_{q_r}(u). \tag{324}$$

Substituting (318) into these two leaves no recursion anywhere: every A_j is a finite expression with rational coefficients in $1/L$ and $\Psi'(u), \dots, \Psi^{(2j)}(u)$. Three structural facts follow immediately. No derivative beyond $\Psi^{(2j)}$ occurs, because the highest jet reached at order $2j$ is $\widetilde{\mathcal{Z}}_{2j-1}$. Every A_j is smooth, one-periodic and bounded, being a polynomial in such functions. And the mean of A_j is its Ψ-free part plus the means of its terms of degree two and higher, which is the content of Warning 14.17.

Remark 14.8. Unlike Theorem 14.3, these two are not machine-checked. They were verified symbolically against the recursions they replace for $j \le 3$, with identically vanishing residuals, and (323) is checked here at $j = 1$: it returns $-\widetilde{\mathcal{Z}}_0^2/2 - \widetilde{\mathcal{Z}}_0 - \widetilde{\mathcal{Z}}_1/2 - 1/12$, which is (321), and $A_1 = a_1$.

For reference, the Ψ-free parts of the first four coefficients — the values obtained by setting $\Psi \equiv 0$, which by Warning 14.17 are close to but not equal to the means — are

j	Ψ-free part of A_j	value
1	$\frac{1}{24} + \frac{1}{2L}$	0.7630141871
2	$\frac{1}{4L^2}$	0.5203422453
3	$-\frac{7}{2880} - \frac{1}{24L} - \frac{1}{4L^2} + \frac{1}{6L^3}$	−0.0824216430
4	$\frac{-L^3 - 72L^2 - 816L + 144}{1152L^4}$	−1.7167954639

The order-two entry is unexpectedly clean: it is exactly $1/(4\log^2 2)$, with no rational term and no term in $1/L$.

14.5 The phase to all orders

The proof of the corrected elementary expansion (344) appeals to the lower-branch Lambert expansion without displaying it, even though the precision of that expansion is what controls the residual. We display it here, in the general variable, and state the all-orders form.

Throughout, $S = -\log x$ and $a = \log L$ as in (342), and we write

$$z = \log S - \log L = m - a, \tag{325}$$

so that z is the second logarithm of the primary scale measured in units of L. Taking logarithms in (274) gives the equation that defines the phase,

$$L\lambda - \log \lambda = S, \qquad \text{equivalently} \qquad L\lambda - \log(L\lambda) = S - a. \tag{326}$$

Theorem 14.9 (Lambert phase expansion)**.** *As $x \downarrow 0$,*

$$L\lambda(x) = S + z + \frac{z}{S} + \frac{z - \frac{1}{2}z^2}{S^2} + \frac{z - \frac{3}{2}z^2 + \frac{1}{3}z^3}{S^3} + \mathcal{O}\left(\frac{(1+|z|)^4}{S^4}\right). \tag{327}$$

More generally, there are unique polynomials p_j, with $\deg p_j = j$ and $p_j(0) = 0$, such that for every $N \geq 1$

$$\boxed{L\lambda(x) = S + z + \sum_{j=1}^{N-1} \frac{p_j(z)}{S^j} + \mathcal{O}\left(\frac{(1+|z|)^N}{S^N}\right).} \tag{328}$$

The first three are

$$p_1(z) = z, \qquad p_2(z) = z - \frac{z^2}{2}, \qquad p_3(z) = z - \frac{3z^2}{2} + \frac{z^3}{3}. \tag{329}$$

Proof. Write $w = L\lambda$. By (326), $w = S + \log w - a$. Substituting $w = S + z + \delta$ and using $\log(S + z + \delta) = \log S + \log(1 + (z+\delta)/S)$ together with $z = \log S - a$ turns the equation into the fixed-point form

$$\delta = \log\left(1 + \frac{z+\delta}{S}\right). \tag{330}$$

Since $\lambda \to \infty$ and $z = o(S)$, one has $\delta \to 0$, and (330) may be solved by successive substitution. Expanding the logarithm,

$$\delta = \frac{z+\delta}{S} - \frac{(z+\delta)^2}{2S^2} + \frac{(z+\delta)^3}{3S^3} - \cdots,$$

and inserting $\delta = p_1(z)S^{-1} + p_2(z)S^{-2} + \cdots$ determines the p_j recursively: comparing coefficients of S^{-1} gives $p_1 = z$; of S^{-2} gives $p_2 = p_1 - z^2/2 = z - z^2/2$; of S^{-3} gives

$$p_3 = p_2 - z^2 + \frac{z^3}{3} = z - \frac{3z^2}{2} + \frac{z^3}{3}.$$

At each stage the coefficient of S^{-j} on the right involves only $p_1, \dots, p_{j-1}$, so the recursion is triangular and the p_j are unique; an induction on j shows $\deg p_j = j$, with leading coefficient $(-1)^{j+1}/j$ and $p_j(0) = 0$. The remainder bound follows by tracking the same recursion one step further, all terms at level N being polynomials of degree N in z. Truncating after $j = 3$ gives (327). □

Remark 14.10 (Closed form of the coefficients). The recursion has the classical solution in unsigned Stirling numbers of the first kind $\mathfrak{s}(j,k)$ familiar from the asymptotic series of Lambert W [11]:

$$p_j(z) = \sum_{m=1}^{j} \frac{(-1)^{m+1}}{m!}\, \mathfrak{s}(j,\, j-m+1)\, z^m. \tag{331}$$

For $j=4$ this gives $p_4(z) = z - 3z^2 + \frac{11}{6}z^3 - \frac{1}{4}z^4$, and for $j=5$, $p_5(z) = z - 5z^2 + \frac{35}{6}z^3 - \frac{25}{12}z^4 + \frac{1}{5}z^5$.

Remark 14.11 (Substituting a truncated phase is a separate step). It is tempting to insert (328) into the all-orders formula (312) and read off a purely elementary logarithmic expansion of $\log F(x)$ to any order. That step needs its own argument, for two reasons that reinforce each other.

First, the main exponent contains $-L\lambda^2/2$, so an error $\delta\lambda$ in the phase propagates to an error of size about $L\lambda\,\delta\lambda$ in the answer; the phase must therefore be known one order beyond the naive count, which is precisely why the fourth term of (327) is required to reach the residual $\mathcal{O}(1/S)$ claimed in (344).

Second, the truncation error in (328) is $\mathcal{O}\big((1+|z|)^N S^{-N}\big)$, not $\mathcal{O}(S^{-N})$; the factor $(1+|z|)^N$ is a power of $\log\log(1/x)$ and is not absorbed by the stated order. When the truncated phase is substituted into Ψ and into the coefficients A_j, these logarithmic factors are inherited and then multiplied by the amplification just described. The periodic function is smooth with bounded derivatives, so it is Lipschitz in the abstract; but the error one is allowed to make in its argument is exactly of the same order as the terms being claimed, so Lipschitz control by the truncation error alone does not close the estimate. A genuine all-orders elementary statement requires composing the two expansions with matched precision and re-collecting the resulting cross terms, and that composition is not assumed silently here. Keeping the exact phase $\lambda(x)$ inside Ψ and inside every A_j, as in (312), avoids the entire issue.

Finally, (327) specializes correctly to the dyadic scale. Put $x = 2^{-t}$, so that $S = Lt$ and, by (325), $z = \log(Lt) - \log L = \log t$. Dividing (327) by L gives

$$\lambda_t = t + \frac{\log t}{L} + \frac{\log t}{L^2 t} + \frac{\log t - \frac{1}{2}\log^2 t}{L^3 t^2} + \frac{\log t - \frac{3}{2}\log^2 t + \frac{1}{3}\log^3 t}{L^4 t^3} + \mathcal{O}\left(\frac{\log^4 t}{t^4}\right), \tag{332}$$

whose first four terms are exactly (348), and whose fourth term is the explicit form of the error term displayed there.

14.6 A numerical check at reciprocal powers

The values $F(2^{-n})$ of (107) are exact rationals, obtained from the half-moment recurrence (97) without any floating-point step, so they give an unusually clean test of (311). For example

$$F(2^{-10}) = \frac{134926369}{1246394851358539387238350848000}, \tag{333}$$

whose logarithm is $-50.5775682823\ldots$, is computed in exact arithmetic and then converted to a high-precision float once, at the end.

Table 1 reports, for $n = 10, 20, 40$: the exact Lambert coordinate λ_n solving (347); the residual $\log F(2^{-n}) - \mathcal{M}(2^{-n})$ against the main term (276); that residual multiplied by λ_n; and the predicted first coefficient $A_1(\lambda_n)$ of (310). All entries were recomputed from the definitions; the periodic function and its derivatives were evaluated from the Gamma–zeta Fourier series (251), and independently from the defining series (246), the two agreeing to fifteen digits.

Three things are visible in the table. First, the unscaled residual does tend to zero, which is (278). Second, the scaled residual does not tend to zero and does not grow: it stays near

n	λ_n	$\log F(2^{-n}) - \mathcal{M}$	$\lambda_n(\log F(2^{-n}) - \mathcal{M})$	$A_1(\lambda_n)$
10	13.78503056	0.05801333439	0.7997155875	0.7629678468
20	24.62186833	0.03182756966	0.7836542295	0.7629268731
40	45.50804986	0.01701388954	0.7742689337	0.7629490545

Table 1: Exact-rational test of the first saddle correction at $x = 2^{-n}$. The third column tends to zero; the fourth, which is the third scaled by the exact Lambert coordinate, tends to the periodic coefficient A_1 of the fifth column rather than to any other constant.

0.78, which is the content of (311), and it converges to the value of the periodic coefficient A_1 evaluated at the exact phase, not to a fitted constant of its own. The differences between the fourth and fifth columns are 0.03675, 0.02073, 0.01132; scaled by λ_n they are 0.507, 0.510, 0.515, confirming the remainder $\mathcal{O}(\lambda^{-2})$ of (311) and showing that the next coefficient A_2 is of size about one half.

Warning 14.12. That last observation is a trap, and it is worth spelling out because the trap is generic to this whole expansion. Extending the same computation along $n = 10, 20, 40, 80, 160, 320$ gives scaled second differences 0.5066, 0.5103, 0.5151, 0.5179, 0.5193, 0.5199, which look like a sequence converging to a constant near 0.520. They are not. The theorem of Section 14 says every coefficient A_j is one-periodic in λ, and A_2 is no exception. Doubling n increases $\log_2 \lambda$ by roughly one, so $\lambda_n \bmod 1$ returns to nearly the same place; a geometric sample of n therefore holds the phase almost fixed and displays only the residual drift in λ. Sampling n densely instead separates the two effects. Over $300 \le n \le 400$, where λ moves by only a third, the quantity $\lambda^2(\log F - \mathcal{M} - A_1(\lambda)/\lambda)$ sweeps from 0.52105 down to 0.51810 and back up, tracking $\lambda \bmod 1$, a spread of 2.95×10^{-3}. Held at one phase instead and varied in magnitude, it moves by only 3.7×10^{-4} from $n = 173$ to $n = 352$, consistent with an $\mathcal{O}(\lambda^{-1})$ correction. So A_2 is a genuinely nonconstant periodic function.

Neither of those sampled numbers should be mistaken for a property of A_2 itself, and the distinction is worth drawing because it is the same trap one level down. The sample covers $0.268 \le \lambda \bmod 1 \le 0.675$, which is 41% of a period and misses the maximum; and it carries an A_3/λ displacement of about -3×10^{-4}. Computed from the coefficient rather than from the sample, A_2 has

$$\max A_2 = 0.522282117, \quad \min A_2 = 0.518402361, \quad \langle A_2 \rangle = 0.5203422413, \tag{334}$$

attained at $\lambda \bmod 1 = 0.101128$ and 0.601131, so the true peak-to-peak swing is 3.880×10^{-3} rather than the 2.95×10^{-3} visible in the window. The measured minimum at 0.602 does agree with the true one.

The amplitude is not an accident, and its growth with j is governed by an exact law, which is the subject of Section 14.7 below. Any numerical extraction of a later coefficient must fix the phase first.

14.7 The highest derivative in each coefficient

The coefficients A_j of (312) are polynomials in L^{-1} and in the derivatives of Ψ. Two facts about them are worth isolating, because together they explain both the size of the oscillation seen above and the ultimate divergence of the series.

Theorem 14.13 (Leading-derivative law)**.** *For every $j \ge 1$ the coefficient A_j involves no derivative of Ψ beyond the $2j$th, that derivative occurs linearly, and its coefficient is exactly*

$$\frac{(-1)^j}{2^j\, j!\, L^{2j}}\, \Psi^{(2j)}(u). \tag{335}$$

Proof architecture. The $(2j)$th derivative reaches A_j through exactly one route. It enters the exponent expansion only in the top exponent polynomial, and there linearly, with coefficient $\mathrm{i}^{2j}/(2j)! = (-1)^j/(2j)!$. The Gaussian contraction replaces v^{2j} by the moment $(2j-1)!!$, and the passage from the saddle mass to its logarithm leaves a linear term untouched. The stated constant is then the identity

$$\frac{(2j-1)!!}{(2j)!} = \frac{1}{2^j\, j!},$$

which is $(2j)! = 2^j j!\,(2j-1)!!$. □

The law is visible in the coefficients that have been computed. For $j=1$ it gives $-\Psi''/(2L^2)$, which is the Ψ'' term of (310); for $j=2$ it gives $+\Psi^{(4)}/(8L^4)$; for $j=3$, $-\Psi^{(6)}/(48L^6)$; for $j=4$, $+\Psi^{(8)}/(384L^8)$.

Remark 14.14 (What is formalized here, and what is not). The substance of Theorem 14.13 is machine-checked for general j, and not merely verified in the four cases just listed. What the formal development proves is the statement about the Gaussian *mass* coefficient: for an arbitrary jet sequence and every j, the order-$(j+1)$ mass coefficient depends on the top jet $\widetilde{\mathcal{Z}}_{2j+1}$ exactly linearly, with coefficient $(-1)^{j+1}/(2^{j+1}(j+1)!)$. The engine of that proof is the observation that the exponential recurrence is affine in its top exponent coefficient, which is what makes the statement provable at once for all orders rather than checkable one order at a time.

The passage from the mass coefficient to the logarithmic coefficient A_j is not formalized. It is immediate — taking the formal logarithm subtracts from the order-j mass coefficient only products of strictly lower mass coefficients, and no such product reaches $\widetilde{\mathcal{Z}}_{2j-1}$, so the linear term survives unchanged — but it is an elementary observation and the article does not claim otherwise.

Corollary 14.15 (Growth of the oscillation)**.** *Differentiating multiplies the first Fourier mode by* $(2\pi)^{2j}$*, so that mode of* $\Psi^{(2j)}$ *has amplitude* $2|\widehat{\Psi}(1)|(2\pi)^{2j}$*, and the leading term* (335) *therefore contributes an oscillation of amplitude*

$$2|\widehat{\Psi}(1)|\,\frac{1}{j!}\left(\frac{2\pi^2}{L^2}\right)^j, \qquad \frac{2\pi^2}{L^2} = 41.0845769104. \tag{336}$$

Numerically this predicts 8.73×10^{-5}, 1.79×10^{-3}, 2.46×10^{-2} and 2.52×10^{-1} for $j=1,2,3,4$.

Warning 14.16. Equation (336) is a growth rate, not the amplitude. It uses only the top derivative; the lower ones contribute in phase and inflate the true swing by a factor that itself grows. Measuring amplitudes — half the peak-to-peak swing, on the same convention as (336) itself — the true values are 8.73×10^{-5}, 1.94×10^{-3}, about 3×10^{-2} and about 0.41 for $j=1,2,3,4$, against the estimates 8.73×10^{-5}, 1.79×10^{-3}, 2.46×10^{-2}, 2.52×10^{-1}. The ratio of truth to estimate drifts 1.00, 1.08, about 1.2, about 1.6. Use (336) to see how fast the oscillation grows, never to size a particular coefficient.

Warning 14.17 (Products of derivatives are not negligible either). It is tempting to argue that since Ψ oscillates by only about 2×10^{-6}, products of its derivatives are of order 4×10^{-12} and may be dropped. That reasoning contradicts the one above and is false. A mean of a product is

$$\langle\Psi^{(a)}\Psi^{(b)}\rangle = \sum_{k\neq0}(2\pi\mathrm{i}k)^a(-2\pi\mathrm{i}k)^b\big|\widehat{\Psi}(k)\big|^2, \tag{337}$$

which carries the same amplification $(2\pi k)^{a+b}$ as everything else, and it vanishes unless $a+b$ is even. Numerically $\langle(\Psi')^2\rangle = 8.917\times10^{-11}$ but $\langle(\Psi''')^2\rangle = 1.390\times10^{-7}$, four decades larger.

The practical consequence is that the Ψ-free part of A_j is *not* its mean. The means themselves are available in closed form. Every mean of a derivative vanishes, and integration by parts over a period kills or pairs the rest:

$$\langle \Psi'\Psi'' \rangle = 0, \qquad \langle (\Psi')^2\Psi'' \rangle = 0, \qquad \langle \Psi'\Psi''' \rangle = -\langle (\Psi'')^2 \rangle, \tag{338}$$

the first two because each integrand is the derivative of a periodic function. Applying these to (310) and to the corresponding expression one order further leaves

$$\langle A_1 \rangle = \frac{1}{24} + \frac{1}{2L} - \frac{\langle (\Psi')^2 \rangle}{2L^2}, \tag{339}$$

$$\langle A_2 \rangle = \frac{1}{4L^2} - \frac{\langle (\Psi')^2 \rangle}{2L^3} - \frac{\langle (\Psi'')^2 \rangle}{4L^4} - \frac{\langle (\Psi')^3 \rangle}{6L^3}. \tag{340}$$

Both are exact. With $\langle (\Psi')^2 \rangle = 8.917038 \times 10^{-11}$, $\langle (\Psi'')^2 \rangle = 3.520305 \times 10^{-9}$ and $\langle (\Psi')^3 \rangle = 1.115006 \times 10^{-21}$ they give $\langle A_1 \rangle = 0.7630141870184$ and $\langle A_2 \rangle = 0.5203422413049$.

The cubic term in (340) is -5.58×10^{-22}, twelve orders below the smallest term beside it, and it is the only one with no partner to cancel against. It is kept because (340) is an identity and not an estimate; dropping it would change the equals sign into an approximation, which is a different claim.

The gap between the mean and the Ψ-free part grows with j like everything else here: it is -9.3×10^{-11}, -3.9×10^{-9} and -1.14×10^{-7} at $j = 1, 2, 3$, so it is invisible at $j = 1$ and reaches the seventh digit by $j = 3$. The Ψ-free part remains the right practical value for the mean, but the reason is a ratio and not an absolute bound: at each order it sits four to five decades below the oscillation of A_j itself.

The same two facts explain why the expansion (312) is asymptotic and not convergent, and the mechanism is worth naming because it is not the usual one. Stirling's estimate for Γ on vertical lines gives

$$|\widehat{\Psi}(k)| \asymp \mathrm{e}^{-\pi^2 k/L}, \qquad \frac{\pi^2}{L} = 14.2385595, \tag{341}$$

so the Fourier modes of Ψ die extremely fast; already $|\widehat{\Psi}(2)|/|\widehat{\Psi}(1)| \approx 6.3 \times 10^{-7}$. But mode k enters A_j amplified by $(2\pi k)^{2j}/(2^j j! L^{2j})$, so its contribution behaves like $\exp(2j \log k - \pi^2 k/L)$, which is maximized at $k \approx 0.14j$. For small j the first mode dominates and the coefficients are minute; past roughly $j = 8$ the higher modes take over and $\sup |A_j|$ grows faster than geometrically. The divergence of the expansion is therefore caused by the *high-frequency* content of the periodic correction, not by the factorial growth one expects from a saddle-point expansion. This paragraph is a size estimate, not a theorem.

Third, and for honesty about what the table can and cannot show at these sizes: the periodic part of A_1 is minute. The constant part $1/24 + 1/(2L) = 0.7630141871$ of (310) is modulated by $-(\Psi'' + (\Psi')^2)/(2L^2)$, whose amplitude is about 8.7×10^{-5} (the second derivative multiplies the first Fourier mode by $(2\pi)^2$), while $\Psi(\lambda_n)$ itself is -1.13×10^{-6}, -2.13×10^{-6} and -1.59×10^{-6} at $n = 10, 20, 40$. These are far below the $\mathcal{O}(\lambda^{-2})$ remainder at $n \le 40$. The table therefore confirms the size and value of the periodic coefficient A_1; it does not, by itself, isolate the oscillation. The proof that the oscillation cannot be dropped is the subsequential-limit argument of (353), not a table. The forward tail plays no role at all here: at $n = 10$ the saddle radius is $r = \lambda_{10} 2^{10} \approx 1.41 \times 10^4$, so by (291) the term $E(r)$ is smaller than 10^{-6100}.

15 Elementary log–log forms and the missing periodic term

Lambert W gives the cleanest formula, but an expansion in elementary logarithms is useful for comparison with published displays. It must, however, retain Ψ at the exact Lambert phase.

15.1 The literal small-x expansion

Put

$$S = -\log x, \qquad m = \log S, \qquad a = \log L. \tag{342}$$

Thus $S \to \infty$ as $x \downarrow 0$. Define the nonperiodic elementary expression

$$\begin{aligned}\mathcal{E}_0(x) = & -\frac{S^2}{2L} - \frac{Sm}{L} + \left(\frac{1}{2} + \frac{1+a}{L}\right) S \\ & -\frac{m^2}{2L} + \frac{am}{L} + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6a^2}{12L} - \frac{7L}{12} - \frac{1}{2}\log\pi \\ & -\frac{m^2}{2LS} + \frac{am}{LS}.\end{aligned} \tag{343}$$

Theorem 15.1 (Corrected elementary expansion)**.** *As $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{E}_0(x) + \Psi(\lambda(x)) + \mathcal{O}\left(\frac{1}{-\log x}\right).} \tag{344}$$

Proof architecture. The standard lower-branch Lambert expansion solves $\lambda L - \log \lambda = S$ by successive substitution. Substituting that expansion into (276), and retaining enough phase terms to control the amplified quadratic contribution, gives (343). A coarse two-term approximation to λ is insufficient: because the main exponent contains $-L\lambda^2/2$, an error $\delta\lambda$ contributes about $-L\lambda\delta\lambda$, so the phase must be known one order more accurately than a naive count suggests. The fourth Lambert phase term in the formal proof is exactly what reduces the residual to $O(1/S)$. ☐

In the original variable $\ell = \log x < 0$, the same expression is

$$\begin{aligned}\mathcal{E}_0(x) = & -\frac{\ell^2}{2L} + \frac{\ell\log(-\ell)}{L} - \left(\frac{1}{2} + \frac{1+\log L}{L}\right)\ell \\ & -\frac{\log^2(-\ell)}{2L} + \frac{\log L\,\log(-\ell)}{L} \\ & +\frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6\log^2 L}{12L} - \frac{7L}{12} - \frac{1}{2}\log\pi \\ & +\frac{\log^2(-\ell)}{2L\ell} - \frac{\log L\,\log(-\ell)}{L\ell}.\end{aligned} \tag{345}$$

This is the literal elementary display formalized in the repository, with the periodic term added separately.

15.2 The dyadic logarithmic scale

Set $x = 2^{-t}$ and let

$$\lambda_t = -\frac{1}{L}W_{-1}(-L2^{-t}). \tag{346}$$

Then λ_t is the unique large solution of

$$\lambda_t - \log_2 \lambda_t = t. \tag{347}$$

Its first terms are

$$\lambda_t = t + \frac{\log t}{L} + \frac{\log t}{L^2 t} + \frac{\log t - \frac{1}{2}\log^2 t}{L^3 t^2} + \mathcal{O}\left(\frac{\log^3 t}{t^3}\right). \tag{348}$$

Define

$$\mathcal{D}_0(t) = -\frac{L}{2}t^2 - t\log t + \left(1 + \frac{L}{2}\right)t - \frac{\log^2 t}{2L} + C_F - \frac{\log^2 t}{2L^2 t}. \tag{349}$$

Theorem 15.2 (Corrected dyadic-scale expansion)**.** *As $t \to \infty$,*

$$\boxed{\log F(2^{-t}) = \mathcal{D}_0(t) + \Psi(\lambda_t) + \mathcal{O}(t^{-1}).} \tag{350}$$

The literal substitution of $x = 2^{-t}$ into (343) differs from $\mathcal{D}_0(t)$ by the explicit term

$$\frac{\log^2 L}{2L^2 t}, \tag{351}$$

which is itself $O(1/t)$. Thus both nonperiodic displays are valid at the stated precision, but they are not algebraically identical.

15.3 Why the uncorrected formula is false at the claimed scale

Let $\mathcal{M}_0(x) = \mathcal{M}(x) - \Psi(\lambda(x))$ be the nonperiodic Lambert main. From (278),

$$\log F(x) - \mathcal{M}_0(x) = \Psi(\lambda(x)) + O(1/\lambda(x)). \tag{352}$$

Since Ψ is nonconstant, the right side does not tend to zero.

Proposition 15.3 (Failure without the periodic correction)**.** *The relation*

$$\log F(x) = \mathcal{M}_0(x) + O(1/(-\log x)) \tag{353}$$

is false. In fact $F(x)/\exp(\mathcal{M}_0(x))$ has distinct subsequential limits.

Proof. Choose $u, v \in [0, 1)$ with $\Psi(u) \neq \Psi(v)$. Set

$$\lambda_n = n + u, \qquad x_n = \lambda_n 2^{-\lambda_n}.$$

Then $x_n \downarrow 0$ and the exact Lambert phase of x_n is λ_n. Hence

$$\log F(x_n) - \mathcal{M}_0(x_n) \longrightarrow \Psi(u).$$

The analogous sequence with phase $n + v$ tends to $\Psi(v)$. Since $1/(-\log x) \to 0$, the bound (353) is impossible. Exponentiating gives distinct ratio limits $\mathrm{e}^{\Psi(u)}$ and $\mathrm{e}^{\Psi(v)}$. $\square$

Numerically the discrepancy is minute because the first Fourier mode has size about 10^{-6}, but asymptotic equivalence is a limit statement, not a graphical tolerance. A nonzero bounded oscillation cannot be hidden inside an error term that tends to zero.

15.4 The periodic correction is unique

The previous subsection shows that the periodic term cannot be deleted. A little more is true and is worth recording, because it removes the last escape route: the periodic term cannot be replaced either. There is exactly one 1-periodic function that makes the sharp formula correct, and it is Ψ.

Theorem 15.4 (Uniqueness of the periodic correction)**.** *Let $\mathcal{M}_0(x) = \mathcal{M}(x) - \Psi(\lambda(x))$ be the nonperiodic Lambert main term, and let $\widetilde{\Psi} : \mathbb{R} \to \mathbb{R}$ be continuous, 1-periodic and of mean zero. Suppose that*

$$\log F(x) = \mathcal{M}_0(x) + \widetilde{\Psi}(\lambda(x)) + o(1) \qquad (x \downarrow 0). \tag{354}$$

Then $\widetilde{\Psi} = \Psi$ identically.

Proof. By (278) one has $\log F(x) = \mathcal{M}_0(x) + \Psi(\lambda(x)) + \mathcal{O}(1/\lambda(x))$, and $\lambda(x) \to \infty$ as $x \downarrow 0$. Subtracting (354) gives

$$(\Psi - \widetilde{\Psi})(\lambda(x)) \longrightarrow 0 \qquad (x \downarrow 0). \tag{355}$$

Fix $t \in \mathbb{R}$ and, for integers n with $n + t > 1/L$, set

$$x_n = (n+t)\, 2^{-(n+t)}. \tag{356}$$

Then $x_n \downarrow 0$, and by (274) the number $n + t$ is a solution of $\lambda 2^{-\lambda} = x_n$ on the branch where $\lambda > 1/L$; since that solution is unique and equals $\lambda(x_n)$, the sequence has exact Lambert coordinate

$$\lambda(x_n) = n + t.$$

This is the point of the construction: the sequence is chosen so that the Lambert coordinate, not the naive scale $\log_2(1/x_n)$, lands on a prescribed residue. By 1-periodicity,

$$(\Psi - \widetilde{\Psi})(\lambda(x_n)) = (\Psi - \widetilde{\Psi})(n+t) = (\Psi - \widetilde{\Psi})(t)$$

for every n, so the left-hand side of (355) is a constant sequence along (x_n). A constant sequence that tends to zero is zero, whence $\Psi(t) = \widetilde{\Psi}(t)$. As $t \in \mathbb{R}$ was arbitrary, the two functions agree everywhere. □

Remark 15.5. The proof uses only 1-periodicity: continuity is never invoked, since the argument produces equality at each real t separately rather than on a dense set. Uniqueness therefore holds in the class of all 1-periodic functions. The mean-zero normalization is likewise not used here, because (354) fixes the nonperiodic main term, including the constant C_F; its role is to remove the trivial ambiguity that arises if one allows the additive constant in $\mathcal{M}_0$ to float, in which case the pair (C_F, Ψ) is determined only up to moving a constant between them, and centering pins it. In particular C_F is itself uniquely determined by (278).

This matters because it changes the logical status of the correction. One might hope that Ψ is an artifact of the Mellin route, and that some other bounded oscillation, or some cleverer nonperiodic reshuffling, would make the elementary formula true. Theorem 15.4 closes that possibility: the order-one periodic term is pinned by the asymptotic statement itself, independently of how it was derived. Any correct sharp formula with this main term must contain exactly the Gamma–zeta function Ψ of (251), evaluated at exactly the Lambert phase. Together with the nonvanishing of every Fourier coefficient, this makes the fluctuation an intrinsic feature of the Fabius law at the endpoint rather than a feature of one proof.

15.5 Symmetry at the right endpoint

The symmetry $F(1-x) = 1 - F(x)$ immediately transfers the left-endpoint expansion to the right:

$$1 - F(1-x) = F(x). \tag{357}$$

Thus the same Lambert phase, periodic fluctuation, and all-orders coefficients describe the survival probability near 1.

16 Consequences of the endpoint expansion

The full sharp formula immediately settles several qualitative questions that are otherwise awkward to read from the differential equation.

Theorem 16.1 (Quadratic logarithmic law)**.** *As $x \downarrow 0$,*

$$\frac{\log F(x)}{(\log x)^2} \longrightarrow -\frac{1}{2\log 2}. \tag{358}$$

Equivalently, on the dyadic scale,

$$\log F(2^{-t}) \sim -\frac{\log 2}{2} t^2. \tag{359}$$

Proof. In (276), $\lambda \sim S/L$ with $S = -\log x$. The quadratic term $-L\lambda^2/2$ dominates the linear term, the logarithmic term, and the bounded periodic correction. Substitution gives (358). □

Corollary 16.2 (Between powers and pure endpoint exponentials)**.** *For every $N > 0$ and every $c > 0$,*

$$F(x) = o(x^N), \tag{360}$$

and

$$\mathrm{e}^{-c/x} = o(F(x)) \tag{361}$$

as $x \downarrow 0$.

Proof. The logarithm of x^N is $-NS$, whereas $\log F(x) \sim -S^2/(2L)$, so $F(x)/x^N \to 0$. On the other hand, $\log \mathrm{e}^{-c/x} = -c\mathrm{e}^S$, which is much more negative than $-S^2/(2L)$; therefore $\mathrm{e}^{-c/x}/F(x) \to 0$. □

This places the endpoint decay in an intermediate regime. The function is flatter than any finite algebraic order, as every compactly supported smooth function must be at its support boundary, but it is vastly larger than the familiar model $\mathrm{e}^{-1/x}$.

Corollary 16.3 (Asymptotics of reciprocal-power values)**.** *For integers $n \to \infty$,*

$$\begin{aligned}\log F(2^{-n}) = &-\frac{L}{2}n^2 - n\log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ &-\frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + O(n^{-1}),\end{aligned} \tag{362}$$

where λ_n is defined by (346). Combining this with (107) gives an equally sharp expansion for the half moments d_n.

The formula shows why a fixed multiplicative constant cannot complete a recurrence-based ansatz for $F(2^{-n})$: the fractional parts of λ_n drift, and the factor $\exp \Psi(\lambda_n)$ is nonconstant.

The corollary above was left at the level of a remark; the resulting expansion of the half moments is worth displaying, because the cancellation is exact and total.

Corollary 16.4 (Sharp asymptotic for the half moments)**.** *Let λ_n be the solution of $\lambda_n - (\log \lambda_n)/L = n$ given by (346). Then, as $n \to \infty$,*

$$\log d_n = \frac{1}{2}\log(2\pi n) + C_F + \Psi(\lambda_n) - \frac{(\log n)^2}{2L} - \frac{(\log n)^2}{2L^2 n} + \mathcal{O}\left(\frac{1}{n}\right), \tag{363}$$

and in particular

$$\boxed{d_n \sim \sqrt{2\pi n}\, \exp\left(C_F + \Psi(\lambda_n) - \frac{(\log n)^2}{2L}\right)}. \tag{364}$$

Proof. By the bridge (107),

$$\log d_n = \log n! + \binom{n}{2} L + \log F(2^{-n}). \tag{365}$$

Insert Stirling's formula $\log n! = n\log n - n + \frac{1}{2}\log(2\pi n) + \mathcal{O}(n^{-1})$, the identity $\binom{n}{2}L = \frac{L}{2}n^2 - \frac{L}{2}n$, and the expansion (362). The three leading blocks cancel identically:

$$\begin{aligned} n^2 \text{ terms:} \quad & \frac{L}{2}n^2 - \frac{L}{2}n^2 = 0, \\ n\log n \text{ terms:} \quad & n\log n - n\log n = 0, \\ n \text{ terms:} \quad & -n - \frac{L}{2}n + \left(1 + \frac{L}{2}\right)n = 0. \end{aligned}$$

What survives is

$$\log d_n = \frac{1}{2}\log(2\pi n) + C_F + \Psi(\lambda_n) - \frac{(\log n)^2}{2L} - \frac{(\log n)^2}{2L^2 n} + \mathcal{O}(n^{-1}),$$

which is (363). The constant is exactly C_F of (263); no separate constant is generated by the cancellation. Since $(\log n)^2/n \to 0$, exponentiating and absorbing the last two terms into a factor tending to 1 gives (364). □

The three cancellations are the reason the half moments look deceptively simple: the superexponentially small factor $F(2^{-n})$ and the superexponentially large factor $n!\,2^{\binom{n}{2}}$ agree through the quadratic, the $n\log n$, and the linear order, and their quotient retains only a Gaussian-type prefactor $\sqrt{2\pi n}$, the endpoint constant, the log-square drift, and the periodic fluctuation. Note that Ψ survives the cancellation untouched. Consequently the normalized sequence $d_n \exp((\log n)^2/(2L))/\sqrt{2\pi n}$ does not converge. Indeed (347) gives $\lambda_n - n = \log_2 \lambda_n$, which tends to infinity with increments tending to zero, so the fractional parts of λ_n are dense in $[0, 1)$; since Ψ is continuous and nonconstant, the set of limit points of the normalized sequence is the full interval

$$\left[\mathrm{e}^{C_F + \min\Psi},\ \mathrm{e}^{C_F + \max\Psi}\right].$$

Any conjecture that this sequence tends to a limit is false, even though its numerical variation is only a few parts in a million.

The comparison (361) is far from sharp. The true decay is governed by $(\log x)^2$, and every negative power of x in the exponent beats that, however small the power.

Proposition 16.5 (Strengthened decay comparison)**.** *For every $c > 0$ and every $\alpha > 0$,*

$$\exp\left(-\frac{c}{x^\alpha}\right) = o(F(x)) \qquad (x \downarrow 0). \tag{366}$$

Proof. Put $S = -\log x \to \infty$. By (358), $\log F(x) = -(1 + o(1))S^2/(2L)$, while

$$\log \exp\left(-\frac{c}{x^\alpha}\right) = -c\,\mathrm{e}^{\alpha S}.$$

Hence

$$\log \frac{\exp(-cx^{-\alpha})}{F(x)} = -c\,\mathrm{e}^{\alpha S} + (1 + o(1))\frac{S^2}{2L} = -c\,\mathrm{e}^{\alpha S}\left(1 - (1 + o(1))\frac{S^2}{2cL\mathrm{e}^{\alpha S}}\right).$$

For every fixed $\alpha > 0$ one has $S^2\mathrm{e}^{-\alpha S} \to 0$, so the bracket tends to 1 and the whole expression tends to $-\infty$. Exponentiating gives (366). □

Taking $\alpha = 1$ recovers (361). The gain is that no exponent is too small: even $\exp(-x^{-1/100})$, which decays extremely slowly by the standards of endpoint models, is still negligible compared with F. Combined with (360), the Fabius endpoint is squeezed between the two scales from both sides, and the squeeze is now tight enough to be decisive for the candidate approximations discussed in section 20. In particular, the quotient of exponentials examined in the subsection on a visually good quotient with the wrong boundary scale has a numerator of the form $\exp(-|x|^{-\alpha})$ with $\alpha = \frac{3}{4}\sqrt{2\pi}$; by Proposition 16.5 that factor alone already forces the candidate to be $o(F)$ at the endpoint, without any appeal to the sharper bound $\exp(-1/(2y))$ used there. The failure is therefore not an artifact of the particular constant in that bound: no expression whose logarithm decays like a negative power of the distance to the endpoint can match a function whose logarithm decays like the square of the logarithm of that distance.

17 The Fourier–Legendre expansion of up

Let P_n be the Legendre polynomial normalized by $P_n(1) = 1$, and define

$$u_n = \frac{4n+1}{2} \int_{-1}^{1} \mathrm{up}(x) P_{2n}(x) \, \mathrm{d}x. \tag{367}$$

Odd Legendre coefficients vanish by `rvachevFullLegendreCoefficient_odd_eq_zero`. The even coefficients have the exact finite formula

$$\begin{aligned} u_n = 4^{-n}(4n+1) \sum_{k=0}^{n} (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} \\ \times (2k)! \, 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \end{aligned} \tag{368}$$

This is `rvachevLegendreCoefficient_eq_fabius_sum` in `FabiusLegendreCoefficients.lean`.

The even series

$$\mathrm{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x) \qquad (-1 \le x \le 1) \tag{369}$$

converges absolutely and uniformly in the formal development. The exact declarations are `summable_abs_rvachevLegendreCoefficient`, `hasSum_rvachevLegendreSeries`, and `hasSum_rvachevLegendreSeries_uniform` in `FabiusLegendreSeries.lean`. Its N-th polynomial partial sum minimizes squared $L^2([-1,1])$ error among all polynomials of degree at most $2N+1$, and every distinct competitor has strictly larger error. These are `rvachevLegendrePartialSum_least_squares` and `rvachevLegendrePartialSum_strict_least_squares` in `FabiusLegendreLeastSquares.lean`. The signed global extension has the corresponding locally finite translated Legendre representation by `globalFabius_eq_tsum_translatedLegendre_formula`.

18 Additional proved summation identities

On the support interval $-1 \le t \le 1$, Poisson summation yields the exact cosine series

$$\mathrm{up}(t) = \frac{1}{2} + \sum_{k=0}^{\infty} \widehat{\mathrm{up}}\left(\frac{2k+1}{2}\right) \cos((2k+1)\pi t). \tag{370}$$

This is `rvachev_cosine_series`. The two-term support specialization

$$\mathrm{up}(t) + \mathrm{up}(t-1) = 1 \qquad (0 \le t \le 1) \tag{371}$$

is `rvachev_add_shift_eq_one`. More general scaled Poisson support identities are `rvachev_poisson_support_specialization_unscaled_of_one_half_le` and `rvachev_poisson_support_specialization_of_one_half_le` in `PoissonSummation.lean`. These supplement the partition formulas (70)–(71); no unformalized derivative partition, probabilistic convolution interpretation, or moment statistic is inferred.

19 Effective computability

The centered spline construction gives a primitive-recursive rational approximation procedure. In the exact coding used by Lean, for every dyadic input code c and precision p,

$$|F(c.\operatorname{value}(p+3)) - \operatorname{Approx}(c,p).\operatorname{value}(p)| \le 5\,2^{-(p+3)}. \tag{372}$$

This is `fabiusSplineApproxPR_error`; the procedure itself is proved primitive recursive by `fabiusSplineApproxPR_primrec`. Together with the sharp Lipschitz bound it proves that every bounded Fabius solution is sequentially computable and effectively uniformly continuous, hence a computable real function in the formal Grzegorczyk-style definition. The declarations are `fabiusReal_sequentiallyComputable`, `fabiusReal_effectivelyUniformContinuous`, and `fabiusReal_isComputableRealFunction` in `FabiusComputableSpline.lean` and `FabiusComputability.lean`.

The same finite spline code also gives a signed-global algorithm. Instead of clamping the positive numerator to the unit grid, it retains the unrestricted centered spline and rounds the signed rational numerator pair. Negative dyadic inputs collapse to the zero grid point, which is exact because $\mathcal{F}(x) = 0$ for $x \le 0$. The resulting primitive-recursive procedure satisfies

$$\Big|\mathcal{F}(c.\operatorname{value}(p+3)) - \operatorname{Approx}_{\mathrm{glob}}(c,p).\operatorname{value}(p)\Big| \le 5\,2^{-(p+3)}. \tag{373}$$

These are `extendedFabiusSplineApproxPR_primrec` and `extendedFabiusSplineApproxPR_error`. Hence, for every F satisfying `IsFabiusF`, its signed extension `extendedFabiusF` is a computable real function, by `extendedFabius_isComputableRealFunction`; the canonical specialization is `globalFabius_isComputableRealFunction`. This is a primitive-recursive correctness statement, not a practical running-time estimate: the unrestricted positive grid numerator sets the length of the finite fold.

The quantitative analytic estimate is stable under moving the evaluation point:

$$|S_p(x) - \mathcal{F}(y)| \le 2^{-p} + 2|x-y|. \tag{374}$$

This is `abs_fabiusUniformSpline_sub_extendedFabius_le_add`. In particular, if $x_p \to x$, then $S_p(x_p) \to \mathcal{F}(x)$, as packaged by `fabiusUniformSpline_tendsto_extendedFabius_of_tendsto`.

20 Formalization boundary and audit conventions

This exposition follows a fail-closed rule. A mathematical assertion remains in the main document only when an exact proved declaration can be identified in the Lean development. A nearby theorem, an executable computation, a numerical check, or an argument that appears straightforward is not treated as a substitute. When the formal theorem has stricter hypotheses, narrower domains, or different endpoint conventions, the prose uses those exact conditions.

That rule explains several choices made above:

- The finite probability approximation uses atomic convolution measures and the exact weak-convergence API, not an unformalized continuous box-convolution density.

- Step cells use the formal half-endpoint indicator, including its point value at a shared boundary.
- The Fourier chapter states integer zeros but neither a complete zero classification nor multiplicities.
- Flatness uses natural powers, and the analytic-locus claims use precisely the domains proved in `NowhereAnalytic.lean`.
- The q-calculus is restricted to the half-base identities actually present in Lean.
- Endpoint asymptotics retain exact phases and the proved remainder scopes; numerical fits and unproved coefficient formulas are absent.

The source drafts have been discharged into two destinations. Proved material was rewritten into the theorem-backed narrative here. The remaining research material is preserved under

`Analysis/FabiusFunction/docs/non-formalized-research-frontiers/`.

In particular, the inverse/saddle, repeated-integration, small-argument, generic q-calculus, and other deferred claims remain available there with an explicit warning that they are not part of the formal corpus.

21 Conclusion

The formal development links three global presentations without confusing their domains. The bounded F supplies the probability and arithmetic viewpoints; the compactly supported up supplies refinement, Fourier, spline, and Legendre viewpoints; and the signed $\mathcal{F}$ supplies the global dilation and binary-translation formulas. Exact bridges connect them to finite atomic laws, weak convergence, centered splines, moments, reciprocal-power values, dyadic algorithms, half-base q-binomial identities, prefix sums, endpoint Laplace analysis, and computable approximation.

The most important methodological conclusion is equally concrete: the primary document is now a map of proved mathematics. Research claims can move back from the frontier only after a matching Lean theorem exists, with the same hypotheses, domain, normalization, endpoint convention, and conclusion.

A Lean module map

Every retained result above points to one or more exact declaration names. The principal source groups are:

Topic	Modules
Definitions, existence, uniqueness	`Basic.lean`, `Existence.lean`, `OriginalCharacterization.lean`, `OriginalUniqueness.lean`
Shape and regularity	`Differential.lean`, `Monotonicity.lean`, `Convexity.lean`, `Regularity.lean`
Probability and finite approximants	`ProbabilityRepresentation.lean`, `EarlyApproximants.lean`, `EarlyMeasureBridge.lean`, `WeakConvergence.lean`, `StepMeasureBridge.lean`, `StepApproximationLimit.lean`, `FabiusUniformSpline.lean`, `FabiusComputability.lean`

Topic	Modules
Fourier analysis	`FourierProduct.lean`, `FourierAnalytic.lean`, `OriginalPaperSupplement.lean`, `PoissonSummation.lean`
Global derivatives and flatness	`GlobalExtension.lean`, `GlobalBounds.lean`, `BoundedDerivatives.lean`, `EffectiveFlatness.lean`, `SharpFlatness.lean`, `FabiusFlatness.lean`
Analyticity, inverse, elementary obstruction	`NowhereAnalytic.lean`, `FabiusInverse.lean`, `ElementaryFunction.lean`, `NotElementary.lean`, `InverseBranch.lean`, `InverseNotElementary.lean`
Moments and exact arithmetic	`Arithmetic.lean`, `AnalyticMoments.lean`, `MomentPowerSeries.lean`, `BernoulliRecurrences.lean`, `ExactInversePower.lean`, `Parity.lean`, `TwoAdic.lean`
Dyadic and q-binomial formulas	`DyadicClosedForm.lean`, `GlobalDyadic.lean`, `DyadicCorrectness.lean`, `FabiusDyadicQBinomialScalar.lean`, `FabiusQBinomialFormula.lean`, `FabiusBinaryReductionSeries.lean`, `FabiusGlobalQBinomialSeries.lean`, `FabiusDiscreteLimitIntegration.lean`
Thue–Morse prefixes	`ThueMorsePrefix.lean`, `ThueMorseGenerating.lean`, `ThueMorseApproximation.lean`
Endpoint asymptotics	`NegativeLaplace.lean`, `PeriodicCorrection.lean`, `PeriodicFourier.lean`, `MellinBose.lean`, `MellinFinitePart.lean`, `BoseFinitePartIntegral.lean`, `FabiusLambertPhase.lean`, `FabiusLambertSaddle.lean`, `FabiusDyadicSharpAsymptotic.lean`, `FabiusSecondSaddleCorrection.lean`, `FabiusSecondSaddleExpansion.lean`
Legendre expansion and computability	`FabiusLegendreCoefficients.lean`, `FabiusLegendreSeries.lean`, `FabiusLegendreLeastSquares.lean`, `FabiusTranslatedLegendreSeries.lean`, `FabiusComputableSpline.lean`

B Notation crosswalk

Article	Lean	Meaning
F	`fabiusReal fabius`	bounded canonical Fabius function
up	`rvachevUp fabius`	compactly supported fold
$\mathcal{F}$	`extendedFabius fabius`	signed global extension
I	`fabiusInv fabius fabius_spec`	totalized inverse
c_n	`moment n`	even moment
d_n	`halfMoment n`	half moment
ε_n	`thueMorseSign n`	signed Thue–Morse value
$\mathrm{wt}_2(n)$	`binaryWeight n`	binary digit sum

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